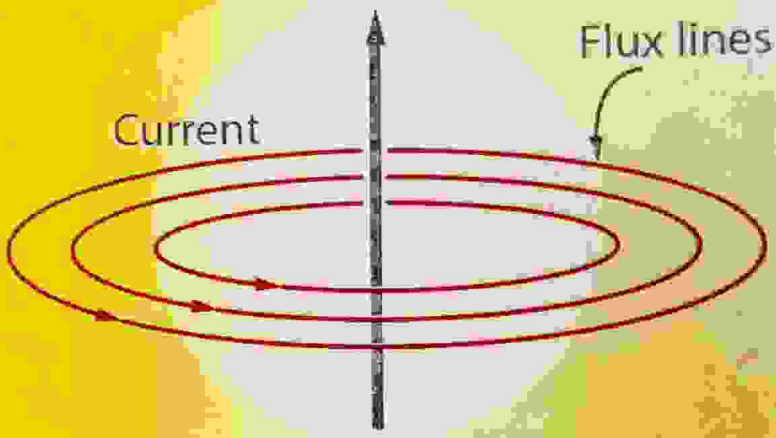
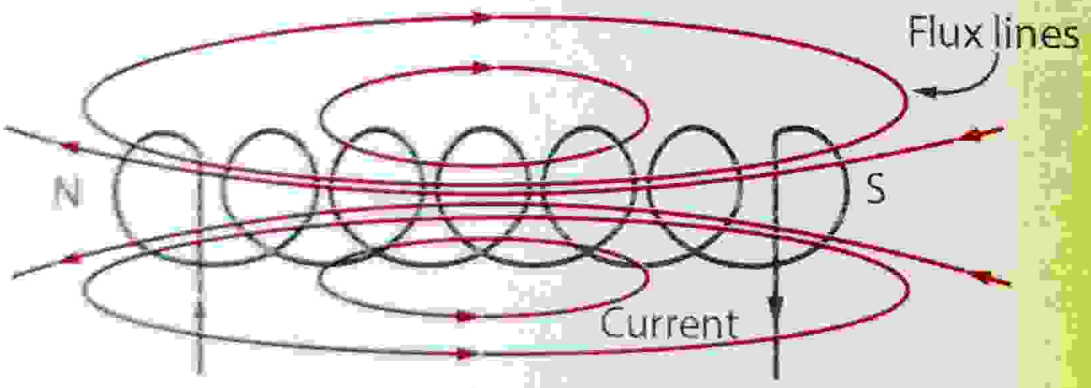
Extreme Series

PHYSICS

GRADE 9-10





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Based on Common Currently Used
National and International Curriculum

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2016 E.C

Abdu Yesuf (Msc)
Arebu Abdella

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UNIT 8

8. Elasticity and Static Equilibrium

8 . 1 Elasticity and Plasticity

8 . 2 Density and Specific Gravity

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8. Elasticity and Static Equilibrium

Dear learner, you are welcome to this wonderful unit where the concepts of material properties such as elasticity, plasticity, density, specific gravity, stress, strain, young's modulus and bulk modulus are clearly defined and explained. what is more, you will also be clear about the first and the second conditions of static equilibrium Have a nice time .

8.1 Elasticity and Plasticity

When a force is exerted on a body, it may cause the body to deform and change its shape and size. And based on their tendency to restore their original shape and size. after the removal of the applied force, materials may have elasticity or plasticity property

Elasticity : Is property of a body by virtue of which it regains its original size and shape after the removal of the applied force,

And,the materials that regain their original size and shape when the deforming force is removed are called *Elastic materials*, and their deformation is called *elastic deformation*.

Spring and rubber band are examples of elastic materials.

Did You know ?

Even an elastic material will no longer behave elasticity after a stress level called an *Elastic limit* and beyond this limit, the material became plastic

Plasticity : Is property of a body by virtue of which it will not regain its original size and shape when the deforming force is removed

And, materials that do not restore their original size and shape when the applied force is removed are called **plastic materials**, and their deformation is called *plastic deformation*.

Mud and wax are examples of plastic materials.

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Illustrative Example

Match the items in column 'A' with those in column 'B'

A	B
1. Deformation	a) Maximum stress level
2. Elasticity	b) Reversible deformation
3. Plasticity	c) Irreversible deformation
4. Elastic material	d) Property to restore size and shape
5. Plastic material	e) Property not to restore size and shape
6. Elastic deformation	f) Restore their size and shape
7. Plastic deformation	g) Can not restore their size and shape
8. Elastic limit	h) Change in size and shape

Answers

1. h
2. d
3. e
4. f
5. g
6. b
7. c
8. a

8.2 Density and Specific Gravity

How particles of a body are packed together is called *density* of the body.

Density : Is the ratio of mass of the body to its volume

$$\text{Density} = \frac{\text{Mass of the body}}{\text{Volume of the body}}$$

$$\rho(\text{rho}) = \frac{m}{v}$$

Density is a scalar quantity and its SI unit is kilogram per meter cube, kg/m^3

And, some times density of a body may be given in gram per centimetre cube, g/cm^3 with the fact that, $1 kg/m^3 = 10^{-3} g/cm^3$

Density is a unique physical property of a material that differentiate it from any other materials

The density of a body as compared to the density of water is

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gravity.

Specific Gravity (Relative Density) : Is the ratio of the density of a body to the density of water at 4^0 c

$$\text{Specific Gravity} = \frac{\text{Density of a body}}{\text{Density of water}}$$

$$SG = \frac{\rho substance}{\rho water}$$

Specific gravity is a unit less scalar quantity. For example, the density of gold is 19.3×10^3 kg/m^3 and its specific gravity is

$$SG = \frac{\rho gold}{\rho water} = \frac{19.3 \times 10^3}{10^3}$$

$$SG = 19.3$$

The density and specific gravity of some substances at 0^0c and 1atm are given below

Material	Density (kg/m^3)	Specific Gravity
Air	1.29	1.29×10^{-3}
Alcohol	7.9×10^2	0. 79
Water	1×10^3	1
Mercury	13.6×10^3	13.6
Glass	2.5×10^3	2.5
Aluminium	2.7×10^3	2.7
Iron	7.86×10^3	7.86
Copper	8.92×10^3	8.92
Silver	10.5×10^3	10.5
Uranium	19.07×10^3	19.07
Gold	19.3×10^3	19.3

Illustrative Examples

9. A body is found to have a mass of 500g and a volume of 47.5cm^3. What

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is

A) Its density

B) Its specific gravity

C) The material

Solution :

A) $\text{Density} = \frac{\text{Mass}}{\text{Volume}} = \frac{0.5}{47.5\times10^{-6}} = 10.5\times10^{3}\,\text{kg/m}^3$

B) $\text{Relative Density} = \frac{\text{Density of body}}{\text{Density of water}} = \frac{10.5\times10^{3}}{10^{3}} = 10.5$

C) It is silver

10. What is the mass of a block of gold with dimension 10cm × 5cm × 4cm

Solution :

$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$, $\text{volume} = (10)(5)(4) = 200\,\text{cm}^3$

$\text{Mass} = (\text{Density})(\text{volume}) = (19.3\times10^{3})(200\times10^{-6}) = 3.86\text{kg}$

11. If the mass of the earth is 6×10^{24} kg and its radius is 6.4×10^{6} m. What is its

A) Density

B) Relative density

Solution :

A) $\text{Density} = \frac{\text{Mass}}{\text{Volume}}$

Since the earth is spherical in shape, its volume is

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$$V = \frac{4}{3}\pi r^3 = \frac{4}{3}(3.14)(6.4\times10^6)^3 = 349.53\times10^{18}\ m^3$$

$$\rho = \frac{M}{V} = \frac{6\times10^{24}}{349.5\times10^{18}} = 17,166\ {}^{kg}/_{m^3}$$

B) $SG = \frac{\rho_{earth}}{\rho_{water}} = \frac{17,166}{1000}$

$SG = 17.166$

Match items in column A with those in column B

A	B
12.Density	a) Ratio of volume to mass
13.Relative Density	b) Ratio of density of water to density of substance
	c) Ratio of density of substance to density of water
	d) Ratio of mass to volume

Answers

12. d 13. c

8.3 Stress and Strain

The magnitude of the force that causes deformation and the amount of deformation caused on the body are described using the concepts stress and strain respectively.

Stress : Is applied force per unit area of the body

$$\text{Stress} = \frac{\text{Force}}{\text{Area}}$$

Stress is a vector quantity and its SI unit is pascal,pa

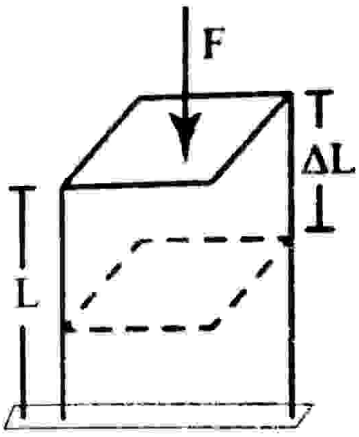
$1pa = 1N/m^2$

Based on the direction of the applied force on the area, stress can be tensile, compressive or shear.

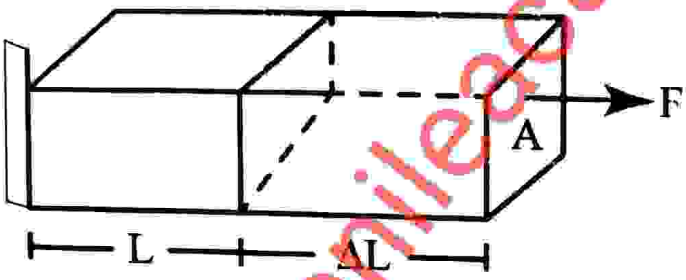
For a tensile and compressive stress, the applied force is perpendicular to

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the cross - sectional area of the body on which the force is exerted.

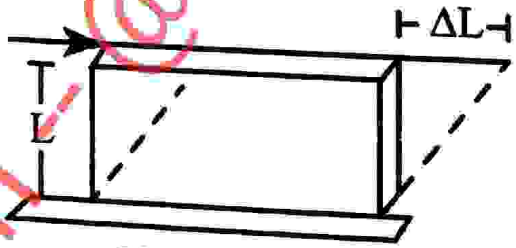


Compressive Stress



Tensile Stress

In a Shear Stress, the applied force is parallel to the cross - sectional area of the body.



Shear Stress

The deformation, that is elongation or compression of the material relative to its original size is described using the concept of *strain*.

Strain : Is the amount of deformation of a body under stress

$$\text{Strain} = \frac{\text{change in size}}{\text{original size}}$$

$$\text{Strain} = \frac{\Delta L}{L}$$

➥ Strain is a unit less quantity and tells us amount of *deformation*.

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When a body is under tensile stress, it gets tensile strain and became thinner and longer. When a body is under compressive stress, it gets compressive strain and became thicker and shorter. When a body is under shear stress, it gets shear strain and shifted side ways

8.4 Young's and Bulk Modulus

The material property that takes into account the stress and strain induced on the material is called *modulus*.

Young's Modulus : Is the ratio of the stress to the strain of a solid material

$$\text{Young's Modulus} = \frac{\text{stress}}{\text{strain}}$$

$$Y = \frac{F/A}{\Delta L/L}$$

$$\boxed{Y = \frac{FL}{A\Delta L}}$$

Young's modulus is expressed in pascal, $1pa = 1N/m^2$

Bulk Modulus : Is the ratio of the stress on a fluid to the strain of the gas or liquid

$$\text{Bulk Modulus} = \frac{\text{Stress}}{\text{Strain}}$$

$$B = \frac{F/A}{\Delta V/V}$$

$$\boxed{B = -\frac{pv}{\Delta v}}$$

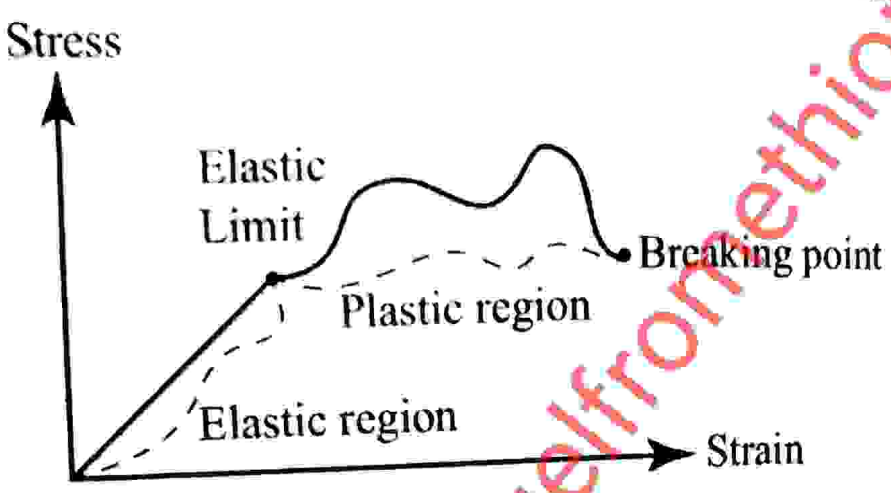
Strain of a fluid is the ratio of the change in volume to its initial volume.

In the above equation of bulk modulus force per area gives pressure, p and the minus sign (-) tells compression.

According to Hook's law, for a ductile material, the stress is proportional to the strain until the elastic limit is reached.

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The relationship between stress and strain is shown in the following graph



After the elastic limit when a force is applied on the material, the material deforms permanently.

Young's modulus of some materials is given below

Material	Young's Modulus (10^{11} pa)
Aluminium	0.70
Brass	0.91
Copper	1.10
Glass	0.72
Quartz	0.56
Steel	2.00
Tungsten	3.50

Illustrative Examples

14. How much load should be suspended on the free end of a 1m copper wire of 10mm diameter so as to stretch it 2mm ?

Solution:

$$Y = \frac{FL}{A\Delta L} \Rightarrow F = \frac{YA\Delta L}{L}, \quad A = \pi r^2$$

$$F = \frac{(1.1\times10^{11})(3.14)(5\times10^{-3})^2(2\times10^{-3})}{1}$$

$$= 17300N$$

15. What will be the increase in length when a 200 kg load is hung on a wire

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of length 4m, thickness $0.2 \times 10^{-5}\ m^2$ and young's modulus 0.8×10^{11} pa ?

Solution:

$$Y = \frac{FL}{A\Delta L} \Rightarrow \Delta L = \frac{FL}{AY}, F = mg$$

$$\Delta L = \frac{(2000)(10)(4)}{(0.2\times10^{-5})(0.8\times10^{11})}$$

$$= 0.5m$$

16. What is the stress induced on a body of cross sectional area $2\times10^{-4}\ m^2$ when a force of 1000N is applied perpendicular to it ?

Solution :

$$\text{Stress} = \frac{F}{A} = \frac{1000}{2\times10^{-4}} =$$

$$\text{Stress} = 5\times10^{6}\text{pa}$$

17. What is the strain of a body of length 4m if it is compressed by 2mm

Solution :

$$\text{Strain} = \frac{\Delta L}{L} = \frac{2\times10^{-3}}{4}$$

$$\text{Strain} = 0.5\times10^{-3}$$

Illustrative Examples

Match the items in column 'A' with those in column 'B'

A	B
18. Stress	a) Ratio of stress to strain in solid
19. Strain	b) Ratio of stress to strain in fluid
20. Young's Modulus	c) Ratio of force to area
21. Bulk Modulus	d) Ratio of deformation to initial size
	e) Ratio of strain to stress

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Answers

18. C 19. d 20. a 21. b

8.5 Static Equilibrium

The word "static" means at rest and the word "Equilibrium" means in balance. Therefore, a body is in static equilibrium if it is not moving and the force and torques acting on it cancel out each other.

Static Equilibrium : Is a condition that occurs when there is no net force and no net torque acting on a body and the body is not moving.

For a body to be in static equilibrium, two conditions must be satisfied. These conditions are the first condition and the second condition of equilibrium.

8.5.1 First Condition of Equilibrium

The First condition of equilibrium also called the *translational equilibrium* prevents the body from linear motion.

The First Condition of Equilibrium states that "the sum of forces acting on the body must be zero"

That is,the sum of the forces acting on the body along the x, and y and the z axes must be zero.

$$\vec{F}_{net} = 0$$

$$\sum F_x = \sum F_y = \sum F_z = 0$$

8.5.2 Second Condition of Equilibrium

The second condition of equilibrium also known as the *rotational equilibrium* prevents the body from rotational motion.

The Second Condition of Equilibrium States that; "the sum of torques acting on the body must be zero."

That is,the sum of the torques acting along the clockwise direction must be equal to the sum of torques acting along the counter clock wise direction.

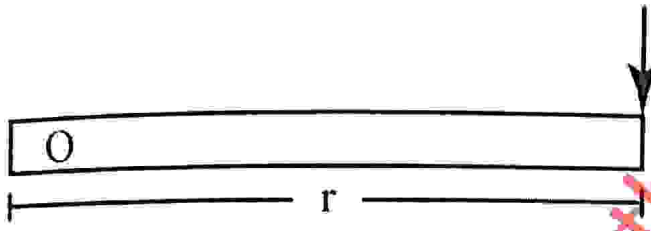
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$\vec{\tau}_{net} = 0$

$\sum \tau$ clockwise $= -\sum \tau$ counter clockwise

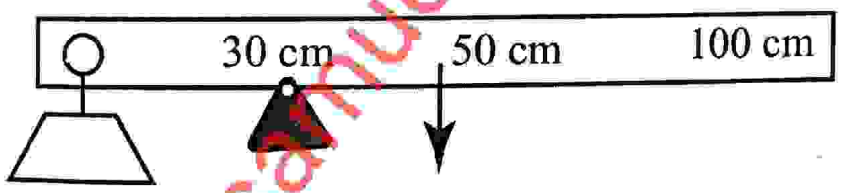
Torque, τ (tau), is the twisting or turning effect of a force and it is the product of the perpendicular components of the force and the lever arm, which is the distance from the pivot point.



$\tau = rF$, Now it is clockwise rotation

Illustrative Examples

22. If a uniform meter stick with a pivot point at the 30cm mark is balanced when a 2kg block is suspended at the 0 cm mark, what is the mass of the stick ?



Solution :

Using the second condition of equilibrium

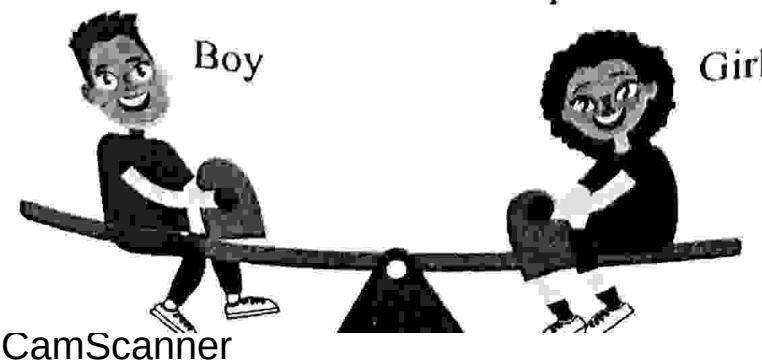
τclockwise $= \tau$counter clockwise

$r_{block} M_{block} = r_{stick} M_{stick}$

$$M_{stick} = \frac{r_{block} M_{block}}{r_{stick}} = \frac{(30)(2)}{20}$$

$= 3kg$

23. At what distance from the pivot should a 32kg girl sit to balance a boy of mass 26kg who sit 1.6m from the pivot of a see saw ?



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Solution :

$$\tau_{clockwise} = \tau_{coonter\ clockwise}$$

$$r_{boy} M_{boy} = r_{girl} M_{girl}$$

$$r_{girl} = \frac{r_{boy} M_{boy}}{M_{girl}} = \frac{(1.6)\,(26)}{32}$$

$$r_{girl} = 1.3m$$

24. Young's Modulus is the

A) Ratio of stress to strain

B) Ratio of force to area

C) Ratio of strain to stress

D) Ratio of force to deformation

Hint : Young's Modulus = stress/strain **Answer : A**

25. When a load of 500kg is hang from a wire of length 3m and cross - sectional area $0.2cm^2$, and young's modulus 2×10^{11} pa, it stretches. What is the amount of elongation ?

A) 3.75 mm B) 2.75 mm C) 1.75 mm D) 0.75 mm

Solution :

$$\text{Young's modulus} = \frac{\text{stress}}{\text{strain}}$$

$$Y = \frac{F/A}{\Delta L/L} = \frac{FL}{A\Delta L} = \frac{mgL}{A\Delta L}$$

$$\Delta L = \frac{FL}{YA} = \frac{(500)(10)(3)}{(2\times10^{11})(0.2\times10^{-4})}$$

$$\Delta L = 3.75\times10^{-3}m$$

$$\Delta L = 3.75mm$$

Answer : A

26. A rope made of steel has a diameter of 5cm, and a breaking strength of 2×10^{8} N. What will be the breaking strength of the wire if its diameter is doubled ?

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A) 2×10^{8} N B) 4×10^{8} N C) 1×10^{8} N D) 8×10^{8} N

Hint: When diameter doubled,cross - sectional area quadrupled breaking strength is proportional to cross - section area .

Answer : D

27. A wire of cross - sectional area $10^{-6}\,m^2$ is increased in length by 0.1 % If the tension is 1000N, What is the young's modulus of the wire ?

A) 10^{12} pa B) 10^{11} pa C) 10^{10} pa D) 10^{9} pa

Solution:

$$Y = \frac{F/A}{\Delta L/L} = \frac{1000/10^{-6}}{0.001}$$

$$Y = \frac{1000}{10^{-9}} = 10^{12}\text{pa}$$

Answer : A

28. A and B are two wires. The radius of A is twice that of B. They are stretched by the same load. Then, the stress on B is

A) Equal to that on A C) Two times that on A

B) Four times that on A D) Half that on A

Hint :

$$\frac{\text{Stress on B}}{\text{Stress on A}} = \frac{F/a_B}{F/a_A} = \frac{r_A^{\,2}}{r_B^{\,2}}$$

$$\frac{\text{Stress on B}}{\text{Stress on A}} = \frac{4r_B^{\,2}}{r_B^{\,2}}$$

$$\text{Stress on B} = 4(\text{Stress on A})$$

Answer : B

29. The extension, of a wire is 3mm. What will be its extension if its radius decreases by half ?

A) 12mm B) 0.75mm C) 6mm D) 15mm

Hint : Same young's modulus

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$$\frac{FL}{A_1\Delta L_1} = \frac{FL}{A_2\Delta L_2}$$

$$A_2\Delta L_2 = A_1\Delta L_1$$

$$\pi r_2^2 \Delta L_2 = \pi r_1^2 \Delta L_1$$

$$\left(\frac{1}{2}r_1\right)^2 \Delta L_2 = r_1^2 (3)$$

$$\Delta L_2 = (3)(4) = 12\text{mm}$$

Answer : A

30. What force is needed to double the length of steel wire of constant cross - sectional area $1cm^2$ ($Y = 2 \times 10^{11}$ pa)

A) 2×10^{11} N B) 2×10^{10} N C) 2×10^{8} N D) 2×10^{7} N

Hint :

$$Y = \frac{\text{Stress}}{\text{Strain}} = \frac{F/A}{\Delta L/L}$$

$$F = YA = (2\times10^{11})(1\times10^{-4}) = 2\times10^{7}\text{N}$$

Answer : D

31. In an experiment, for the determination of Young's modulus of a wire, when the length and the applied force are doubled, the Young's modulus will?

A) Remain constant
B) Become doubled
C) Become four times
D) Become sixteen times

Hint : Young's modulus is material property.

Answer : A

32. Copper wire and a steel wire of the same diameter and length are connected end to end and a force caused their combined length to strech by 1cm Then, the two wires will have

A) Same stress and strain
B) Same stress, different strain
C) Same strain, different stress
D) Different stress and strain

Hint : Different deformation level.

Answer : B

33. The Young's modulus of a wire of length L and radius R is Y. What will

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be the young's modulus of the wire if its length is doubled and its radius is halved ?

A) 2 Y B) Y C) 0.5 Y D) 8 Y

Hint : Young's modulus do not depend on material geometry

Answer : B

34. A wire is stretched by 1mm, what will be its stretch if the applied force is doubled and the radius is halved ?

A) 8mm B) 4mm C) 2mm D) 1mm

Hint : $\text{Strain} = \frac{\Delta \ell}{\ell}$

Answer : A

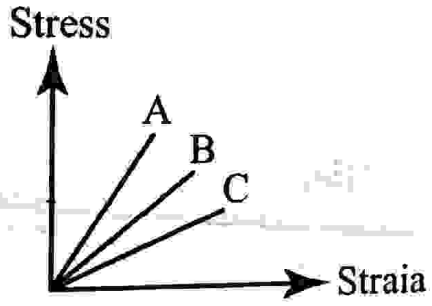
35. The stress - strain graph for brass, steel and rubber are given. The lines A, B and C are respectively for

A) Steel, brass and rubber

B) Brass, steel and rubber

C) Steel, rubber and brass

D) Rubber, steel and brass



Hint : Compare the strain for the same stress

Answer : A

36. Two wires of the same radius and material have lengths in the ratio 1:2. what will be the ratio of their strains when subjected to the same force ?

A) 1:1 B) 1:2 C) 2:1 D)1:4

Hint : $\text{Strain} = \frac{\Delta \ell}{\ell}$

Answer : A

37. What force is needed to stretch a 2m wire of cross - sectional area 2 mm^2 by 0.5mm ($Y = 2.2 \times 10^{11}$ pa)

A) 1.1×10^5 N B) 1.1×10^4 N C) 1.1×10^3 N D) 1.1×10^2 N

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Solution :

$$Y = \frac{FL}{A\Delta L} \Rightarrow F = \frac{YA\Delta L}{L}$$

$$F = \frac{(2.2\times10^{11})(2\times10^{-6})(0.5\times10^{-3})}{2}$$

$$= 1.1\times10^{2}\,N$$

38. A Cable is replaced by another one of the same length and material but of twice the diameter. The elongation would now become ; **Answer : D**

A) Double
C) One - fourth
B) One - half
D) One - sixth

Hint : $Y = \frac{FL}{A\Delta L} \Rightarrow \Delta L \sim \frac{1}{A}$

$$\Delta L \sim \frac{1}{d^2}$$

Answer : C

39. Two wires of the same material and length are stretched by the same force. If their masses are in the ratio 3:2 Their deformations are in ratio;

A) 3:2 B) 9:4 C) 2:3 D) 4:9

Hint : $\Delta L \sim \frac{1}{A}$, $mass = LA\rho$

$$\Delta L \sim \frac{1}{m} \Rightarrow \frac{\Delta L_1}{\Delta L_2} = \frac{m_2}{m_1} = \frac{2}{3}$$

Answer : C

40. If the stress applied on a wire is equal in magnitude to the Young's modulus. The deformation of the wire will be equal to ;

A) Its original length
C) Twice its length
B) Half of its length
D) Four times its length

Hint : $Y = \frac{F\Delta L}{AL} = \frac{stress}{\Delta L/L} = \frac{\Delta L}{L} = 1$

$$\Delta L = L$$

Answer : A

41. The dimensions of four wires of the same material and stress are given

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below. Which wire will have maximum deformation ?

A) Length 100cm, Diameter 1mm

B) Length 200cm, Diameter 2mm

C) Length 300cm, Diameter 3mm

D) Length 50cm , Diameter 0.5mm

Hint : $Y = \frac{\text{stress}}{\text{strain}} = \frac{F/A}{\Delta L/L} = \frac{FL}{A\Delta L}$

$$\Delta L = \frac{FL}{AY} = \frac{\Delta FL}{\pi d^2 Y} = \Delta L \sim \frac{L}{d^2}$$

$$\frac{100}{(1)^2} = 100, \frac{200}{(2)^2} = 50, \frac{300}{(3)^2} = 33.33, \frac{50}{(0.5)^2} = 200$$

Answer : D

End of unit Questions and problems

I. Give Short Answers to the Following Questions

1. What is Elasticity ?
2. What is Plasticity ?
3. What is Stress ?
4. What is Strain ?
5. What is Elastic Limit ?
6. What is Youngster's modulus ?
7. What is Bulk modulus
8. State Hooke's Law
9. What is Static Equilibrium ?
10. Explain the two conditions of equilibrium
11. What is Torque ?

II. Solve the following problems

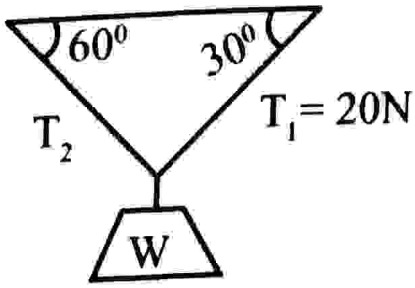
12. A block is suspended by two ropes as shown find

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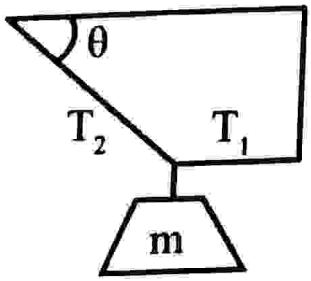
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A) Tension, T_2

B) Weight, W



13. What are the tensions in the two ropes given that $\theta = 30^0$ and $m = 200\,kg$



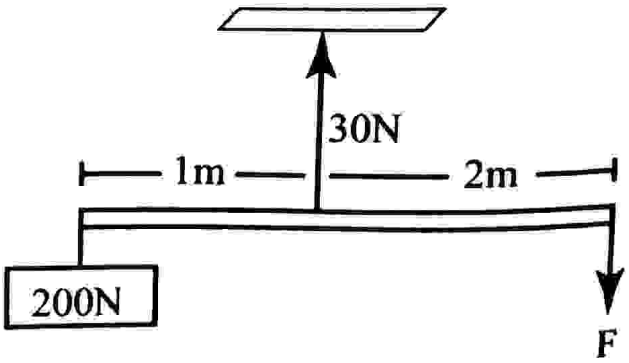
14. What force is required to double the length of a steel wire of cross - sectional area $0.1cm^2$

15. A uniform beam of 1m length is supported by a cable of tension force 500N at a distance of 20cm from one end. What is the weight of the beam ?

16. A wire of young's modulus $13.78\times10^{10}\,N/m^2$, length 4m and diameter 4 mm is stretched when a 4kg mass is suspended on it. What is its deformation ?

17. Two wires A and B are made of the same material. If wire A has twice the length and twice the diameter of wire B. What will be the ratio of their deformations when subjected to the same load ?

18. What should be the value of F so that the system will be in equilibrium ?



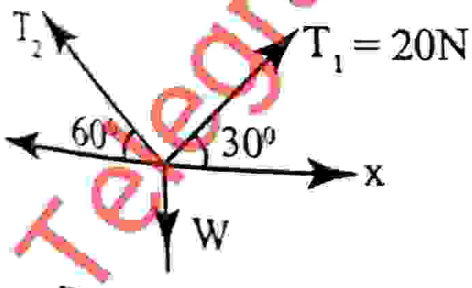
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Answers to the End of unit Questions

1. Elasticity is the property of a body by virtue of which it regains its original size and shape when the deforming force is removed.
2. Plasticity is the opposite of elasticity.
3. Stress is force per unit area .
4. Strain is deformation per original size.
5. Elastic limit is the maximum stress level beyond which the body deform permanently.
6. Young's modulus is the ratio of stress to strain.
7. Bulk modulus is the ratio of stress to volume strain.
8. Hooke's law states that;stress is proportional to strain until the elastic limit.
9. Static equilibrium is a condition for which the sum of forces and torques acting on a body are zero
10. First condition, which states that all forces acting on a body add up to zero. And second condition states that clockwise torque acting on a body is equal to the counter clock wizen torque acting on the body;
11. Torque is the turning or twisting effect of a force acting on a body.

Solution to End of unit problems

12.



A) $\Sigma F_x = 0$

$T_1\cos 30^0 - T_2\cos 60^0 = 0$

$(20)(0.867) = T_2(0.5)$

$T_2 = 34.6N$

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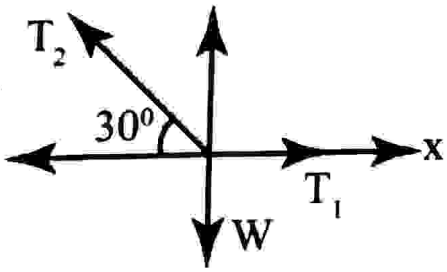
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B) $\sum F_y = 0$

$T_1 \sin 30^0 + T_2 \text{sine} 60^0 = W$

$W = 20(0.5) + 34.6(0.867) = 40N$

13.



$\sum F_x = 0$

$T_1 - T_2 \cos 30^0 = 0$

$T_1 = 0.867 T_2 \ldots\ldots\ldots\ldots (*)$

$\sum F_y = 0$

$T_2 \text{sine} 30^0 - W = 0$

$T_2 (0.5) = mg = (200)(10) = 4000N$

Substitute for T_2 into equation(*)

$T_1 = 0.867(4000) = 3468N$

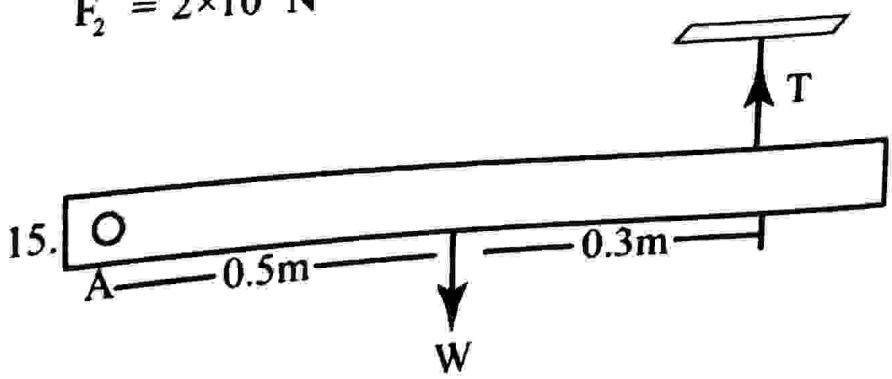
14. $Y = \dfrac{F_1 L}{A \Delta L_1} \Rightarrow F_1 = \dfrac{Y A \Delta L_1}{L}$

$F_2 = \dfrac{Y A \Delta L_2}{L}$ $= , \Delta L_2 = L$

$F_2 = YA = (2 \times 10^{11})(0.1 \times 10^{-4})$

$F_2 = 2 \times 10^6 \, N$

15.



Let point A be pivot point

$0.5 W = 0.8T$

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$W = 1.6T = 1.6\ (500) = 800N$

16. $Y = \dfrac{FL}{A\Delta L} = \Delta L \Rightarrow \dfrac{FL}{AY}, F = mg$

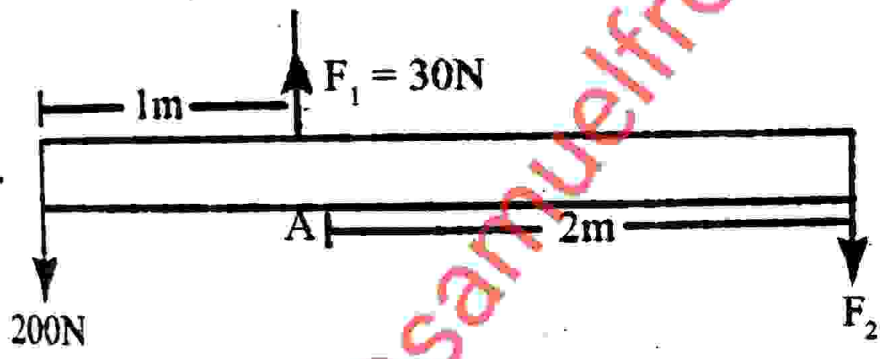
$$\Delta L = \frac{mgL}{AY} = \frac{4(10)(4)}{\pi(2\times10^{-3})^2(13.78\times10^{10})}$$

$$\Delta L = 0.09\times10^{-3}m = 0.09mm$$

17. $Y = \dfrac{FL_A}{\pi r_A^{\,2}\,\Delta L_A}$ and $Y = \dfrac{FL_B}{\pi r_B^{\,2}\,\Delta L_B}$

$$\frac{\Delta L_A}{\Delta L_B} = \frac{1}{2}$$

18.



Let A be pivot point

$F_2(2) = 200(1) \Rightarrow F_2 = F = 100N$

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UNIT 9

9 . Static and Current Electricity

9.1 Charges in Nature

9.2 Methods of Charging a Body

9.3 The Electroscope

9.4 Electrical Discharge

9.5 Coulomb's Law

9.6 Electric Field

9.7 Electric Circuits

9.8 Current, Voltage and Ohm's Law

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9. Static and Current Electricity

You are welcome to this incredible unit where you will be able to know charges in nature, describe methods of charging, explain the working principle of electroscope, state coulomb's law, examine the nature of current, voltage and resistance in series and parallel combinations. Know how to use Ammeter and Voltmeter in an electric circuit and understand the electrical safety rules. Enjoy the lesson!

9.1 Charges in Nature

In nature, a body is neutral since it, contains equal number of electrons and protons carrying negative and positive charges respectively.

A body is then said to be *charged* when it gained or lost electrons.

A body is said to be negatively charged when it gained electrons and positively charged when it lost electrons.

Charge is a scalar quantity and its SI unit is the coulomb(C).

From the principle of quantization of charge, the smallest charge that exists so far is the charge carried by an electron or proton and its magnitude is;

$$e = 1.6 \times 10^{-19}\,c$$

The amount of charge a body possessed is then obtained by the relation.

$$\boxed{Q = ne}$$

Where: Q = Charge carried by the body.

n = number of electrons gained or lost.

e = charge of an electron, = 1.6×10^{-19}C

Just like the mass, energy and momentum of an isolated system, charge is also conserved.

For an isolated system, charge is neither created nor destroyed but transferred from one body to the other.

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☑ An isolated system is a system that does not interact with the surrounding. Any form of exchange like energy and charge is within itself.

Illustrative Example

1. What is an electrically neutral body?

Explanation:

A neutral body is a body with equal number of electrons and protons.

2. What are the types of charges exit in nature?

Explanation:

Negative charge carried by electrons and positive charge carried by protons.

3. When do we say that a body is positively charged?

Explanation:

When it lost electrons.

4. State the law of conservation of charge

Explanation:

Charge is neither created nor destroyed it only transferred from one body to other.

5. What does it mean when we say charge is quantized

Explanation:

It means, charge exists in a discrete amount as an integer multiple of the charge of an electron.

6. State the fundamental law of electrostatics

Explanation:

Like charge repel each other and unlike charges attract each other.

7. What is electrostatics or static electricity?

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Explanation

It is the study of charges at rest.

8. How many electrons are there in one coulomb of charge?

Solution:

$$Q = ne$$

$$n = \frac{Q}{e} = \frac{1}{1.6\times10^{-19}}$$

$$= 6.25\times10^{18} \text{ electrons}$$

9. How many charges are carried by a body that lost 1000 electrons

Solution:

$$Q = ne$$

$$Q = (1000)(1.6\times10^{-19})$$

$$= 1.6\times10^{-16} \text{c}$$

10. Body A carrying a 500c charge is made in contact with body B carrying -1500c charge, after contact, what will be;

A. Charge carried by each body

B. Number of electrons lost or gained by each body.

Solution

a. $$Q = \frac{Q_A + Q_B}{2} = \frac{(500)(-1500)}{2}$$

$$= -500\text{c}$$

b. $$Q = ne \Rightarrow n = \frac{Q}{e} = \frac{500}{1.6\times10^{-19}}$$

$$= 3.125 \times 10^{17} \text{ electrons}$$

Keep in mind that; For charges at rest:

☑ Charges of the same sign repel each other and charges of opposite

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sign attract each other

- ☑ Total charge of an isolated system is conserved.
- ☑ Charge is quantized and hence, net charge on a body is determined as the integer multiple of the basic charge. $e = 1.6 \times 10^{-19}$c

$$Q = ne$$

n = number of electrons gained or lost by the body

$e = 1.6 \times 10^{-19}$c, charge carried by a proton or an electron.

- ☑ The electric filed lines are radically outward from a positive charge and radially inward towards a negative charge.

9.2 Methods of Charging a Body

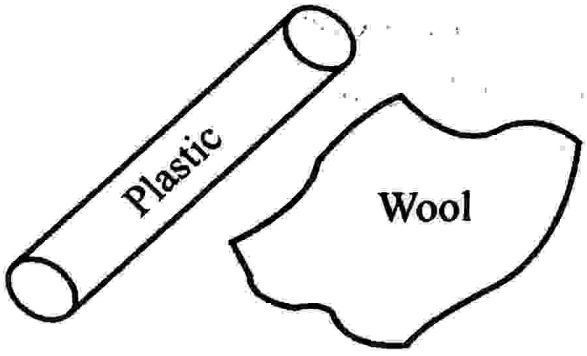
The process of adding or removing electrons from a body is called *charging or electrifying*.

The well known mechanisms of charging a body are rubbing, conduction and induction.

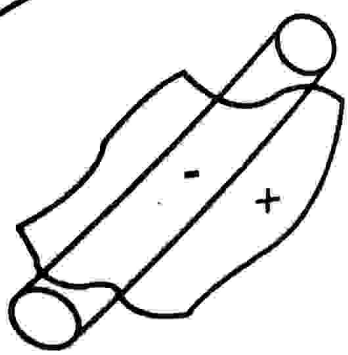
Charging by Rubbing

It involves the transfer of electrons from one body to the other when two *neutral* bodies are rubbed together.

For example, when a neutral plastic rod and a neutral wool cloth are rubbed together, they became charged.

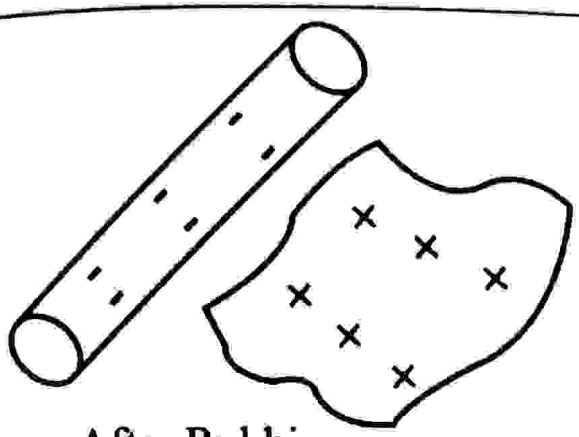


Before Rubbing



DuringRubbing

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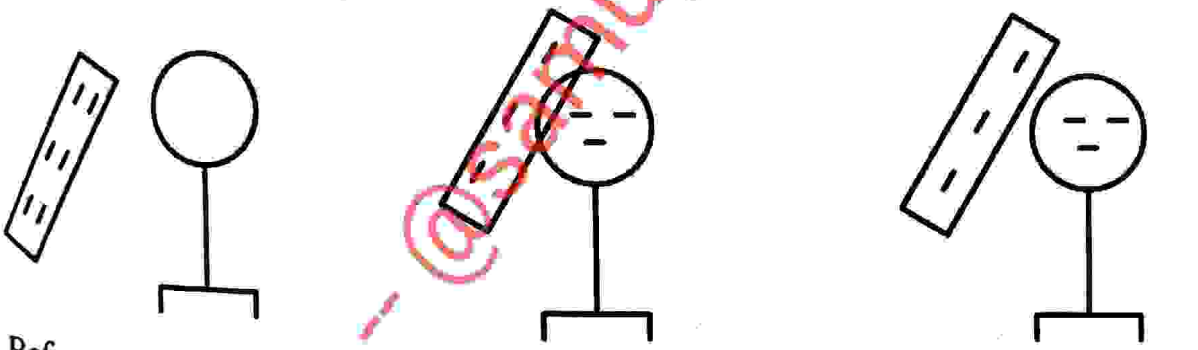
After Rubbing

☑ After the process of charging by rubbing the two bodies become *oppositely* charged

Charging by Conduction

It involves the sharing of charges between a *charged* body and a *neutral* body when they are made in contact.

For example, when a negatively charged rubber rod is made in contact with a metal sphere,the sphere become charged.



Before contact During Contact After Contact

☑ After the process of charging by conduction, the two bodies carry charges of the *same sign*.

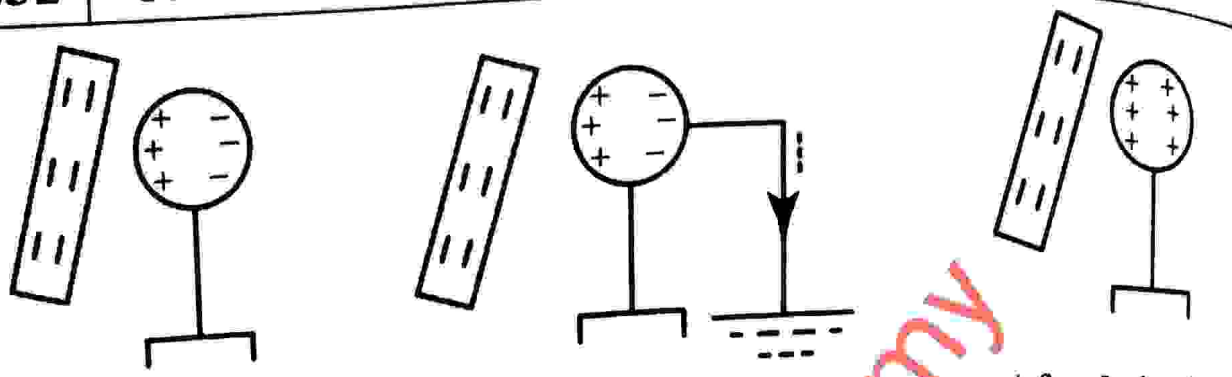
Charging by Induction

It involves the formation of electric polarities when a charged body is brought closer to a neutral body.

For example, when a negatively charged glass rod is brought very close but not in physical contact to a neutral metal sphere, electric poles are formed on the sphere.

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Polarization Grounding After Induction

☑ After the process of charging by induction, the two bodies become *oppositely* charged.

Illustrative Example

11. What is charging?

Explanation:

It is the process of adding to or removing from a body electrons.

12. What are the typical methods of charging?

Explanation:

Rubbing or friction, conduction or sharing and induction.

13. In which process of charging will bodies carrying the same type of charge?

Explanation:

Charging by conduction gives rise to same type of charge. However, charging by rubbing and induction left the bodies oppositely charged.

9.3 The Electroscope

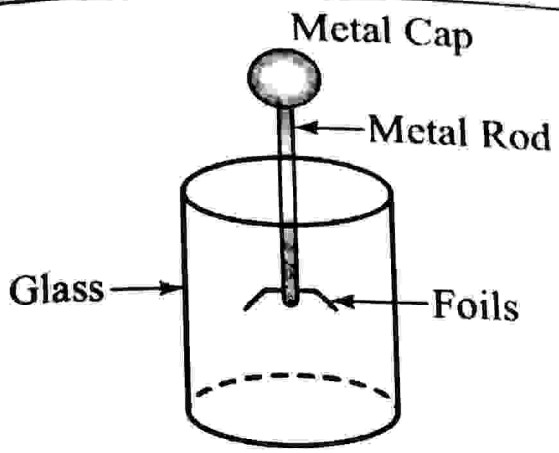
To know whether a body is electrically charged or not we use a device called an *electroscope.*

Electroscope: Is a device that is used to detect the presence of an electric charge on a body.

A typical electroscope consist of a metal cap, a metal rod, foils of gold or aluminium in a glass jar as shown.

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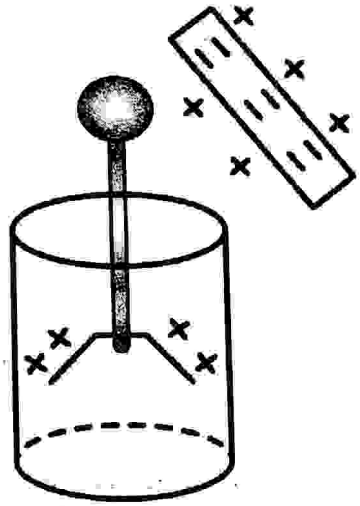
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Typical Electroscope

The collapsing or diverging of the leaves of the foils indicates the presence of charge on body.

When a positively charged rod is brought closer with out touching the metal cap, positive charge is induced in the foils and they diverge.



Foils Diverge

Typical Electroscope

9.4 Electrical Discharge

A charged body can be made neutral by the process of *discharging*.

Discharging: Is the process of removing charges from a charged body.

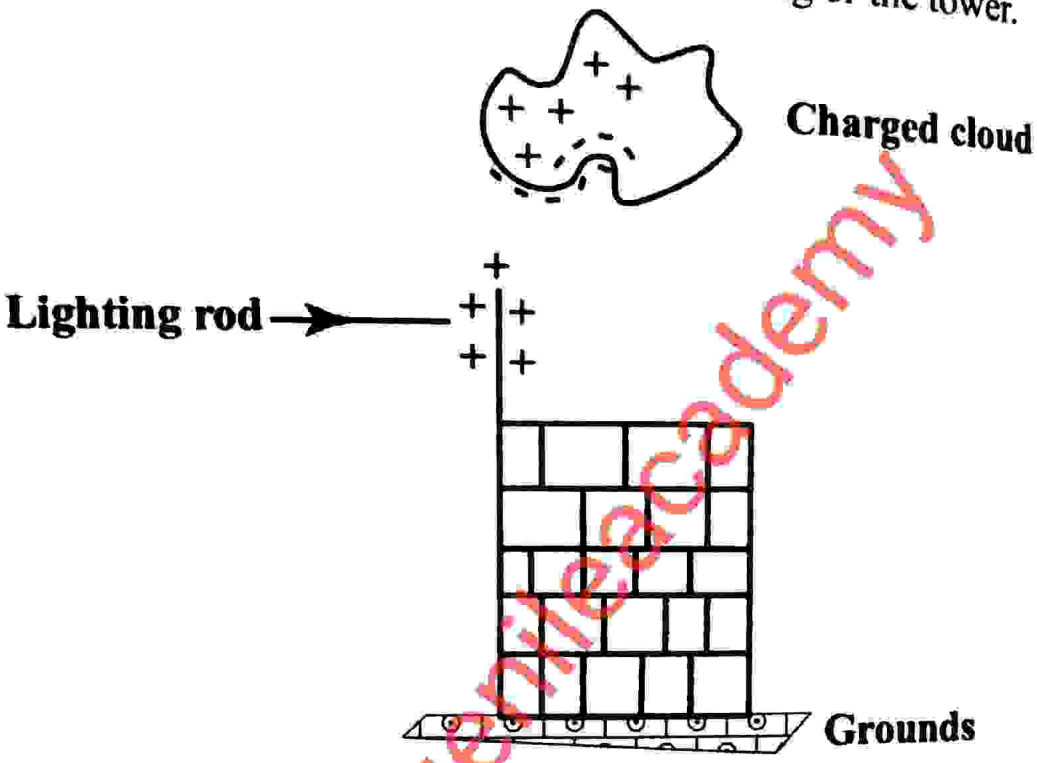
The production of sparks is an examples of discharging a charged body.

The Lighting Rod

The rod that prevents buildings and towers from the damage of lighting is known as lighting rod. The working principle of a lighting rod is that, when

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a charged cloud approaches it, opposite charge to that of the cloud will be induced on its tips and it fight back the charges of the cloud or allow a path way to the ground with out damaging the building or the tower.



☑ Lighting is an electromagnetic wave with huge amount of heat caused by the discharging of oppositely charged cloud layer followed by a thunder storm of terrific sound caused by the difference in pressure of air column. If lightning falls on a body it may trigger a huge damage.

Illustrative Examples

14. What is an electroscope?

Explanation:

It is a device that is used to, detect charge on a body and identify the type of charge carried by the body.

15. What is discharging?

Explanation:

It is the process of removing charge from a charged body.

16. How is lighting formed?

Explanation:

It is formed when two oppositely charged cloud layers made contact.

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17. What is lighting rod?

Explanation:

It is a metal rod that is fixed at the top of a building or a tower so as to keep it safe from the lighting damage.

18. What are the dos and don't dos during lighting ?

Explanation:

Things adviced during heavy rain are; get shelter inside a house or a car, and avoid yourself from sources of electromagnetic wave such as phone, and don't stand under a tree, don't expose yourself to a place where there is a potential of lighting strike.

9.5 Coulomb's Law

The magnitude of the force of attraction or repulsion between two point charges was discovered by Charles Coulomb

Coulomb's Law: States that ; the magnitude of the electrostatic force between two point charges is directly proportional to the product of the magnitude of the charges and inversely proportional to the square of the distance between the charges.

That is, if we have two charges Q_1 and Q_2 a distance r apart,

Q_1 ——————— Q_2
r

Then, $F \sim Q_1Q_2$ and

$$F \sim \frac{1}{r^2}$$

$$\Rightarrow F \sim \frac{Q_1Q_2}{r^2},$$

$$F = \frac{1}{4\pi e_{\circ}} \frac{Q_1Q_2}{r^2}$$

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$$k = \frac{1}{4\pi e_o}$$

$$\boxed{F = \frac{kQ_1Q_2}{r^2}}$$

Where: $e_o = 8.85\times10^{-12}c^2/Nm^2$ and is called *permittivity of free space*

$K = 9\times10^9 Nm^2/c^2$ and is called the *electrostatic constant.*

Illustrative Examples

19. What is the magnitude and nature of the electrostatic force between two point charges, $Q_1 = 2\times10^{-6}c$ and $Q_2 = -4\times10^{-3}c$ that are 3cm apart?

Solution:

$$F = \frac{kQ_1Q_2}{r^2} = \frac{(9\times10^9)(2\times20^{-6})(4\times10^{-3})}{(3\times10^{-2})^2}$$

$F = 8\times10^4 N$, attractive

20. If the electrostatic force between two point charges is 400N. What will be the force between these point charges if their separation is

A. Doubled

B. Halved

Solution:

$$F_1 = \frac{kQ_1Q_2}{r_1^2} = 400N$$

A. $F_2 = \frac{kQ_1Q_2}{r_2^2}$, $r_2 = 2r_1$

$$F_2 = \frac{kQ_1Q_2}{(2r_1)^2} = \frac{1}{4}\frac{kQ_1Q_2}{r_1^2}$$

$$= \frac{1}{4}(F_1) = \frac{400}{4} = 100N$$

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B. $F = \frac{kQ_1Q_2}{r_1^2} = 400N$

$$F_2 = \frac{kQ_1Q_2}{(2r_2)^2}, r_2 = \frac{1}{2}r_1$$

$$F_2 = \frac{kQ_1Q_2}{\left(\frac{1}{2}r_1\right)^2} = 4\frac{kQ_1Q_2}{r_1^2}$$

$$4(F_1) = (400)4 = 1600N$$

21. What is the separation between an electron and a proton if the force of attraction between them is 0.01N

Solution:

$$F = \frac{kQ_1Q_2}{r^2} = \frac{(9\times10^9)(1.6\times10^{-19})(1.6\times10^{-19})}{r^2} = 0.01$$

$$r^2 = 23.04\times10^{-27}$$

$$r = 4.5\times10^{-13}m$$

22. What is the repulsive force between two electrons that are 3mm apart?

Solution:

$$F = \frac{kQ_1Q_2}{r^2} = \frac{(9\times10^9)(1.6\times10^{-19})(1.6\times10^{-19})}{(3\times10^{-3})^2}$$

$$F = 2.56\times10^{-23N}$$

23. The electrostatic force between two point charges of 100c and -200c is 50N. What will be the force between these charges if they are made to contact and then placed at their initial position?

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Solution:

$$F_1 = \frac{kQ_1Q_2}{r^2} = \frac{(9\times10^9)(100)(200)}{r^2} = 50N$$

And, after contact, the charges are going to be

$$Q_1 = Q_2 = \frac{Q_1+Q_2}{2} = \frac{(100)+(-200)}{2} = -50C$$

$$F_2 = \frac{kQ_1Q_2}{r^2} = \frac{(9\times10^9)(50)(50)}{r^2}$$

$$F_2 = \frac{\frac{(9\times10^9)(2500)}{(9\times10^9)(20000)}}{50}, \quad r^2 = \frac{(9\times10^9)(20000)}{50}$$

$$F_2 = 6.25N$$

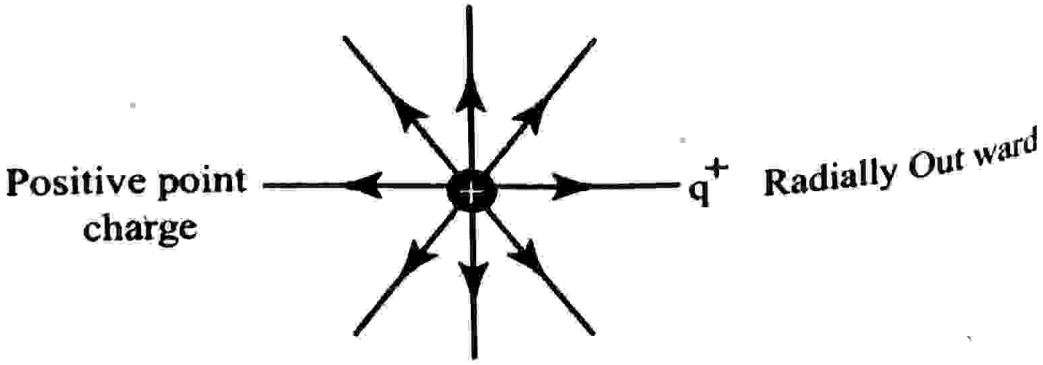
9.6 Electric Field

The field where an electric force is felt by a charged particle is called an *electric field.*

Electric Field: Is the region around a charged body where a test charge experiences an electrostatic force.

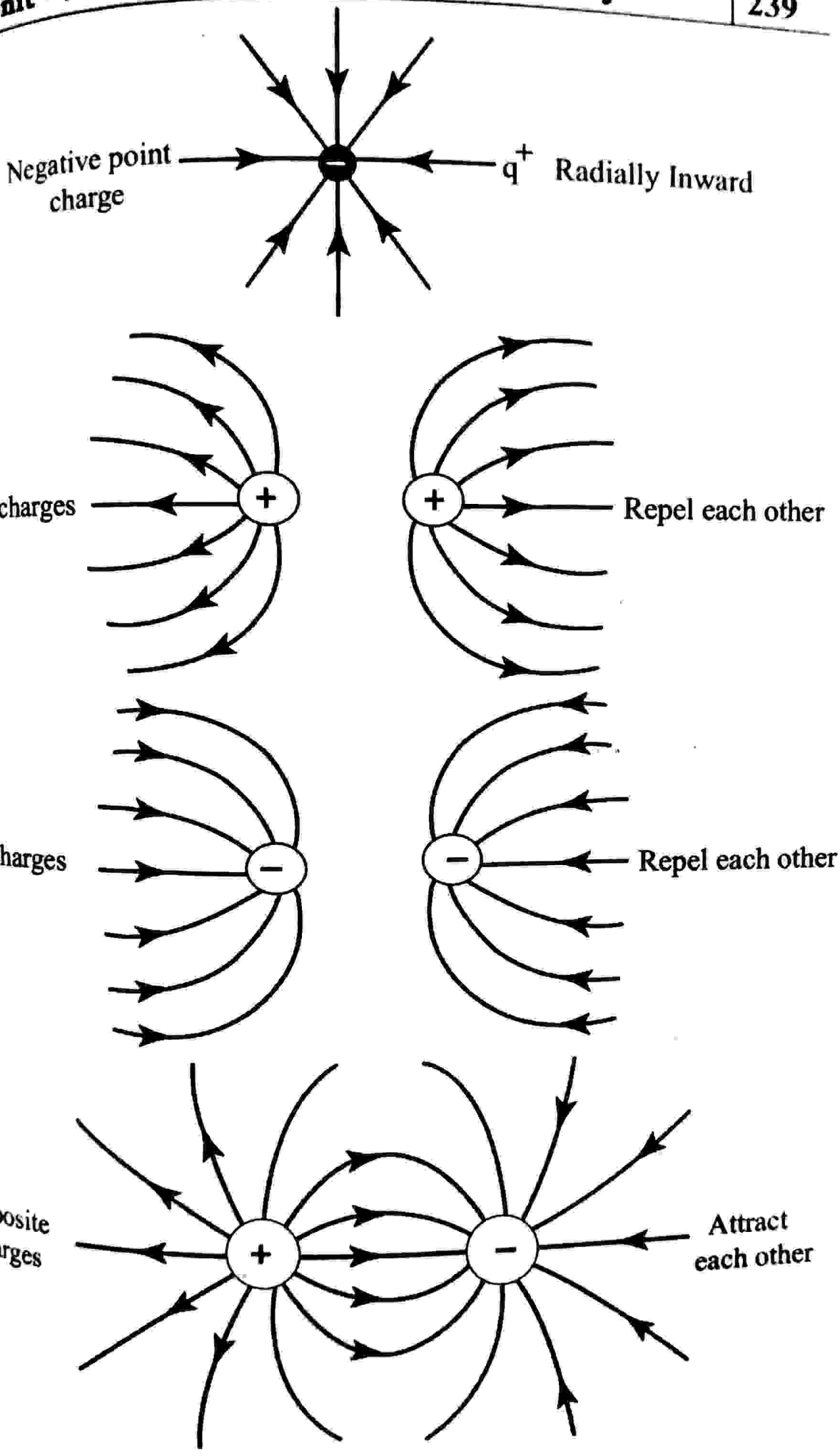
To visualize the electric field of a point charge, we use imaginary lines called *electric field lines.* And, to detect the presence of a point charge, we use a small positive charge called a *test charge,* q^+

Now, let us examine the nature of the field lines for different charge arrangements.

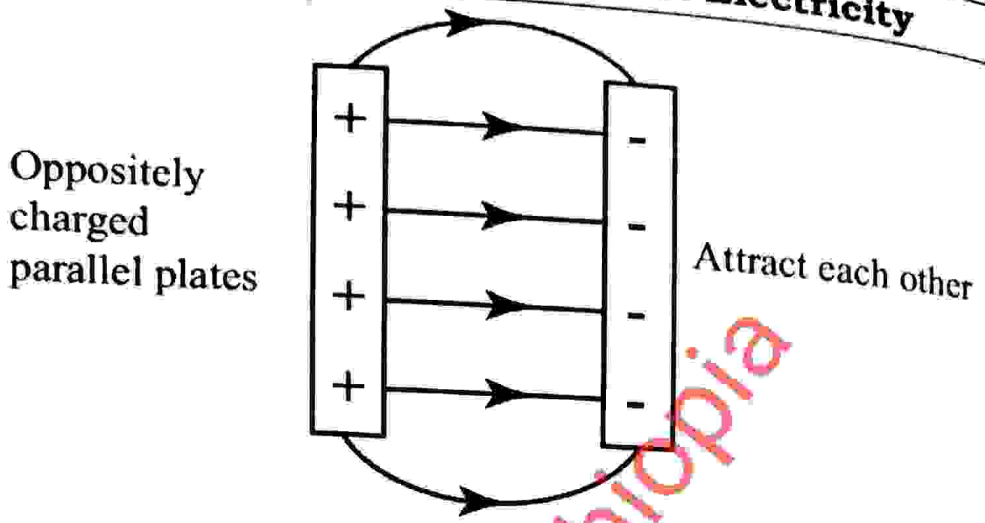


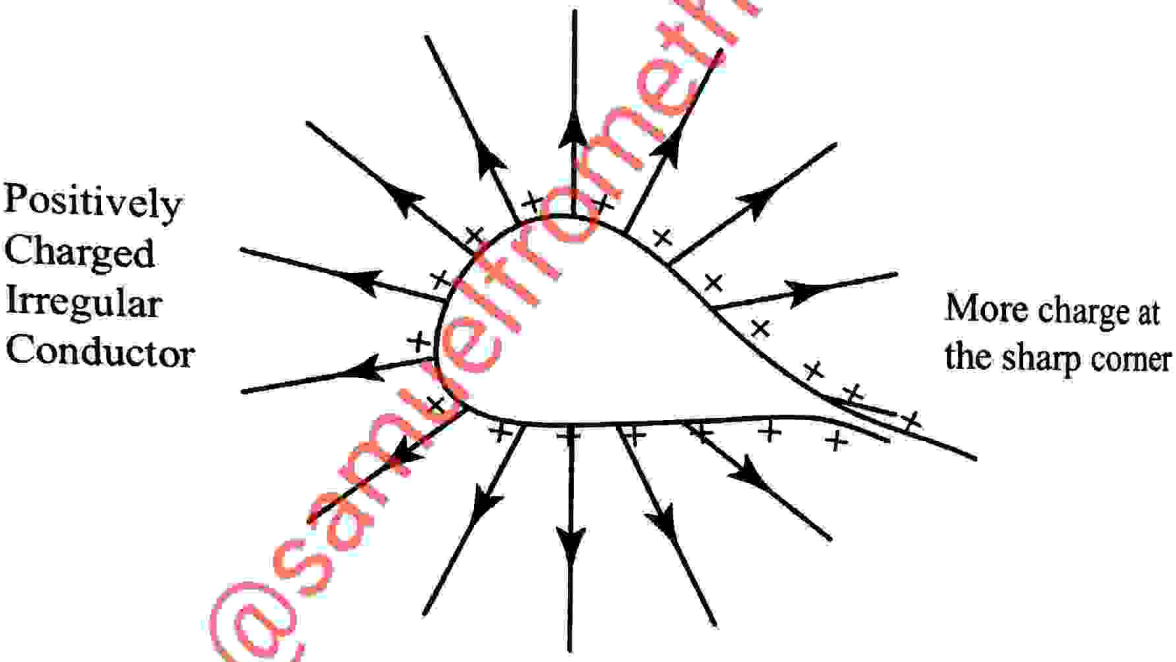
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Characteristics of Electric Field Lines

Electric field lines do have the following properties

- ☑ They run from a positive point charge to a negative point charge.
- ☑ They are symmetrical to each other
- ☑ They never cross each other
- ☑ Where they are closer together the field is strong
- ☑ They are denser at sharp points of a conductor

Electric Field Strength

The electric field strength or intensity at a point due to a charged body is defined as follows

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Electric Field Strength: Is the electrostatic force per unit positive charge exerted on a test charge.

$$\text{Electric Field Strength} = \frac{\text{Electric Force}}{\text{Charge}}$$

$$\boxed{\vec{E} = \frac{\vec{F}}{Q}}$$

Electric field strength is a vector quantity and its SI unit is Newton per Coulomb,(N/C)

For a point charge, the electric field strength can be obtained as follows

Q ——————— P
r

At point P, the electric field strength is $E = \frac{F}{Q^+} = \frac{kQQ^+}{r^2/Q^+}$

$$\boxed{E = \frac{kQ}{r^2}}$$

And,the net electric field strength at a point due to many point charges is obtained using the super-position principle of vector addition

☑ The direction of the electric field is radially outward from a positive charge and radially inward from a negative charge.

Illustrative Example

24. What is the electric field strength of a 3μc point charge at distance of 6cm

Solution:

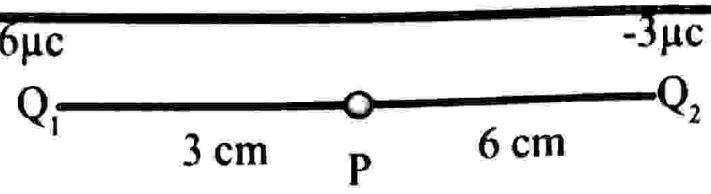
$$E = \frac{kQ}{r^2} = \frac{(9\times10^9)(3\times10^{-6})}{(6\times10^{-2})^2}$$

$$E = 7.5\times10^6\ \text{N/c}$$

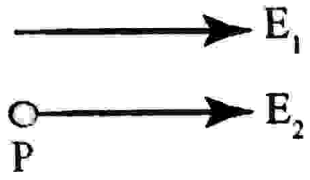
25. What is the net electric field at point P due to the two charges?

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Solution:



At point P, we assume a test charge and the nature of the electric field at that point is as shown below,



$$E_P = E_1 + E_2$$

$$E_p = \frac{kQ_1}{r_1^2} + \frac{kQ_2}{r_2^2}$$

$$E_P = \frac{(9\times10^9)(6\times10^{-6})}{(3\times10^{-2})^2} + \frac{(9\times10^9)(3\times10^{-6})}{(6\times10^{-2})^2}$$

$$E_P = 6.75\times10^7\, N/c$$

26. What is the electric force exerted on a 3.5μc charge by charge by a 250N/C electric field directed due east?

Solution:

$$E = \frac{F}{Q} \Rightarrow F = EQ$$

$$F = (250)(3.5\times10^{-6})$$

$$F = 8.75\times10^{-4}\,N$$

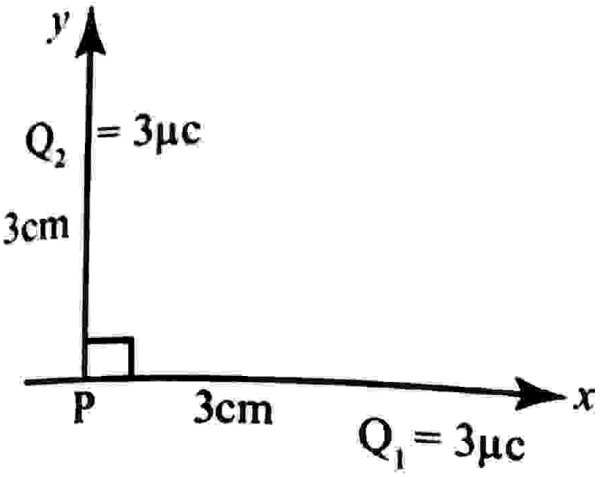
27. What is the net electric field strength at point P

A)



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B)



Solution:

A. $E_P = E_1 - E_2 = \frac{kQ_1}{r_1^2} - \frac{kQ_2}{r_2^2}$

$$E_P = 9\times10^9\left[\frac{3\times10^{-6}}{(3\times10^{-2})^2} - \frac{3\times10^{-6}}{(3\times10^{-2})^2}\right] = 0$$

B. $E_P = \sqrt{E_1^2 + E_2^2} = \sqrt{\left(\frac{kQ_1}{r_1^2}\right)^2 + \left(\frac{kQ_2}{r_2^2}\right)^2}$

$$E_P = \sqrt{\frac{9\times10^9(3\times10^{-6})}{(3\times10^{-2})^2} + \frac{9\times10^9(3\times10^{-6})}{(3\times10^{-2})^2}}$$

$= 7.5\times10^{7}$ N/C

28. If the electrostatic force between two identical point charges is F, then what will be the force between them if their charges are doubled and their separation is halved?

A. 16F B. 8F C. 4F D. 2F

Solution:

$$F_1 = \frac{kQ_1Q_2}{r_1^2} = \frac{kQ^2}{r_1^2} = F$$

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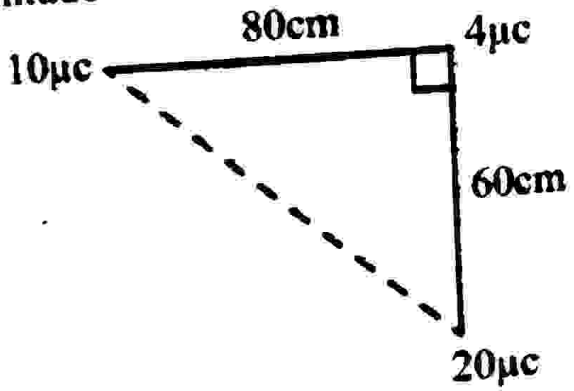
$$F_2 = \frac{kQ_1Q_2}{r_2^2}, Q_1 = 2Q,\ Q_2 = 2Q \text{ and } r_2 = \frac{r_1}{2}$$

$$F_2 = \frac{k(2Q)(2Q)}{\left(\frac{r_1}{2}\right)^2} = \frac{4kQ^2}{r_1^2/4}$$

$$F_2 = \frac{16kQ^2}{r_1^2} = 16F$$

Answer: A

29. What is the magnitude of the electrostatic force on the 20μc charge



A. 5N B. 4N C. 3.4N D. 2.4N

Solution:

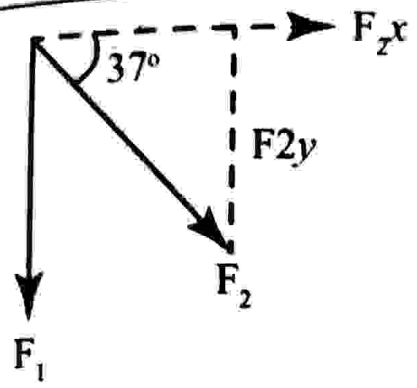
Force between the 4μc and 20μc is,

$$F_1 = \frac{kQ_1Q_2}{r_1^2} = \frac{(9\times10^9)(4\times10^{-6})(20\times10^{-6})}{(60\times10^{-2})^2} = 2N$$

And,the force between the 10μc and 20μc is,

$$F_2 = \frac{kQ_2Q_2}{r_2^2} = \frac{(9\times10^9)(10\times10^{-6})(20\times10^{-6})}{(100\times10^{-2})^2} = 1.8N$$

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$$F_x = F_{1x} + F_{2x} = 0 + (1.8)\cos 37^o = (1.8)(0.8) = 1.44N$$

$$F_y = F_{1y} + F_{2y} = 2 + (1.8)\sin 37^o = 2 + (1.8)(0.6) = 3.08$$

$$F = \sqrt{F_x^{\,2} + F_y^{\,2}} = \sqrt{(1.44)^2 (3.08)^2}$$

$$F = 3.4N$$

Answer: C

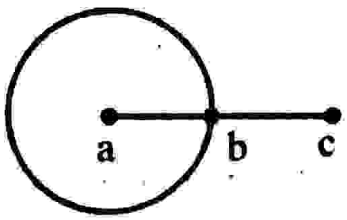
30. A spherical conductor of radius 5cm carries a charge of 10μc. What are the electric field strength at 1cm, 5cm and 10cm from the centre of the sphere respectively in N/c?

A. $36\times10^8, 9\times10^8, 0$
B. $9\times10^6, 36\times10^6, 0$
C. $0, 9\times10^8, 36\times10^6$
D. $0, 36\times10^6, 9\times10^6$

Solution:

$$E_a = \frac{kQ}{r_a^{\,2}} = 0$$

$$E_b = \frac{kQ}{r_b^{\,2}} = \frac{(9\times10^9)(10\times10^{-6})}{(5\times10^{-2})^2} = 36\times10^6\ N/c$$



$$E_c = \frac{kQ}{r_c^{\,2}} = \frac{(9\times10^9)(10\times10^{-6})}{(10\times10^{-2})^2} = 9\times10^6\ N/c$$

Answer: D

31. Which one of the following statement is in disagreement with coulomb's law

A. The electrostatic force between two point charges is directly proportional to the square of the distance between then

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B. The electrostatic force between two point charges is proportional to the product of their charges

C. Bodies of the like charge repel each other

D. Bodies of unlike charge attract each other

Hint: $F = kQ_1Q_2/r^2$ **Answer: A**

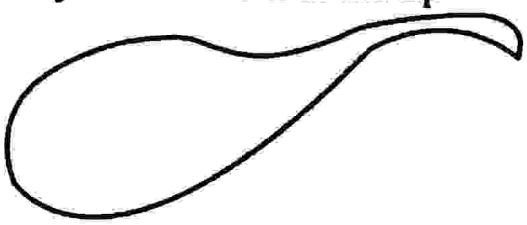
32. If the conductor shown below is charged positively then;

A. The charge is uniformly distributed inside

B. The charge is uniformly distributed outside

C. The electric field is maximum inside

D. The charge density is maximum at the tip



Hint: The change distributed outside and its density is maximu at the sharp edge. **Answer: D**

33. If a positively charged plastic rod charged a neutral metal sphere by conduction, the sphere will be charged

A. Positively

B. Negatively

C. Positively and Negatively

D. Neutral

Hint: Conduction given rise to similar charged bodies **Answer: A**

34. If charge Q_1 attracts charge Q_2 and charge Q_2 repels charge Q_3, Then, their signs would be?

	Q_1	Q_2	Q_3
A.	+	+	+
B.	−	+	+
C.	+	−	+
D.	−	+	−

Hint: Use the fundamental law of electrostatics **Answer: B**

35. The electrostatics force between two point charge is 4×10^{-3}N when they are 3m apart. For what separation will the force between them became 1.6×10^{-2}N?

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A. 2.5m B. 1.5m C. 2m D. 4m

Solution:

$$F_1 = \frac{kQ_1Q_2}{r_1^2}$$

$$4\times10^{-3} = \frac{kQ_1Q_2}{(3)^2} \Rightarrow kQ_1Q_2 = (4\times10^{-3})(9)$$

$$1.6\times10^{-2} = \frac{(4\times10^{-3})(9)}{r_2^2}$$

$$r_2^2 = \frac{36\times10^{-3}}{1.6\times10^{-2}} = 2.25$$

$$r_2 = 1.5m$$

Answer: B

36. A particle of mass 1.5×10^{-4}kg is released at a place where there is an electric field of 2.4×10^{4}N/c. If the particle experiences an acceleration of 4×10^{2}m/s^2, then what is the charge carried by the particle?

A. 4μc B. 3μc C. 2.5μc D. 2μc

Solution:

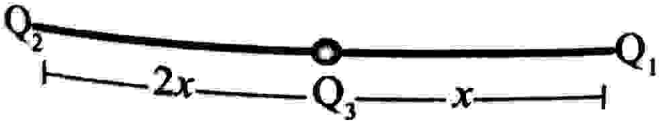
$$EQ = ma$$

$$Q = \frac{ma}{E} = \frac{(1.5\times10^{-4})(4\times10^{2})}{2.4\times10^{4}} = 2.5\times10^{-6}c$$

$$Q = 2.5\mu c$$

Answer: C

37. For the net force on Q_3 to be zero, the relation between Q_1 and Q_2 must be



A. $Q_2 = -4Q_1$ B. $Q_2 = 4Q_1$ C. $Q_2 = 2Q_1$ D. $Q_2 = -2Q_1$

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Solution:

$$\frac{kQ_1Q_3}{x^2} = \frac{kQ_2Q3}{(2x)^2}$$

$$\frac{Q_1}{x^2} = \frac{Q_2}{4x^2}$$

$$Q_2 = 4Q_1$$

Answer: B

38. What is the relationship between Q_1 and Q_2 if their resultant electric field is zero at x = 16cm



A. $Q_1 = 4Q_2$
B. $Q_1 = 2Q_2$
C. $Q_2 = 0.24Q_1$
D. $Q_1 = -0.25Q_2$

Solution:

$$\frac{kQ_1}{r_1^2} = \frac{kQ_2}{r_2^2}, r_1 = 2r_2$$

$$\frac{Q_1}{(2r_2)^2} = \frac{Q_2}{r_2^2}$$

$$\frac{Q_1}{4} = Q_2$$

$$Q_1 = 4Q_2$$

Answer: A

39. The electric field strength at a point is defined by the force exerted on:
A. A point charge
B. A test charge
C. An electron
D. A proton

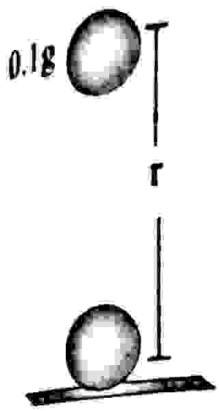
Answer: B

40. If the charge on each pith ball is 2nc then, what should the distance between them be for the electrostatic force to balance the weight of the upper ball ?
A. 6cm
B. 6mm
C. 1,8cm
D. 1.8mm

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Solution:



$$\frac{kQ_1Q_2}{r^2} = \frac{kQ^2}{r^2} = mg$$

$$r = Q\sqrt{\frac{k}{mg}} = (2\times10^{-9})\sqrt{\frac{9\times10^{9}}{(0.1\times10^{-3})(10)}}$$

$$r = 6\times10^{-3}m = 6mm$$

Answer: B

41. Two point charges repel each other with a force of 0.16N when their separation is 1.5m. What will be the electrostatic force between them if their separation is 1m?

A. 0.36N B. 0.64N C. 0.54N D. 0.24N

Solution:

$$F_1 = \frac{kQ_1Q_2}{r_1^2}$$

$$0.16 = \frac{kQ_1Q_2}{(1.5)^2} \Rightarrow kQ_1Q_2 = (0.16)(1.5)^2$$

$$F_2 = \frac{kQ_1Q_2}{r_2^2} = \frac{(0.16)(1.5)^2}{(1)^2} = 0.36N$$

Answer: A

42. A girl observed that a comb rubbed through her hair attracted piece of papers. What is the cause of this phenomena?

A. Rubbing C. Conduction

B. Induction D. Radiation

Hint: The rubbing combo is charged but the papers are neutral and the attraction is due to induction.

Answer: B

43. Which one of the following can be the charge carried by a body in terms of charge of an electron,e?

A. 0.5e B. 5e

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C. 1/2e

D. $6\frac{3}{4}e$

Hint: Charge exists as an integer multiple of the basic change, e. Since it is quantized.

Answer: B

44. If a positively charged glass rod is brought closer to a neutral metallic sphere with out physical contact, what charge will be observed on the closest side of the sphere?

A. Negative

B. Positive

C. Neutral

D. Can not be know

Hint: Induction gives rise to opposite charge.

Answer: A

45. When a glass rod is rubbed with a silk cloth, 0.5×10^{14} electrons are transferred from the silk to the rod. What charge is acquired by each body?

A. 6μc B. 8μc C. 3μc D. 5μc

Solution:

$$Q = ne$$

$$Q = \left(0.5\times10^{14}\right)\left(1.6\times10^{-19}\right)$$

$$Q = 0.8\times10^{-5}c = 8\mu c$$

Answer: B

46. Two metal spheres initially carrying charges of -3μc and 7μc are made to contact and then separated. What is the number of electrons gained or lost after the process?

A. 3×10^5 B. 2×10^5 C. 1×10^5 D. 1.2×10^5

Solution:

$$Q = \frac{Q_1 + Q_2}{2} = \frac{-3+(7)}{2} = 2\mu c$$

$$\text{and, } Q = ne \Rightarrow n = \frac{Q}{e}$$

$$n = \frac{2\times10^{-6}}{1.6\times10^{-19}} = 1.25\times10^5 \text{ electros}$$

Answer: D

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47. If a 3.2×10^{-14}kg oil drop is balanced by an electric field of 2×10^{6}N/c, then what is the charge on the drop

A. -1.6×10^{-18}c B. 1.6×10^{-18}c C. -1.6×10^{-19}c D. 1.6×10^{-19}c

Solution:

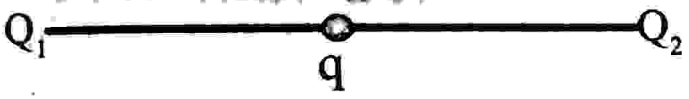
$$Eq = mg$$

$$q = \frac{mg}{E} = \frac{(3.2\times10^{-14})(10)}{2\times10^{6}}$$

$$= 1.6\times10^{-19}\,c$$

Answer: D

48. If the net electrostatic force on q is zero, what can be said about Q_1 and Q_2



A. Q_1 and Q_2 are unequal in magnitude and opposite in sign

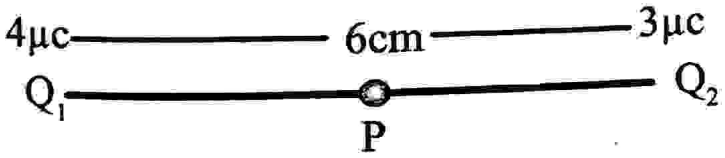
B. Q_1 and Q_2 are unequal in magnitude and the same in sign

C. Q_1 and Q_2 are equal in magnitude and the same in sign

D. Q_1 and Q_2 are equal in magnitude and opposite in sign

Hint: Use law of electrostatic and coulomb's law **Answer: B**

49. What is the net electric field strength mid way between the two point charges shown below



A. 1×10^{7}N/c to the right C. 1×10^{7}N/c to the left

B. 2×10^{7}N/c to the right D. 2×10^{7}N/c to the left

Solution:

$$E_P = E_1 - E_2 = \frac{kQ_1}{r_1^2} - \frac{kQ_2}{r_2^2} = \frac{k}{r^2}[Q_1 - Q_2]$$

$$E_P = \frac{9\times10^{9}}{(3\times10^{-2})^2}\left[4\times10^{-6} - 3\times10^{-6}\right]$$

$$E_P = 1\times10^{7}\,N/c, \text{ to the right.}$$

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Answer: A

50. If the electric field strength 3cm from a point charge is 50N/c, What will be the field strength at a distance of 10cm from this charge?

A. 5N/c B. 4.5N/c C. 500N/c D. 15N/c

Solution:

$$E_1 = \frac{kQ}{r_1^2} = 50$$

$$kQ = (50)(r_1^2) = (50)(3\times10^{-2})^2 = 45\times10^{-3}$$

$$E_2 = \frac{kQ}{r_2^2} = \frac{45\times10^{-3}}{(10\times10^{-2})^2} = 4.5\text{N/c}$$

Answer: B

51. What is the electrostatic force between two identical point charges of each 1c separated by a distance of 1cm?

A. 0.09×10^9N C. 9×10^9N

B. 0.9×10^9N D. 90×10^9N

Solution:

$$F = \frac{kQ_1Q_2}{r^2} = \left(\frac{9\times10^9(1)(1)}{(1)^2}\right)$$

$$= 9\times10^9\,\text{N}$$

Answer: C

52. The electrostatic force between two point charges is 8×10^{-5}N. What will be the force between these point charges if their separation is halved?

A. 8×10^{-5}N B. 3.2×10^5N C. 3.2×10^{-4}N D. 8×10^{-4}N

Solution:

$$F_1 = \frac{kQ_1Q_2}{r_1^2} = 8\times10^{-4}\,\text{N}$$

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$$F_2 = \frac{kQ_1Q_2}{r_1^2}, r_2 = \frac{1}{2}r_1$$

$$F_2 = \frac{kQ_1Q_2}{\left(\frac{1}{2}r_1\right)^2} = 4\left(\frac{kQ_1Q_2}{r_1^2}\right) = 4F_1 = 4\left(8\times10^{-5}\right)$$

$$F_2 = 3.2\times10^{-4}\,N$$

Answer: C

53. What is the electric field strength at a distance of 3mm from a 3μc point charges?

A. 3×10^9 N/c B. 6×10^9 N/c C. 9×10^9 N/c D. 4×10^9 N/c

Solution:

$$E = \frac{kQ}{r^2} = \frac{\left(9\times10^9\right)\left(3\times10^{-6}\right)}{\left(3\times10^{-3}\right)^2}$$

$$= 3\times10^9\,N/c$$

Answer: A

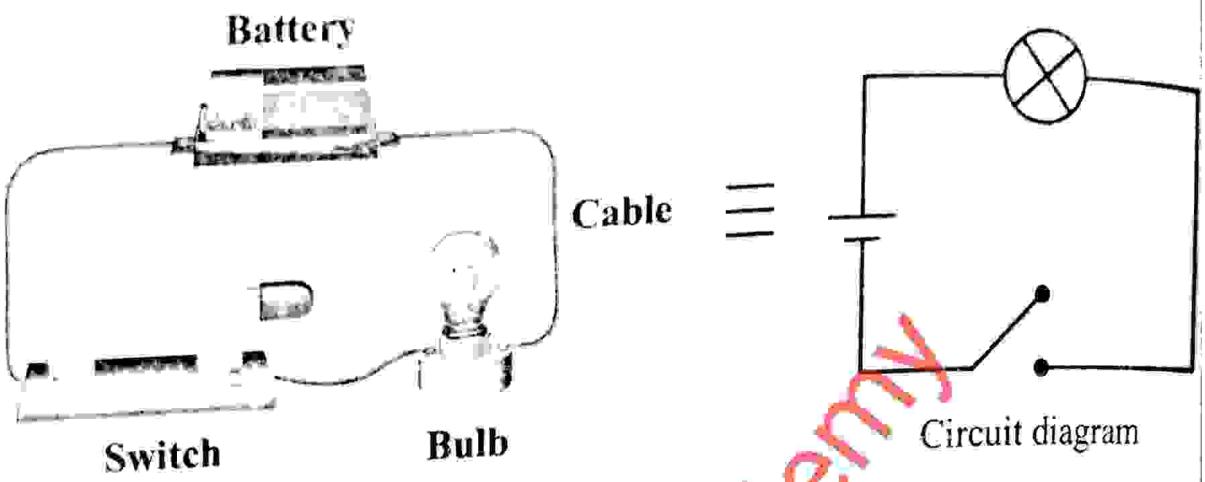
9.7 Electric Circuits

For the flow of electrons through a conductor, there should be a complete path, and this path is known as *circuit*.

Electric Circuit: Is a path way through which charged particles can flow to give a defined purpose.

A typical electric circuit consists of connecting cables, voltage source, load resistor such as a light bulb to dispate energy and a switch to close or open the system as shown below:

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Physical Circuit

☑ When the switch is ON, charge flows through the circuit and hence there is current. And, when the switch is OFF, the circuit will not have current.

Some of the circuit elements with their symbols and purposes are given below

Circuit Element	Symbol	Use
Bulb		Give Light
Electric Cell		Drive Charges
Switch		ON, OFF
Resistor		Disppate Energy
Ammeter	A	Measure Current
Voltmeter	V	Measure Voltage

Illustrative Examples

54. What is an electric circuit?

Explanation:

It is a complete path for the flow of electrons through a conductor to get

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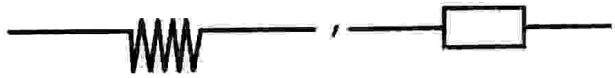
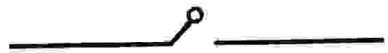
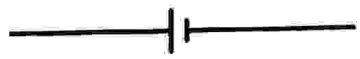
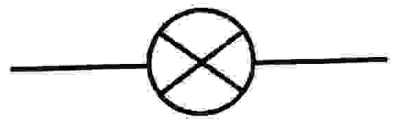
light, heat, sound,...etc. from electric energy.

55. What are the basic circuit, elements?

Explanation:

Cable, electric source such as dry cell, switch, external resistor such as bulb, heater...etc.

Match the items in column 'A' with those in column 'B'

A	B
56. Bulb	a. 
57. Dry Cell	b. 
58. Swith	c. 
59. Resistor	d. 
60. Ammeter	e. 
61. Voltmeter	f. 

Answers

56.f 57.e 58.d 59.c 60.b 71.a

9.8 Current, Voltage and Ohm's Law

Here, we are going to define current and voltage and state ohm , law

Electric current

The flow of charged particles through a conductor is called *current*

Electric Current: Is the flow of charge across a given cross-sectional area of a conductor per unit time

$$\text{Electric Current} = \frac{\text{Flow of Charge}}{\text{Time Taken}}$$

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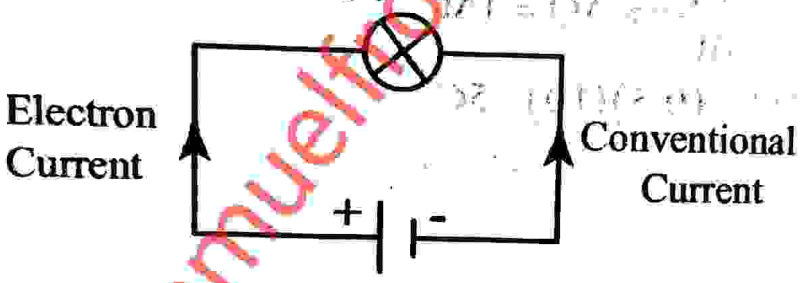
$$I = \frac{\Delta Q}{\Delta t}$$

Electric current is a scalar quantity and its SI Unit is the ampere ,(A).

$$1A = \frac{1C}{s}$$

Current is of two types, electron current and conventional current

☑ The flow of electrons from the negative terminal to the positive terminal of a battery is called *electron current*. And, the "flow" of protons from the positive terminal to the negative terminal of battery is called *conventional current* or simply current.



Illustrative Example

62. What is electric current?

Explanation:

It is the flow of charges per unit time

63. What is the difference between electron current and conventional current?

Explanation:

Electron current is the flow of electrons from negative to positive terminal of a battery. While, conventional current is the "flow" of protons from the positive to the negative terminal of the battery.

64. What device is used to measure electric current?

Explanation:

Ammeter, Galvanometer and Multimeter can measure current in a circuit.

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65. What is the current in a circuit where 60μc charge flows in 0.006seconds?

Solution:

$$I = \frac{Q}{t} = \frac{60\times10^{-6}}{6\times10^{-3}}$$
$$= 0.01A$$

66. What is the amount of charge in a wire if 0.5A current is recorded flowing for 10 seconds?

Solution:

$$I = \frac{\Delta Q}{\Delta t} \Rightarrow \Delta Q = I\Delta t$$
$$\Delta Q = (0.5)(10) = 5C$$

67. If 5×10^{22} electrons pass through a wire in 200second, What is the current in the wire?

Solution:

$$I = \frac{\Delta Q}{\Delta t} = \frac{ne}{\Delta t}$$
$$I = \frac{(5\times10^{22})(1.6\times10^{-19})}{200}$$
$$= 40A$$

Voltage

Voltage, also known as *potential difference* is the agent that drives electrons to flow through a conductor.

Voltage: Is the work done per unit charge needed to move a test charge from one point to the other.

$$\text{Voltage} = \frac{\text{Work done}}{\text{Charge}}$$

$$\boxed{V = \frac{W}{Q}}$$

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Voltage or potential difference is a scalar quantity, and its SI unit is the volt(V) $1V = 1\frac{J}{C}$

☑ Voltage is measured using an instrument called *Voltmeter.*

Illustrative Examples

68. What is the potential difference in moving a 3c charge with a 360J of work done?

Solution:

$$V = \frac{W}{Q} = \frac{360}{3}$$

$$= 120V$$

69. What charge is moved across a potential difference of 12V using a work done of 600J?

Solution:

$$V = \frac{W}{Q} \Rightarrow Q = \frac{W}{V}$$

$$Q = \frac{600}{12}$$

$$Q = 50C$$

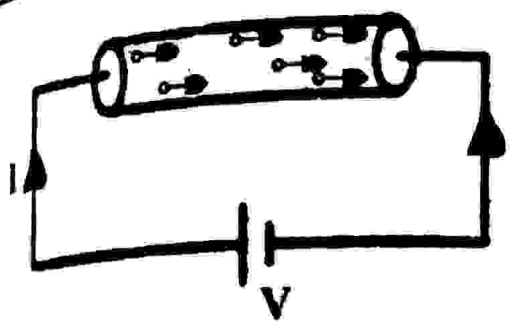
Ohm's Law

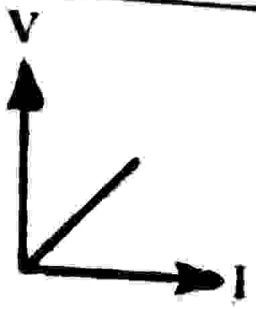
The relationship between voltage and current was discovered by George Simon Ohm and is stated as follows,

Ohm's Law States that: The current flowing through a conductor is directly proportional to the potential difference supplied across the ends of the conductor.

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Voltage versus current graph

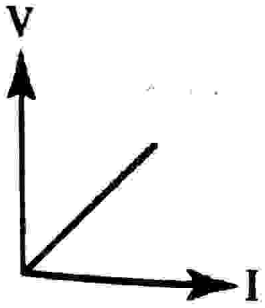
$\boxed{V = IR}$ ⟶ **Ohm's Law**

And, The slope of the V - I current is resistance,R. $R = \frac{V}{I}$

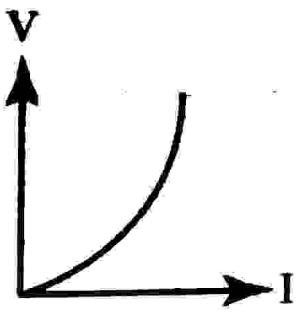
Where: R = Resistance of the conductor and its SI unit is the ohm(Ω)

$$1\Omega = \frac{1V}{A}$$

Materials that obey ohm's law are said to be ohmic materials and their V - I graph is straight line. While non-ohmic materials have no linear V - I graph as shown below:



Ohmic

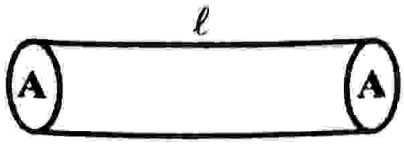


Non - ohmic

- Resistance is the measure of difficultly electrons face to flow through a material. The resistance of a metallic conductor is found to be dependent on its geometrical features.
- Resistance is directly proportional to the length and inversely proportional to the cross sectional area of wire

$R \sim \frac{\ell}{A}$

$\boxed{R = \rho \frac{\ell}{A}}$



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Where: ρ(rho) = Is called resistivity of the conductor and it depends on t type of material the wire is made of and its unit is ohm-meter(Ωm)

Environmental Factor on Resistance

The resistance of a metallic conductor is found to depend on the temperatu of the environment.

☑ Resistance of a metallic conductor is directly proportional to t temperature of the environment, ΔT and its original resistance, R_o

$$\Delta R \sim \Delta T R_o$$

$$\boxed{\Delta R = \alpha R_o \Delta T}$$

Where: α(alpha) = Is coefficient of thermal resistivity.

Illustrative Examples

70. What electric current flows through a 200Ω light bulb subjected to a 220V voltage?

Solution:

$$I = \frac{V}{R} = \frac{220}{200} = 1.1A$$

71. What is the resistance of a wire if a 2A current flows through it when pluged to a 110V electric supply?

Solution:

$$R = \frac{V}{I} = \frac{110}{2}$$

$$R = 55\Omega$$

72. A current of 2A flows through an ohmic wire of resistance 10Ω. What current will flow through the wire if the potential difference is doubled?

Solution:

$$I_1 = \frac{V}{R} = 2$$

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$$I_2 = \frac{V_2}{R}, V_2 = 2V_1$$

$$I_2 = 2\frac{V_1}{R} = 2(I_1)$$

$$= 2(2) = 4A$$

73. What is the resistance of a 2m metallic wire of diameter 10mm and resistively $1.84\times10^{-4}\Omega m$?

Solution:

$$R = \frac{\rho\ell}{A}, A = \pi r^2 = (3.14)(5\times10^{-3})^2$$

$$R = \frac{(1.84\times10^{4})(2)}{7.85\times10^{-5}}$$

$$R = \frac{3.68\times10^{-4}}{7.85\times10^{-5}} = 4.7\Omega$$

74. What is the length of a copper wire with a diameter of 20mm and resistance 110Ω ($\rho = 1.7\times10^{-8}\Omega m$)

Solution:

$$R = \frac{\rho\ell}{A}, A = \pi r^2 = (3.14)(10\times10^{-3})^2 = 314\times10^{-6}$$

$$\ell = \frac{RA}{\rho} = \frac{(110)(3.14\times10^{-4})}{1.7\times10^{-8}}$$

$$= 2\times10^{6}\,m$$

75. What will be the resistance of a wire of initial resistance 100Ω. After its
A. Length is halved and diameter doubled
B. Length is double and diameter is halved
C. Both length and diameter are halved
D. Both length and diameter are doubled

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Solution:

use, $R_1 = \frac{\rho \ell_1}{A_1}, A = \pi r_1^2$

A. $R_2 = \frac{\rho \ell_2}{A_2}, A_2 = \pi r_2^2 = 4\pi r_1^2$

$$R_2 = \frac{\rho\left(\frac{1}{2}\ell_1\right)}{4\pi r_1^2} = \frac{1}{8}R_1 = \frac{100}{8} = 12.5\Omega$$

B. $R_2 = \frac{\rho \ell_2}{A_2}, A_2 = \pi r_2^2 = \frac{1}{4}\pi r_1^2$

$$R_2 = \frac{\rho\left(\frac{1}{2}\ell_1\right)}{\frac{1}{4}\pi r_1^2} = 8R_1 = 8(100) = 800\Omega$$

C. $R_2 = \frac{\rho \ell_2}{A_2}, A_2 = \pi r_2^2 = \frac{1}{4}\pi r_1^2$

$$R_2 = \frac{\rho(2\ell_1)}{\frac{1}{4}\pi r_1^2} = 2R_1 = 2(100) = 200\Omega$$

D. $R_2 = \frac{\rho \ell_2}{A_2}, A_2 = \pi r_2^2 = 4\pi r_1^2$

$$R_2 = \frac{\rho(2\ell_1)}{4\pi T_1^2} = \frac{1}{2}R_1 = \frac{100}{2} = 50\Omega$$

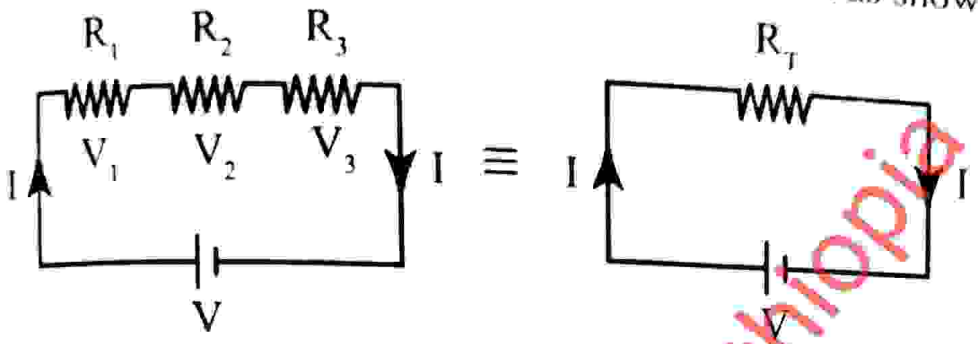
9.9 Combination of Resistors

In a circuit, resistors may be connected in series,in parallel or in both based on the requirement.

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Resistors in Series

When resistors are connected in series, they are joined one after the other and the current do have only one single path in the circuit as shown below;



The current that passes through each resistor is the same as the total current in the circuit.

$$I_1 = I_2 = I_n = I_T = I$$

And, the total potential drop in the circuit is the sum of the potential drops across each resistor.

$$V_T = V_1 + V_2 + \ldots + V_n$$

From ohm's law, $V = IR$

$$I_T R_T = I_1 R_1 + I_2 R_2 + \ldots\ldots + I_n R_n$$

Since current is the same , we have

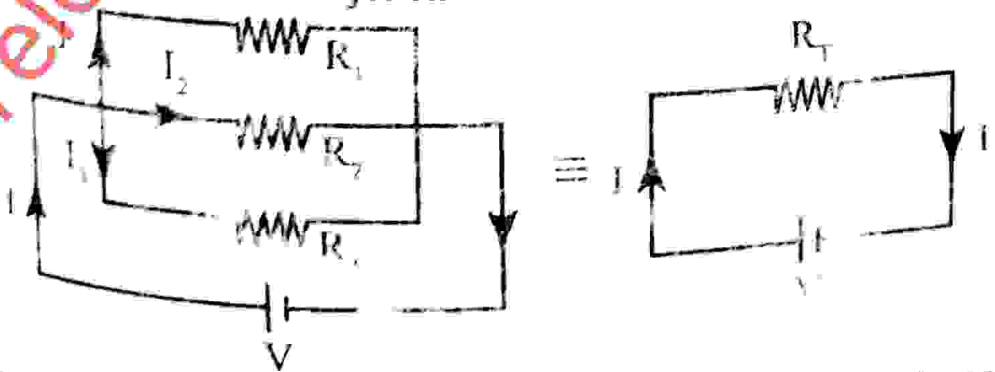
$$I(R_T) = I(R_1 + R_2 + \ldots\ldots + R_n)$$

Therefore, the total or equivalent resistance when resistors are connected in series is obtained by

$$\boxed{R_T = R_1 + R_2 + \ldots + R_n}$$

Resistors in Parallel

When resistors are connected in parallel, they are joined face to face and the current branches out at a junction as shown below



The potential drop in the network is the same as the potential drops across each resistor.

$$V_1 = V_2 = V_n = V_T = V$$

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And, the total current in the system is the sum of the current that flow through each resistor,

$$I_T = I_1 + I_2 + ... + I_n$$

From ohm's law, I = V/R

$$\frac{V_T}{R_T} = \frac{V_1}{R_1} + \frac{V_2}{R_2} + ... + \frac{V_n}{R_n}$$

Since voltage is the same,we have;

$$\frac{V}{R_T} = V\left(\frac{1}{R_1} + \frac{1}{R_2} + ... + \frac{1}{R_n}\right)$$

Therefore, the total or equivalent resistance in parallel connection is obtained by:

$$\boxed{\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + ... + \frac{1}{R_n}}$$

Illustrative Example

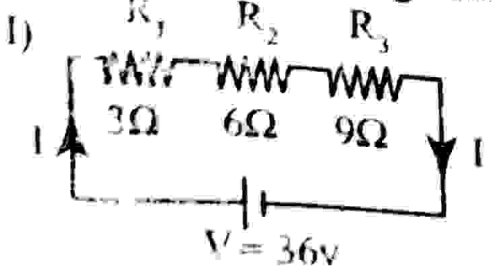
76. In which combination of resistors will the total resistance
    A. Increase
    B. Decrease

Explanation:

A. In series combination
B. In parallel combination

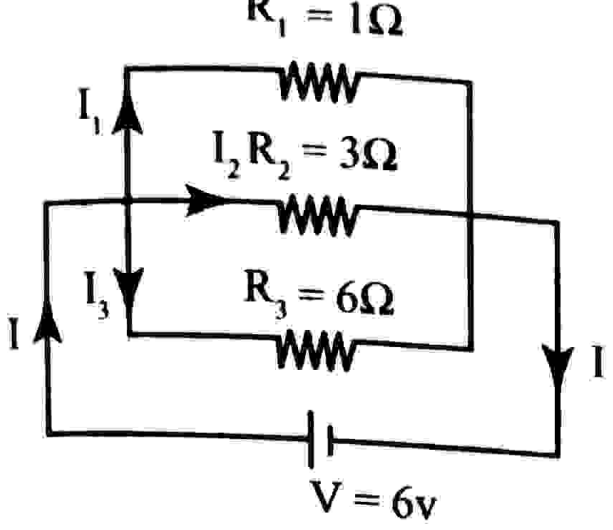
77. For the circuits shown below, find
    A. The total resistance
    B The total current
    C. The potential drop across each resistor
    D. The current flowing through each resistor

I)

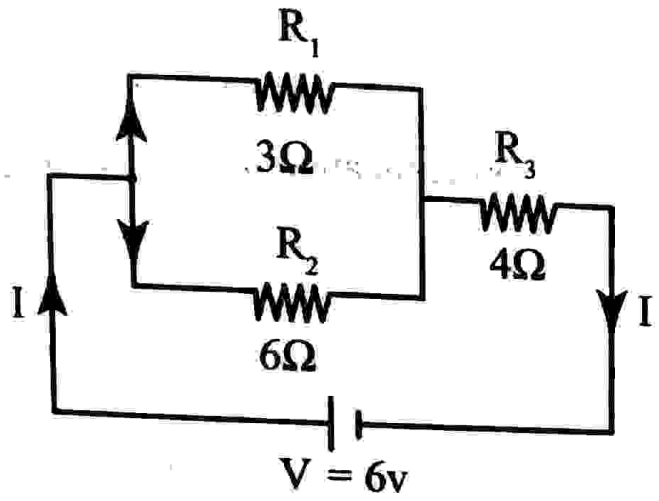


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II)



III)



Solution:

(i) A) $R_T = R_1 + R_2 + R_3$

$R_T = 3 + 6 + 9 = 18\Omega$

B) $I = \frac{V}{R} = \frac{36}{18} = 2A$

C) $V_1 = IR_1 = (2)(3) = 6V$

$V_2 = IR_2 = (2)(6) = 12V$

$V_3 = IR_3 = (2)(9) = 18V$

D) $I_1 = \frac{v_1}{R_1} = \frac{6}{3} = 2A$

$I_2 = \frac{v_2}{R_2} = \frac{v_2}{R_2} = \frac{12}{6} = 2A$

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$$I_3 = \frac{V_3}{r_3} = \frac{18}{9} = 2A$$

(ii) A) $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{1} + \frac{1}{3} + \frac{1}{6} = \frac{9}{6}$

$$\frac{1}{R_T} = \frac{3}{2} =$$

$$R_T = \frac{2}{3}\Omega$$

B) $I = \frac{V}{R} = \frac{6}{2/3} = 9A$

C) $V_1 = V_2 = V_3 = 6V$

D) $I_1 = \frac{V_1}{R_1} = \frac{6}{1} = 6A$

$$I_2 = \frac{V_2}{R_2} = \frac{6}{3} = 2A$$

$$I_3 = \frac{V_3}{R_3} = \frac{6}{6} = 1A$$

(iii) A) R_1 and R_2 are in parallel

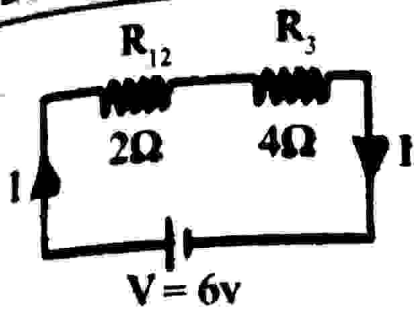
$$\frac{1}{R_{12}} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{3} + \frac{1}{6} = \frac{1}{2} = 2\Omega$$

R_{12} and R_3 become in series

B) $I = \frac{V}{R} = \frac{6}{6} = 1A$

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$$R_T = R_{12} + R_3 = 2 + 4 = 6\Omega$$

C) Now, let us first find the potential drop across R_3

$$V_3 = I_3R_3 = IR_3 = (1)(4) = 4V$$

$$V_1 = V_2 = V - V_3 = 6 - 4 = 2V$$

$$V_1 = 2V \text{ and } V_2 = 2V$$

D) $I_1 = \frac{V_1}{R_1} = \frac{2}{3}A$

$$I_2 = \frac{V_2}{R_2} = \frac{2}{6} = \frac{1}{3}A$$

$$I_3 = \frac{v_3}{R_3} = \frac{4}{4} = 1A$$

78. The equivalent resistance of two resistors is 7Ω when connected in series and it is 10/7Ω. When they are in parallel. What is the resistance of each resistor?

Solution:

$R = R_1 + R_2 = 7$, in series.

$R = \frac{R_1R_2}{R_1 + R_2} = \frac{10}{7}$, in parallel.

from, $R_1 + R_2 = 7 \Rightarrow R_2 = 7 - R_1$ Substitute this into

$$\frac{R_1R_2}{R_1 + R_2} = \frac{10}{7}$$

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$$\frac{R_1(7-R_1)}{7}=\frac{10}{7}$$

$$7R_1-R_1^{\,2}=10$$

$$R_1^{\,2}-7R_1+10=0$$

Use the quadratic equation to solve for R_1

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

$$R_1=\frac{7\pm\sqrt{(-7)^2-4(1)(10)}}{2(1)}=\frac{7\pm3}{2}$$

$$R_1=\frac{7+3}{2}=5\Omega \text{ or } R_1=\frac{7-3}{2}=5\Omega$$

When, $R_1=5\Omega$, $R_2=2\Omega$, and when $R_1=2\Omega$, or $R_2=5\Omega$

79. A wire of resistance R is cut into five equal pieces,If the pieces equivalent resistance when connected in parallel is 2Ω, what was the resistance of the original wire?

Solution:

We know , $R=\frac{\rho\ell}{A}$ And, when length decreases, resistance also decreases

So, each piece will have a resistance of R/5

$$\frac{1}{R_T}=\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}+\frac{1}{R_4}+\frac{1}{R_5}$$

$$\frac{1}{2}=\frac{1}{\frac{R}{5}}+\frac{1}{\frac{R}{5}}+\frac{1}{\frac{R}{5}}+\frac{1}{\frac{R}{5}}+\frac{1}{\frac{R}{5}}=\frac{25}{R}$$

$$R=50\Omega$$

80. Two resistors of $R_1=2\Omega$ and $R_2=8\Omega$ are connected in series with a 12V battery. What is the potential drop across each resistor when the switch is on?

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Solution:

$$R = R_1 + R_2 = 2 + 8 = 8\Omega$$

$$I_1 = I_2 = \frac{V}{R} = \frac{12}{10} = 1.2A$$

$$V_1 = I_1R_1 = (1.2)(2) = 2.4V$$

$$V_2 = I_2R_2 = (1.2)(8) = 9.6V$$

81. Suppose a resistor is connected to a constant voltage supply. When will the current in the resistor decrease

A. When length of the resistor increases

B. When length of the resistance decreases

C. When cross-sectional area of the resistor decreases

D. When the resistance decreases

Solution: $I = \frac{\rho\ell}{A}$ and $I = \frac{V}{R}$ **Answer: A**

82. Consider a circuit with an emf of 4.5V and a load resistance of 40Ω and a current of 0.1A. What will be the current if additional cell of emf 4.5V is connected to the previous cell in series?

A. 0.18A B. 0.05A C. 0.20A D. 0.21A

Solution: $E_1 = I(R + r)$

$$R + r = \frac{E_1}{I_1} = \frac{4.5}{0.1} = 45$$

$$r = 45 - R = 45 - 40 = 5\Omega$$

$$E_2 = 4.5 + 4.5 = 9V$$

$$R + r + r = \frac{E_2}{I_2}$$

$$I_2 = \frac{E_2}{R + r + r} = \frac{9}{40 + 5 + 5} = 0.18A$$

Answer: A

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83. The correct way of connecting an Ammeter and a voltmeter in a circuit is?

A. Both are connected in series with the resistor

B. Both are connected in parallel with the resistor

C. Ammeter is connected in series and voltmeter in parallel with the resistor

D. Ammeter is connected in parallel and voltmeter in series with the resistor.

Answer: C

84. You are given three 1Ω resistors. All of the following equivalent resistance can be produced using all three EXCEPT?

A. 1/3Ω B. 1.5Ω C. 1Ω D. 3Ω

Hint: All three in series, R = 3Ω

All three in parallel, R = 1/3Ω

Two in parallel, one series, R = 1.5Ω

The in serics, one parallel = $\frac{2}{3}\Omega$

Answer: C

85. When a 15V dry cell is connected to a load resistor of 144Ω, a current of 0.1A flows through the resistor. What is the internal resistance of the dry cell?

A. 300Ω B. 150Ω C. 10Ω D. 6Ω

Solution:

$$R + r = \frac{E}{I} = \frac{15}{0.1} = 150\Omega$$

$$r = 150 - R = 150 - 144$$

$$= 6\Omega$$

Answer: D

86. A battery of emf 5V and internal resistance 10Ω is connected in series with an external resistor of resistance 240Ω. What is the potential drop on the load resistor?

A. 0.2V B. 5.2V C. 4.8V D. 5V

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Solution:

$$E = I(r + R)$$

$$I = \frac{E}{r + R} = \frac{5}{10 + 240}$$

$$= 0.02A$$

$$V_{external} = IR = (0.02)(240)$$

$$= 4.8V$$

Answer: C

87. In an experiment four resistors 3Ω, 4Ω, 12Ω, and 16Ω are required at different time to give 1.7Ω. But, the lab technician told you to use only two resistors. Which two resistos will you choose to do the experiment?

A. 3Ω and 6Ω
C. 4Ω and 12Ω
B. 3 Ω and 4Ω
D. 4Ω and 16Ω

Hint:

$$\frac{1}{R} = \frac{1}{3} + \frac{1}{4} = \frac{7}{12}$$

$$R = \frac{12}{7} = 1.7\Omega$$

Answer: B

88. The terminal p.d of a battery is maximum when

A. The emf is minimum
B. The battery is being discharged
C. There is no current flowing
D. There is a steady current flow

Hint: $E = V_{im} + V_{ext}$

Answer: C

89. A lighting protection provided to a building is

A. An overhead grid of insulators extended to the ground
B. An overhead grid of wire extended to the ground
C. A system of lighting rod extended to the building
D. Stem of lighting rods connected to the clouds

Answer: B

90. What is the resistance of copper wire of length 30m length and 0.8cm radius ($\rho cu = 1.7 \times 10^{-8}\Omega m$)

A. 2Ω
B. 1Ω
C. 0.5Ω
D. 0.25Ω

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Solution:

$$R = \frac{\rho\ell}{A} = \left(1.7\times10^{-8}\right)\left(\frac{30}{201\times10^{-8}}\right)$$

$$= 0.25\Omega$$

Answer: D

91. What length of wire of resistivity $8\times10^{-7}\Omega m$ and cross-sectional area $0.2mm^2$ is required to get a resistance of 10Ω?

A. 50m B. 25m C. 10m D. 5m

Solution:

$$R = \frac{\rho\ell}{A} \Rightarrow \ell = \frac{RA}{\rho} = \frac{(10)\left(0.2\times10^{-6}\right)}{8\times10^{-7}} = 25m$$

Answer: B

92. A metal rod of 1mm diameter has a resistance of 0.01Ω. What would be the resistance of the rod if the diameter were doubled?

A. 0.1Ω B. 0.5Ω C. 0.0025Ω D. 0.25Ω

Hint: $$R_1 = \frac{\rho\ell}{A_1} = \frac{\rho\ell}{\frac{\pi d_1^{\,2}}{4}} = 0.01$$

$$R_2 = \frac{\rho\ell}{A_2} = \frac{\rho\ell}{\frac{\pi d_2^{\,2}}{4}} = \frac{\rho\ell}{\frac{4\pi d_1^{\,2}}{4}} = \frac{R_1}{4} = \frac{0.01}{4}$$

$$= 0.0025\Omega$$

Answer: C

93. A potential difference of 5.1V is applied across a 2m long copper wire of cross-sectional area $0.01mm^2$. What is the current flowing through the wire?($\rho = 1.7\times10^{-8}\Omega$.)

A. 5.1A B. 4.1A C. 3.1A D. 1.5A

Solution:

$$I = \frac{V}{R} = \frac{V}{\frac{\rho\ell}{A}} = \frac{AV}{\rho\ell}$$

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$$I=\frac{(10^{-8})(5.1)}{(1.7\times10^{-8})(2)}=1.5A$$

Answer: D

94. When a potential difference of 6V is applied across a brass wire of cross-sectional area $0.02cm^2$, a current of 12A flows through it. What is the length of the wire? ($\rho = 6.6\times10^{-8}\Omega m$)

A. 15.2m B. 5.2m C. 10.2m D. 6.2m

Solution:

$$R=\frac{V}{I}=\frac{6}{12}=0.5\Omega$$

$$R=\frac{\rho\ell}{A}\Rightarrow\ell=\frac{RA}{\rho}$$

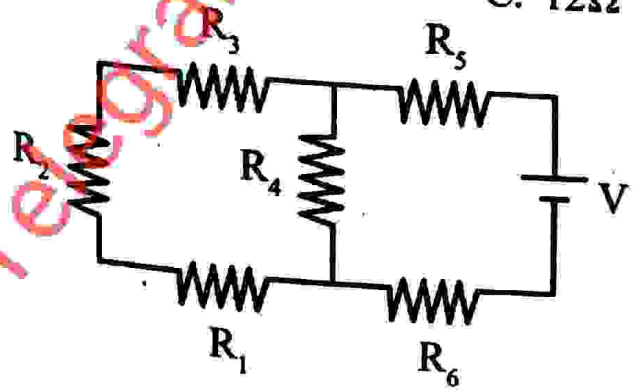
$$\ell=\frac{(0.5)(0.02\times10^{-4})}{6.6\times10^{-8}}$$

$$=15.2m$$

Answer: A

95. Six identical resistors of each 4Ω are connected in a circuit as shown below, What is the total resistance of the circuit?

A. 6Ω B. 10Ω C. 12Ω D. 11Ω



Solution:

R_1,R_2 and R_3 are in series, hence

$R_{123}=R_1+R_2+R_3=4+4+4=12\Omega$

And, R_{123} is in parallel with R_4

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$$\frac{1}{R_{1234}} = \frac{1}{R_{123}} + \frac{1}{R_4} = \frac{1}{12} + \frac{1}{4} = \frac{1}{3}$$

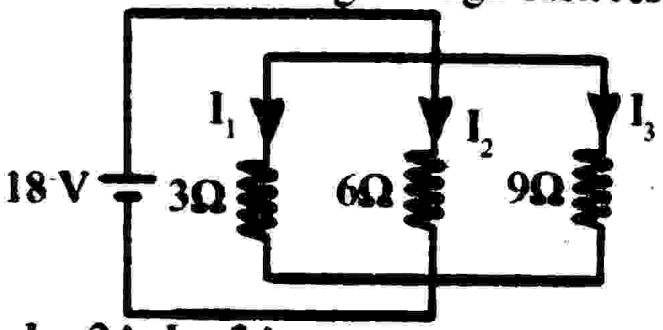
$$R_{123} = 3\Omega$$

And, R_{1234}, R_5 and R_6 are in series.

$$R = R_{1234} + R_5 + R_6 = 3 + 4 + 4 = 11\Omega$$

Answer: D

96. What are the currents flowing through each resistor shown below



A. $I_1 = 6A, I_2 = 2A, I_3 = 3A$
B. $I_1 = 6A, I_2 = 3A, I_3 = 2A$
C. $I_1 = 3A, I_2 = 6A, I_3 = 2A$
D. $I_1 = 3A, I_2 = 2A, I_3 = 6A$

Solution:

$$I_1 = \frac{18}{3} = 6A \quad , \quad I_3 = \frac{18}{6} = 3A \quad , \quad I_3 = \frac{18}{9} = 2A$$

Answer: B

97. A current in a torch bulb is 0.1A. How long will it take for one coulomb of charge to pass through the bulb?

A. 5sec B. 10sec C. 0.1sec D. 1sec

Solution:

$$t = \frac{Q}{I} = \frac{1}{0.1} = 10\,\text{sec}$$

Answer: B

98. A current of one ampere corresponds to the flow of one coulomb of charge

A. In one second
B. Across a one ohm resistor
C. In a uniform electric field
D. Across one volt P.d

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Hint:

$$\text{Current} = \frac{\text{Charge}}{\text{Time}}$$

$$1A = \frac{1C}{S}$$

99. For an electric current in a wire **Answer: A**

A. The direction of current and direction of flow of electrons are opposite

B. The electric current in the wire is due to the random motion of its electron

C. The electric current in the wire is due to the perpendicular motion of its electrons.

D. Electrons flow in the same direction as the conventional current.

Hint: Conventional current is opposite in direction to the electron current.

Answer: A

100. A wire of length 1m and 2mm diameter has a resistance of 6Ω. What will be the resistance of this wire if its length become 5m, keeping all other factors constant?

A. 6Ω B. 5Ω C. 30Ω D. 60Ω

Hint: $R_1 = \frac{\rho \ell_1}{A} = 6$

$$R_2 = \frac{\rho \ell_2}{A} = 5\left(\frac{\rho \ell_1}{A}\right) = 5(6)$$

$$= 30\Omega$$

Answer: C

101. A resistor wire of length ℓ and cross sectional area A is connected in a simple circuit. If this wire is folded into two and connected in the same circuit, then what will happen to the circuit resistance?

A. Doubled

B. Quartered

C. Halved

D. Remains the same

Hint: $R_1 = \frac{\rho \ell_1}{A_1}$

$$R_2 = \frac{\rho \ell_2}{A_2} = \frac{\rho\left(\frac{1}{2}\ell_1\right)}{(2A_1)} = \frac{1}{4}\frac{\rho \ell_1}{A_1}$$

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$$= \frac{1}{4} R_1$$

Answer: B

102. An electric cell supplies 0.9A current through a 2Ω load resistor and a current of 0.3A through a 7Ω load resistor. What is the internal resistance of the cell?

A. 2Ω B. 1Ω C. 1.2Ω D. 0.5Ω

Solution:

$$E_1 = E_2$$
$$I_1(R_1 + r) = I_2(R_2 + r)$$
$$0.9(2 + r) = 0.3(7 + r)$$
$$1.8 + 0.9r = 2.1 + 0.3r$$
$$0.6r = 0.3$$
$$r = 0.5\Omega$$

Answer: D

103. Potential difference of an electric cell in an open circuit is 6V but it falls to 4V when a current of 2A is flowing through it. what is internal resistance of the cell?

A. 2Ω B. 1Ω C. 3Ω D. 4Ω

Solution:

$$E = V_{int.} + V_{ext.}$$
$$6 = V_{int.} + 4$$
$$V_{int.} = 2$$
$$1r = 2$$
$$2r = 2$$
$$r = 1\Omega$$

Answer: A

104. Ten identical resistors each having a resistance of 1Ω are connected in parallel. what is the equivalent resistance of the network?

A. 10Ω B. 1Ω C. 0.1Ω D. 0.01Ω

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Solution:

$$\frac{1}{R_T}=\frac{1}{R}+\frac{1}{R}+\frac{1}{R}+\frac{1}{R}+\frac{1}{R}+\frac{1}{R}+\frac{1}{R}+\frac{1}{R}+\frac{1}{R}+\frac{1}{R}=\frac{10}{R}$$

$$R_T=\frac{R}{10}=\frac{1}{10}$$

$$R_T=0.1\Omega$$

Answer: C

105. A wire of resistance R is cut into n equal parts. These parts are then connected in parallel. Then equivalent resistance of the combination is

A. nR B. R/n C. n/R D. R/n^2

Solution:

$$\frac{1}{R_T}=\frac{1}{R/n}+\frac{1}{R/n}+....+\frac{1}{R/n}$$

$$\frac{1}{R_T}=\frac{n}{R/n}=\frac{n^2}{R}$$

$$R_T=\frac{R}{n^2}$$

Answer: D

106. When both the length and cross-section area of a wire are doubled, its resistance will?

A. Be halved
B. Be double
C. Remain constant
D. Be quartered

Hint: $R_1=\frac{\rho\ell_1}{A_1}$, and $R_2=\frac{\rho\ell_2}{A_2}=\frac{2\rho\ell_2}{2A_1}=\frac{\rho\ell_1}{A_1}=R_1$

$R_2=R_1$

Answer: C

107. Three equal resistors are given. How many different combinations can be made from them?

A. Three B. Four C. Five D. Six

Hint: They may be connected:

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- All in series
- All in parallel
- Two in series and parallel with the third
- Two in parallel and series with the third

Answer: B

108. The resistance of a wire of uniform diameter 'd' and length ℓ is R. The resistance of another wire of the same material but diameter 2d and length 4 ℓ will be?

A. 2R B. R C. $\frac{R}{2}$ D. $\frac{R}{4}$

Hint: $R_1 = \rho\frac{\ell}{A} = \frac{\rho\ell}{\pi\left(\frac{d}{2}\right)^2} = 4\frac{\rho\ell}{\pi d^2}$

$$R_2 = \frac{\rho(4\ell)}{\pi\left(\frac{2d}{2}\right)^2} = \frac{16}{4}\frac{\rho\ell}{\pi d^2} = \frac{4\rho\ell}{\pi d^2}$$

$$R_1 = R_1 = R$$

Answer: B

109. How many 2.1V electric cells with internal resistance of 0.2Ω each are to be connected in series to obtain a battery so that a conductor of resistance 6Ω would carry a current of 1.5A?

A. 4 B. 5 C. 6 D. 7

Hint:

$$nE = I(nr + R)$$

$$2.1n = 1.5(0.2n + 6) = 0.3n + 9$$

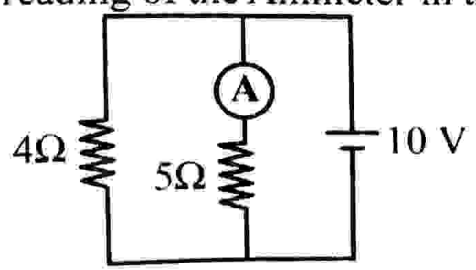
$$1.8n = 9$$

$$n = 5$$

Answer: B

110. What is the reading of the Ammeter in the circuit shown below?

A. 4A
B. 5A
C. 3A
D. 2A



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Hint: Since the resistors are in parallel, the potential drop across each resistor is the same.

$$I = \frac{V}{R} = \frac{10}{5} = 2A$$

Answer: D

11. Two resistors x and y are connected in parallel. Resistor x carries twice the current then that of resistor y , then:

A. The potential drop across x is twice that of y

B. The potential drop across x is half that across y

C. Resistor x has twice resistance that of y

D. Resistor x has half resistance that of y

Hint: I = V/R and from the nature of combination of resistors.

Answer: D

12. When 5 resistance of identical resistance R are connected in parallel, the equivalent resistance will be

A. 5R B. R/5 C. 5/R D. 0.5R

Solution:

$$\frac{1}{R_T} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R} + \frac{1}{R} + \frac{1}{R}$$

$$\frac{1}{R_T} = \frac{5}{R}$$

$$R_T = \frac{R}{5}$$

Answer: B

13. How many identical resistors should be connected in parallel to give an equivalent resistance of 60Ω

A. 6 Five ohm's

B. 12 six ohm's

C. 12 five ohm's

D. 12 four ohm's

Hint: $R_T = R + R + ... + R$

$$R_T = nR$$

$$60 = nR$$

$$60 = (12)(5)$$

12 five ohms

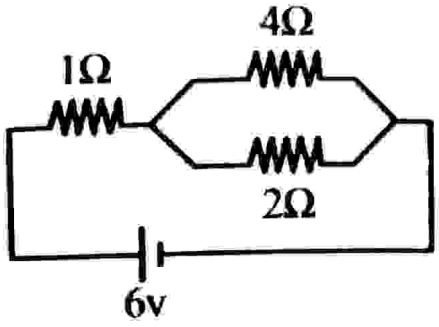
Answer: C

14. What is the current through the 2Ω resistor shown below.

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A. 7/12A
B. 12/7A
C. 12/5A
D. 5/12A



Hint: The 4Ω and the 2Ω are in parallel

$$\frac{1}{R_1}=\frac{1}{4}+\frac{1}{2}=\frac{3}{4}\Rightarrow R_1=\frac{4}{3}\Omega$$

And, R_1 is in series with the 1Ω resistor.

$$R=R_1+1=\frac{4}{3}+1=\frac{7}{3}\Omega$$

The current through the 1Ω is:

$$I_1=\frac{V}{R}=\frac{6}{7/3}=\frac{18}{7}A$$

The p.d across the 1Ω is:

$$V_1=I_1R_1=\left(\frac{18}{7}\right)1=\frac{18}{7}V$$

The p.d across the 4Ω and the 2Ω is equal to:

$$V_2=V_4=6-V_1=6-\frac{18}{7}=\frac{24}{7}A$$

The current through the 2Ω is:

$$I_2=\frac{V_2}{R_2}=\frac{24}{7/2}=\frac{24}{14}=\frac{12}{7}A$$

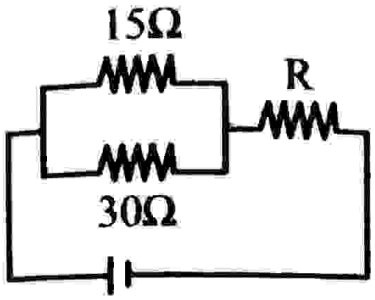
Answer: B

115. Two resistors of 15Ω and 30Ωare connected in parallel what should be the value of to be connected on series with the two so that the net resistance will be 20Ω?

A. 5Ω B. 10Ω C. 15Ω D. 6Ω

Hint: The circuit is of the form:

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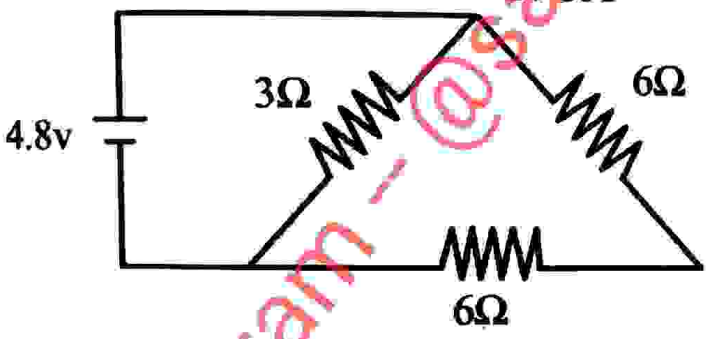
$$\frac{1}{R_1}=\frac{1}{15}+\frac{1}{30}=\frac{1}{10}\Rightarrow R_1=10\Omega$$

$$R_T=R_1+R \Rightarrow 20=10+R=10\Omega$$

Answer: B

16. What is the current flowing in the circuit shown below

A. 1A B. 2A C. 3A D. 4A



Hint: The two 6Ω resistors are in series

$$R_1=6+6=12\Omega$$

R_1 and 3 resistors are in parallel

$$\frac{1}{R}=\frac{1}{3}+\frac{1}{R_1}=\frac{1}{3}+\frac{1}{12}=\frac{5}{12}$$

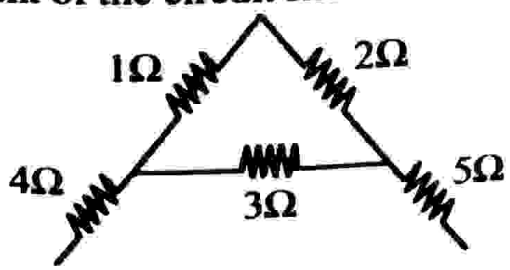
$$R=\frac{12}{5}=2.4\Omega$$

$$I=\frac{V}{R}=\frac{4.8}{2.4}=2A$$

Answer: B

17. What is the equivalent of the circuit shown below :

A. 5Ω
B. 10.5Ω
C. 5.1Ω
D. 6.5Ω

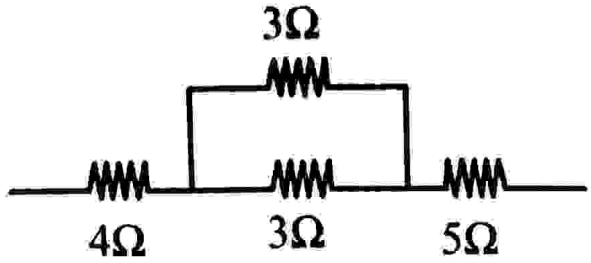


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Hint: The 1Ω and 2Ω resistors are in series

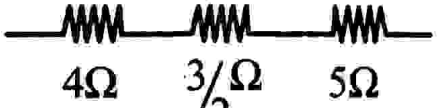
$$R_1 = 1 + 2 = 3\Omega$$



So we have :

The two 3Ω resistors are in parallel

$$\frac{1}{R_2} = \frac{1}{3} + \frac{1}{3} = \frac{2}{3} \Rightarrow R_2 = \frac{3}{2}\Omega$$



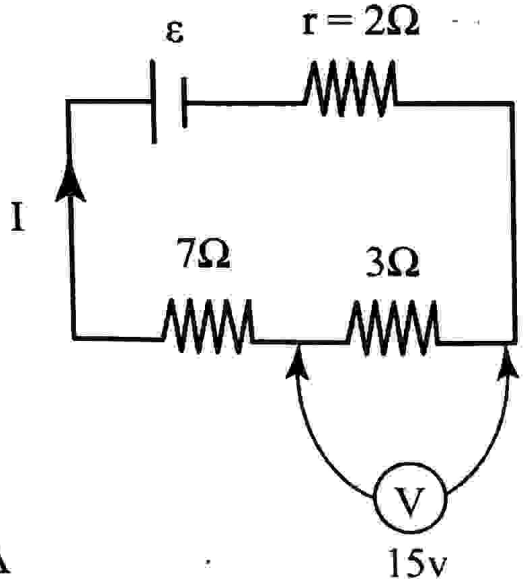
$$R_T = 4 + \frac{3}{2} + 5 = \frac{21}{2}$$

$$= 10.5\Omega$$

Answer: B

118. What are the circuit current and the emf of the battery respectively

A. 60A, 5V
B. 3A, 60V
C. 5A, 60V
D. 6A, 60V



Hint: $I = \frac{15}{3} = 5A$

$$E = I(r + 7 + 3)$$

$$E = 5(2 + 7 + 3)$$

$$= 60V$$

Answer: C

119. A battery has an emf of 3V and an internal resistance of 1Ω. What current will it give if it is connected to a circuit of 4Ω resistor?

A. 0.6A B. 0.3A C. 0.2A D. 0.7A

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Hint: $E = I(r+R)$

$$I = \frac{E}{r+R} = \frac{3}{1+4} = \frac{3}{5}$$
$$= 0.6A$$

Answer: A

120. By what factor will the current in a circuit change when a battery of 12V initially connected to a 4Ω resistor is subjected to a load resistor of 48Ω?

A. 8 B. 4 C. 1/8 D. 1/12

Solution:

$$I_1 = \frac{V}{R} = \frac{12}{4} = 3A$$

$$I_2 = \frac{V}{R_2} = \frac{12}{48} = \frac{1}{4}A$$

$$\frac{I_2}{I_1} = \frac{1/4}{3} = \frac{1}{12}$$

Answer: D

121. If the equivalent resistance of two identical resistors is 4Ω when they are connected in parallel, then what will be their equivalent resistance when they are connected in series?

A. 16Ω B. 0.5Ω C. 0.25Ω D. 8Ω

Solution:

In series, $R_T = R + R = 8 + 8 = 16\Omega$

In parallel, $\frac{1}{R_T} = \frac{1}{R} + \frac{1}{R} = \frac{2}{R}$

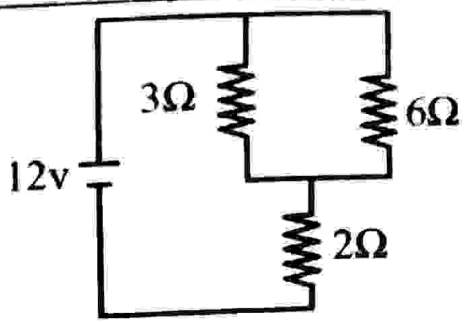
$$\frac{1}{4} = \frac{2}{R}$$
$$R = 8\Omega$$

Answer: A

122. For the circuit shown below, what is the current flowing through the 3Ω resistor?

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A. 1A B. 2A C. 3A D. 4A

Hint: The 3Ω and the 6Ω are in parallel and their resultant is 2Ω and this is in series with the 2Ω. The total resistance of the circuit is 4Ω.

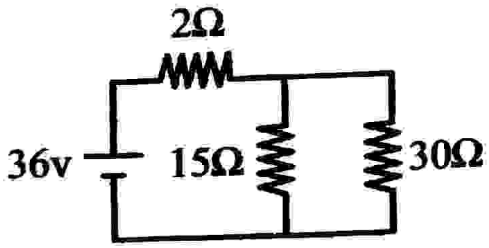
$$\text{Total current} = \frac{12}{4} = 3A$$

Voltage across the 2Ω is then 6V, which leaves 6V to the 3Ω and 6Ω resistors.

$$I_3 = \frac{V_3}{R_3} = \frac{6}{3} = 2A$$

Answer: B

123. For the circuit shown below, what is the current flowing through the 30Ω resistor?



A. 1A B. 2A C. 3A D. 4A

Hint: The 15Ω and 30Ω resistors are in parallel and their equivalent resistance is 10Ω and it gives a total circuit resistance of 12Ω.

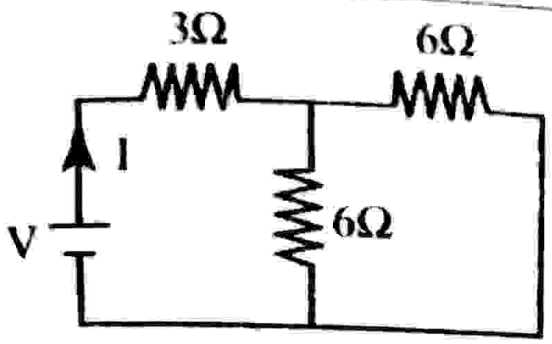
$$\text{Total current} = \frac{36}{12} = 3A$$

Potential drop across the 2Ω is 6V and leaves the 30V to the 15Ω and 30Ω resistors $I_{30} = \frac{V_{30}}{R_{30}} = \frac{30}{30} = 1A$

Answer: A

124. For the circuit shown below, if the current in the circuit is 2A. What is the voltage supplied to the circuit?

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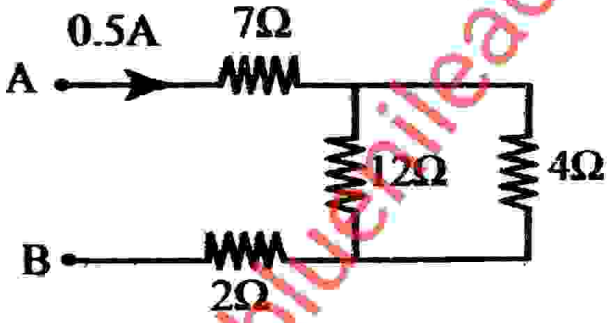


A. 3V B. 6V C. 9V D. 12V

Hint: The two 6Ω resistors are in parallel and their equivalent resistance is 3Ω and gives a total circuit resistance of 6Ω when added to the 3Ω resistor in series. V = IR = (2)(6) = 12V

Answer: D

125. What is the potential difference between point A and B



A. 3V B. 6V C. 9V D. 12V

Hint: The 4Ω and the 12Ω are in parallel and their equivalent resistance is 3Ω which is then in series with the 7Ω and 2Ω resistors. And, gives a total resistance of 12Ω.

V = IR = (0.5)(12) = 6V

Answer: B

9.10 Voltmeter And Ammeter

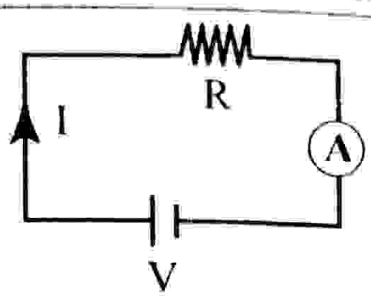
The most common electric property measuring devices are ammeter and voltmeter.

Ammeter: Is a current measuring device with a very small resistance.

Ammeter has a circuit symbol of —Ⓐ— and is connected in series with the resistor

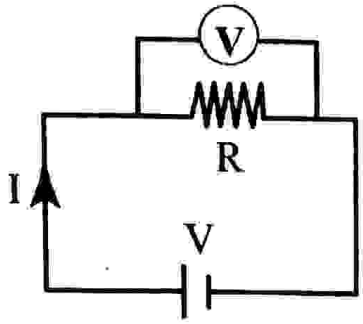
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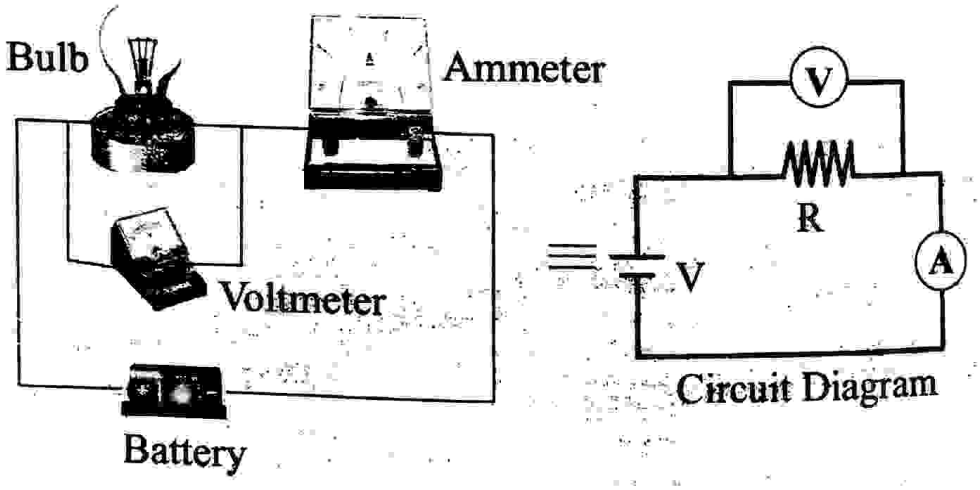


Voltmeter: Is a voltage measuring device with a very large resistance.

Voltmeter has a circuit symbol of —(V)— and is connected in parallel with the resistor



When we want to measure current and voltage at the same time, we connect ammeter and voltmeter as shown below



Conceptual Example

126. What will happen if ammeter is connected in parallel with the resistor?

Explanation:

It would draw most of the current and will get damage.

127. What will happen is voltmeter is connected in series with the

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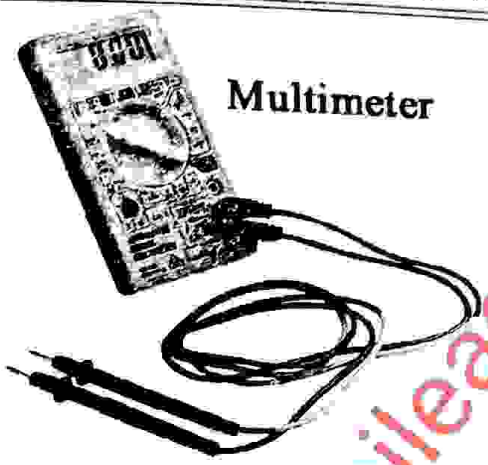
resistor?

Explanation:

It will increase the total resistance of the circuit and no current will flow in the circuit.

To measure different electric properties, we use multimeter.

Multimeter: Is a current, voltage and resistance measuring device.



Multimeter

Illustrative Example

Match the item in column A with those in column B

A	B
128. Measure current	A. Ohmmeter
129. Measures voltmeter	B. Ammeter
130. Measures resistance	C. Voltmeter
131. Measures current, voltage resistance	D. Tape meter
	E. Multimeter

Answers

128.B 129.C 130.A 131.E

9.11 Electric Safety

We humans are good conductors of electricity and electric shock can result in fatal burns and causes the heart to malfunction.

In fact, the degree of damage to the body depends on the magnitude of the current and the time of exposure.

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For example, current of 5mA or less can cause a sense of shock with no damage and a current of 10mA cause muscle to contract.

A current of 100mA and larger can be fatal.

What is more, the current that passes through a 60W bulb is about 0.5A and this current entering our body could be deadly.

We need to protect our selves and our electric devices from over dose of current. To protect devices from damaging against short circuits and overloading of circuits we use fuse.

Fuse: Is a metal of low melting point inserted in a circuit as a protective device



Fuse

132. How will a fuse provide protection against excessive current during short circuits?

Explanation:

When the current is beyond the safe limit, the fuse melts and disconnects the current flow and keep the device safe from damaging.

End of Unit Questions and Problems

I. Give short Answer to the following Questions.

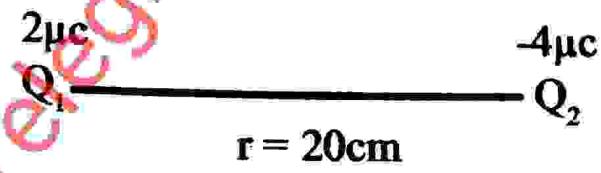
1. What are the types of charges exist in nature?
2. State the fundamental law of electrostatics
3. What is charging?
4. What are the mechanisms of charging a body?
5. State the law of conservation of charge
6. What is an electroscope?
7. What is a lighting rod?
8. State coulomb's law
9. What is electric filed strength?
10. Describe the characteristics of electric field lines
11. What is an electric current?

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12. State ohm's law
13. Describe the factors that affect resistance of metallic wire
14. What is a multimeter?
15. List electric safety rules
16. Why isn't a bird sitting on a high-voltage power line electrocuted?
17. How will the current change when the voltage is doubled keeping the resistance constant?
18. If aluminium and copper wires of the same length have similar resistances, which one will have larger diameter?
19. Why is the series combination not used for domestic circuits?
20. Why do we use copper and aluminium wires for electricity transmission?
21. What determines the severity of an electric shock?
22. If all the light in the room go out when an electric heater is plugged. Explain what must have happened?

II. Solve the following Problems

23. Determine the magnitude and direction of the electrostatic force that one charge exerts on the other .



24. Two spheres that are 4cm apart attract each other with a force of 1.2×10^{9}N. What is the magnitude of the charge on each sphere if one has twice the charge than the other?
25. If the electrostatics force between two identical charged bodies of each charges 1.1×10^{7}c is 4.2×10^{4}N, what is their separation?
26. A copper wire of resistivity $1.6\times10^{-8}\Omega$m and 0.5mm diameter has a resistance of 10Ω. What its length?
27. Plot the graph and calculate the resistance of a metallic wire whose current and voltages are following

V (v)	1.5	3	4.5	6	7.5
I (A)	0.1	0.2	0.3	0.4	0.5

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28. What resistor should be connected to a 12v battery to product a current of 2.5mA in the circuit?

29. If a battery of 9v is connected in series with 2Ω,3Ω,4Ω,5Ω and 12Ω resistors, then what current flows through the 5Ω resistor?

30. How many176Ω resistors should be connected in parallel to get a 5A current on a 220v line?

31. How should three resistors of each 6Ω resistance be connected so that their combination has a resistance of

    A. 9A

    B. 4Ω

32. Why is it dangerous to use a fuse that is rated 30A on a circuit calling for a 15A fuse?

I. Answers to the End of Unit Questions

1. There are two types of charges, Negative and Positive.
2. Like charge repel each other and unlike charges attract each other.
3. Charging is the process of transferring charge from one body to the other.
4. Rubbing, conduction and Induction.
5. Charge is neither created nor destroyed, but transferred from one body to the other
6. Electroscope is a device used to detect charge
7. Lighting rod is a metallic rod mounted at the top of a building to keep the structure safe from lightning.
8. Coulomb's law state that, the electrostatic force between two point charges is directly proportional to the product of their charges and inversely proportional to the squire of their separation.
9. Electric field strength is the electric force per unit charge acting on a test charge placed in the vicinity.
10. Field lines are:-Symmetrical
    - ➥ Never cross - each other
    - ➥ Perpendicular to the charged body

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- Closer where the field is stronger
- Run from positive to negative charge

11. Electric circuit is a closed path for the flow of electric current.

12. Ohm's law states that, current flowing in a circuit increases as the voltage across the ends of the conductor increases.

13. Length, cross-sectional area, temperature and the material from which the wire is made are some of the factors that affect resistance of a metallic conductor

14. Multimeter is a device that is used to measure the current, resistance and voltage in a circuit

15. Use fuse, use grounding, use insulators, handles, switch off power while working on a live wire.

16. Its leg is insulator and the circuit is open as long as the bird sit on a single wire

17. $I_1 = \frac{V_1}{R}$

$I_2 = \frac{V_2}{R}, V_2 = 2V_1$

$I_2 = \frac{2V_1}{R} = 2\left(\frac{V_1}{R}\right)$

$I_2 = 2I_1$, doubled

18. $R = \frac{\rho\ell}{A} = 4\frac{\rho\ell}{\pi d^2}$

$R_{Al} = \frac{4\rho_{Al}\ell}{\pi d_{Al}^{\ 2}}$ and $R_{cu} = \frac{4\rho_{cu}\ell}{\pi d_{cu}^{\ 2}}, R_{Al} = R_{cu}$

$\frac{R_{Al}}{R_{cu}} = \frac{d_{cu}^{\ 2}\rho_{Al}}{d_{Al}^{\ 2}\rho_{cu}} = 1$

$\frac{d_{Al}^{\ 2}}{d_{cu}^{\ 2}} = \frac{\rho_{Al}}{\rho_{cu}}$, But $\rho_{Al} > \rho_{cu}$

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Therefore, Aluminium has larger diametr

19. This is so because, in series connection, if the line fails at some point, all the electric devices stop working.
20. Because, they have very small resistance as compared to other available conductors.
21. It depends on the magnitude of the current, the length of time of exposure and the part of the body involved.
22. Because, electricity transfers from the outlet into the bulbs and causes a rapid draw on.

II. Solution to the end of Unit Problems

23. $F = \frac{kQ_1Q_2}{r^2} = \frac{(9\times10^9)(2\times10^{-6})(4\times10^{-6})}{(20\times10^{-2})^2}$

$F = 6\times10^{-4}\,N$, Attractive

24. $F = \frac{kQ_1Q_2}{r^2}, Q_2 = 2Q_1$

$F = \frac{2kQ_1^2}{r^2} = Q_1^2 = \frac{Fr^2}{2k}$

$Q_1 = \sqrt{\frac{Fr^2}{2k}} = \sqrt{\frac{(1.2\times10^9)(4\times10^{-2})^2}{2(9\times10^9)}}$

$Q_1 = 1\times10^{-11}\,C$ and $Q_2 = 2Q_1 = 2\times10^{-11}\,C$

25. $F = \frac{kQ_1Q_2}{r^2} = \frac{kQ^2}{r^2}$

$r^2 = \frac{kQ^2}{F}$

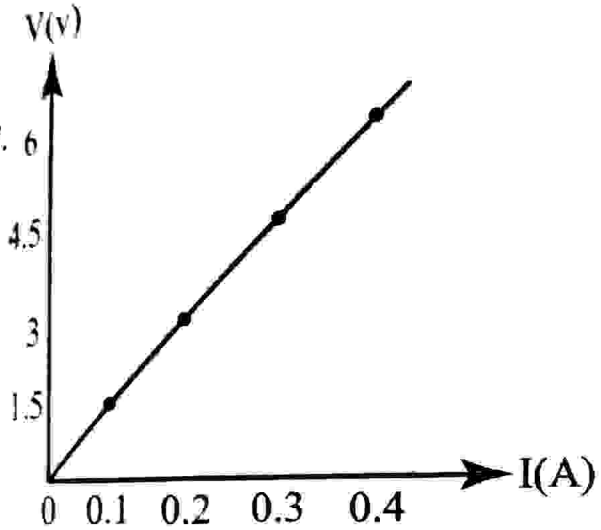
$r = \sqrt{\frac{kQ}{F}} = (1.1\times10^7)\sqrt{\frac{9\times10^9}{4.2\times10^4}} = 0.5m$

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26. $R = \rho\frac{\ell}{A} = \frac{4\rho\ell}{\pi d^2}$

$$\ell = \frac{R\pi d^2}{4\rho} = \frac{(10)(3.14)\left(0.5\times10^{-3}\right)^2}{4\left(1.6\times10^{-8}\right)} = 123\text{m}$$

27.



$$R = \frac{V}{R} = \frac{7.5-1.5}{0.5-0.1} = \frac{6}{0.4} = 15\Omega$$

28. $V = IR$

$$R = \frac{V}{I} = \frac{12}{2.5\times10^{-3}} = 4.8\times10^{3}\Omega$$

29. $V = IR$

$$I = \frac{V}{R} = \frac{12}{2+3+4+5+12} = 0.5\text{A}$$

30. $R_T = \frac{V}{I} = \frac{220}{5} = 44\Omega$

$$\frac{1}{R_T} = \frac{n}{176}$$

$$n = \frac{176}{44} = 4\,\text{Resistors}$$

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UNIT 10

10. Magnetism

10.1 What is a Magnet ?

10.2 Magnetic Field

10.3 Earth's Magnetic Field

10.4 Magnetic Force on a Moving Charge

10.5 Magnetic Force on a Current Carrying Wire

10.6 Magnetic Field Current Carrying Wire

10.7 Magnetic Force Between two Parallel Current Carrying Wires

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10. Magnetism

Dear learner,how do you do ? Welcome to this impressive unit where you will be able to define magnetic field, explore the cause of Earth's magnetic nature, determine the magnetic force acting on a moving charged particle and on a current carrying wire, describe the nature and magnitude of the magnetic field produced by a current carrying wire,find out the magnetic force between two parallel current carrying wires and appreciate the applications that magnets have in our every day life.

10.1 What is a Magnet?

The attraction or repulsion property of some materials due to the spinning of their electrons is called magnetic effect.

Materials such as Nickel, cobalt and steel are said to be magnetic. And,the natural occurring stone with magnetic property is called *lodestone.*

Types of Magnets

Magnets are categorized as permanent, temporary and electromagnet.

i. **Permanent Magnet**

is made from specialized iron ores,steel ... etc. and have a prolonged magnetic property.

Permanent magnets are available in different shapes and sizes as shown below.



Bar magnet



Horse shoe magnet



U shape mag

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Arc magnet Ring magnet Cylinderical magnet

Permanent magnets can demagnetized or lose their magnetic property if they are over heated, hammered or stroked.

ii. Temporary Magnets

Soft irons such as nail and paper - clip gain magnetic property when immersed in an external magnetic field and lose their magnetic property when the magnetic field is removed.

iii. Electromagnet

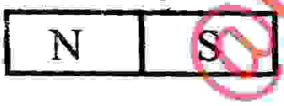
Is a coil of wire with an iron core inside of it and acts as a magnet when current passes through it.

These magnets are used in generators, motors and transformers

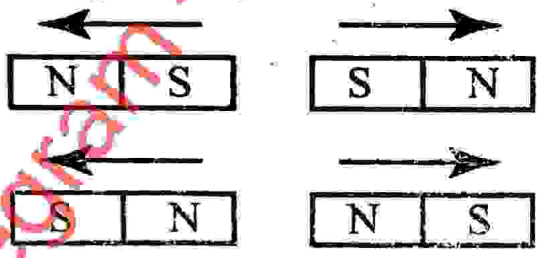
Properties of Magnets

The basic properties of magnets are

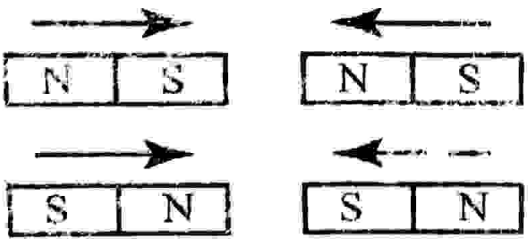
- Every Magnet has two poles. North and south



- Like poles repel each other and unlike poles attract each other



Like poles repel

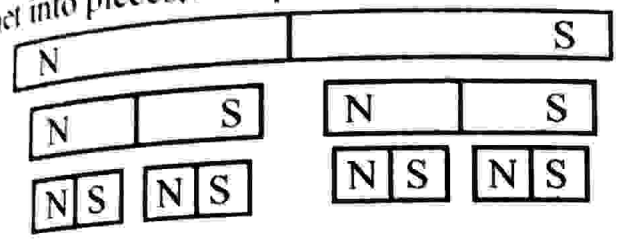


unlike poles attract

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Magnetic poles exist in pairs. There is no isolated single pole If you cut a magnet into pieces,each piece will have both north and south poles.



10.2 Magnetic Field

Magnetic field is produced by a magnet and a current carrying conductor

Magnetic field : Is a region around a magnet and a current carrying conductor Where a magnetic force is felt.

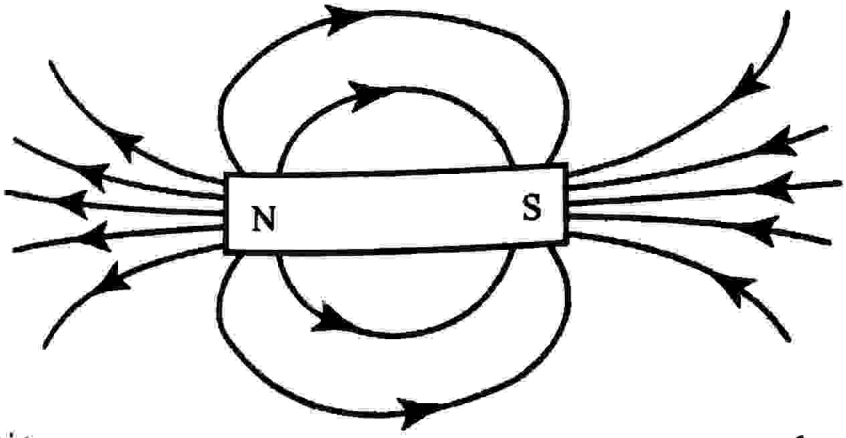
The direction and intensity of a magnetic field is visualized using imaginary lines called *magnetic field lines.*

- The direction of a magnetic field is determined using the deflection direction of a compass needle placed near the magnet.

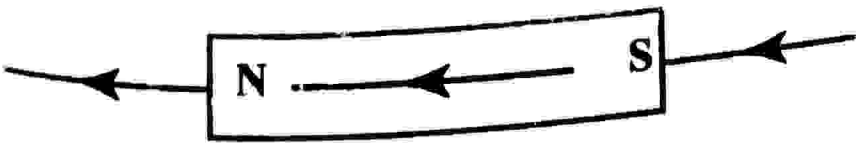
Characteristics of Magnetic Field Lines

Some of the properties of magnetic field lines are the following.

- Out side the magnet, magnetic field lines run from the North pole to the South pole.

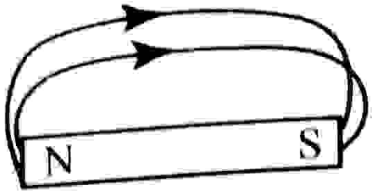


In side the magnet, magnetic field lines run from the south pole to the North pole



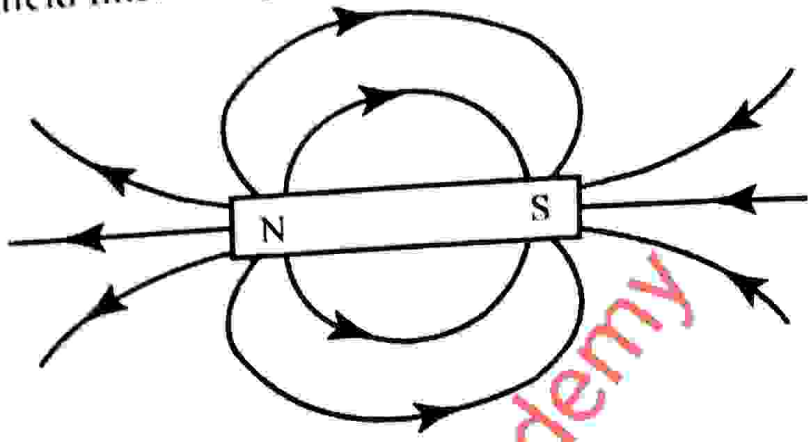
Magnetic field lines never cross each other

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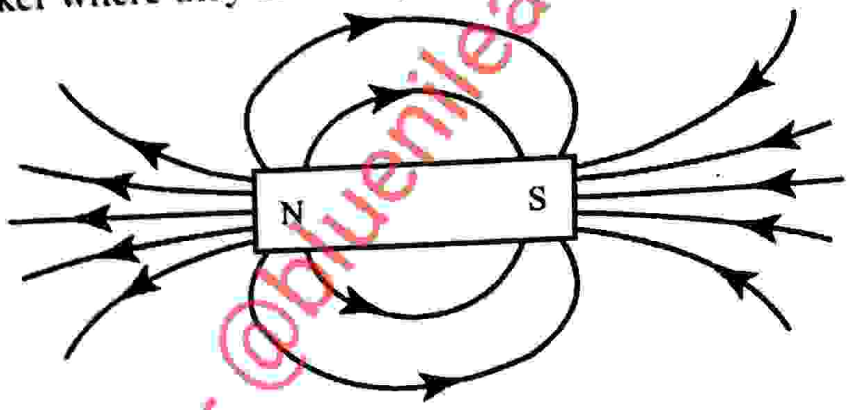


Not possible

- Magnetic field lines are symmetrical and from a closed loop.



- Magnetic force is stronger where the field lines are closer together and weaker where they are far apart



Strong field

weak field

- Magnetic field lines are denser at the poles than they are at the middle of the magnet as shown above.

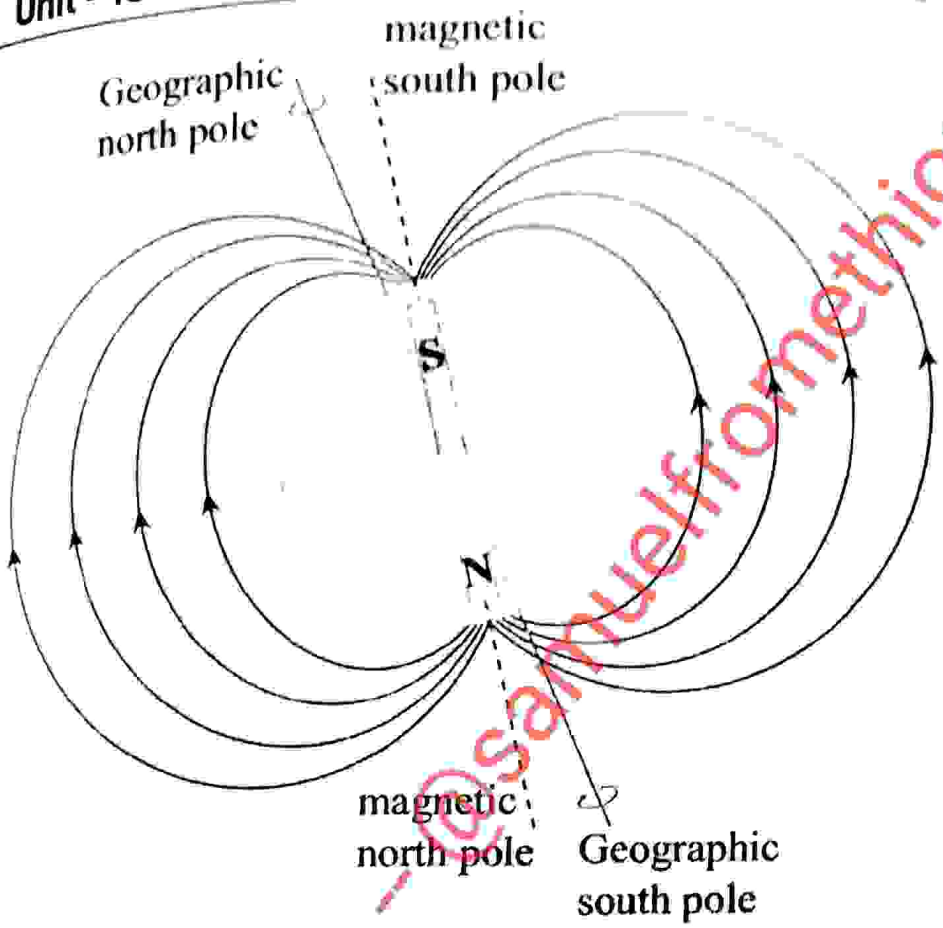
10.3 Earth's Magnetic Field

The earth acts as a huge bar magnet and the cause of earth's magnetic field is thought to be the circulation of molten materials in the outer core layer of the earth.

- The orientation of earth's magnetic field is like a bar magnet tilted about 11° from the geographic North pole as shown below

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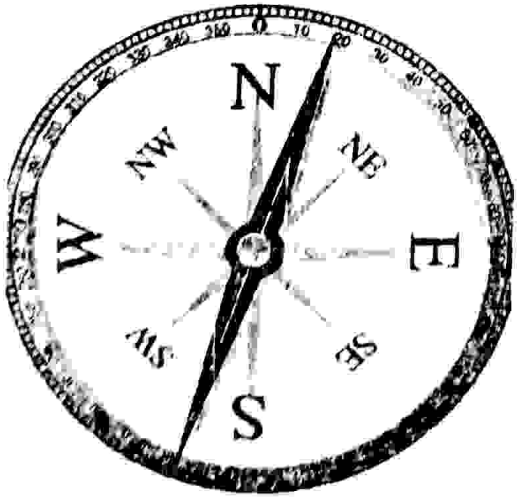
- Earth's magnetic South pole is near the geographic north pole and the earth's magnetic North pole is near the geographic south pole

A) Compass

Compass is used to know direction when we are on the land, on the ocean and in the air

Compass : Is a material with a magnetic needle that is free to rotate.

The compass needle north pole always points in the geographic North pole



A compass

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Since Compass has a magnetic needle,earth's magnetic field causes it to rotate and align itself in opposite directions of earth magnetic poles.

Illustrative Examples

1. What a Magnet ?

Explanation

A magnet is a material that has an attraction property to some materials due to the motion of its charging particles .

2. What is the naturally occurring magnet ?

Explanation

The naturally occurring magnet is known as lodestone

3. What is magnetic field

Explanation

Magnetic field is a region around a magnet or a current wire where magnetic force is felt

4. What are the properties of magnets ?

Explanation

- Magnets have north and south poles
- Magnetic poles always exist in pairs. There is no single pole
- Like poles repel and unlike poles attract each other

5. What are the types of magnets ?

Explanation

Permanent magnet

Temporary magnet

Electromagnet

6. What are magnetic field lines ?

Explanation

Magnetic field lines are imaginary lines used to visualize the

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direction and magnitude of magnetic field.

7. How will a magnet lose its magnetic property ?

Explanation

A permanent magnet can lose its magnetic property if it experiences ;

- Over heating
- Stroking
- Hammering effects

8. List some properties of magnetic field lines ?

Explanation

Characteristics of magnetic field lines are

- They move from the north pole to the south pole outside the magnet and from the south to the north pole inside the magnet
- They never cross each other
- They are symmetrical
- They form closed loop
- The field is stronger where the field lines are closer

9. State the fundamental law of magnetism

Explanation

The basic law of magnetism stakes that ;

Like poles repel each other and unlike poles attract each other.

10. What is the evidence that supports the idea that earth has a magnetic field ?

Explanation

One possible evidence that supports the existence of earth's magnetic field is the alignment of a compass or a bar magnet in the North - South direction when suspended in air

11. What is the cause of earth's magnetic field ?

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Explanation

The cause of earth's magnetic field is assumed to be the circulation of molten materials inside the earth.

12. What is a compass ?

Explanation

A compass is a magnetic needle that is free to rotate and shows us direction.

13. Why does the needle of a compass point in the North - South direction ?

Explanation

Because,the earth's magnetic field interacts with the magnet of the compass and the alignment is based on the fundamental law of magnetism.

14. Is there any difference between the geographic and magnetic poles of the earth ?

Explanation

Yes. The earth's magnetic south pole is near the geographic North pole and the earth's magnetic north pole is near the geographic south pole

15. What is the angle of inclination of the earth's magnetic south pole from the geographic north pole ?

Explanation

The earth's magnetic south pole is tilted about 11^0 towards west from the geographic north pole.

16. Can we expect a huge solid bar magnet in side the earth ?

Explanation

No. The inner layer of the earth,the core is of extreme temperature and hence, materials exist in a molten state. Therefore, a solid huge bar magnet could not exist in the interior of the earth.

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10.4 Magnetic Force on a Moving Charge

When a moving charge is subjected to a uniform external magnetic field,it experiences a force. And, the magnitude of the magnetic force acting on a charged particle moving in a uniform magnetic field is obtained by;

$$F = QvB \sin \theta$$

Where: F magnetic force on the particle

V Speed of the particle

Q charge of the particle

B magnetic field strength

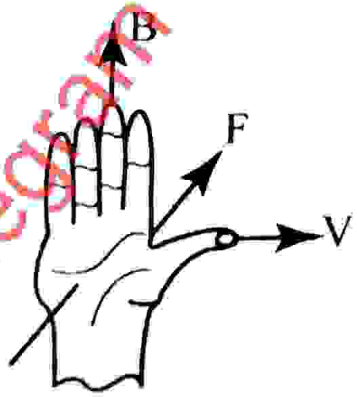
θ Angle between direction of motion of the particle and the magnetic field

Magnetic field strength, B is a vector quantity and its SI unit is the Tesla (T)

Moreover, the direction of the force acting on the moving charged particle is determine using the right hand rule.

Right Hand Rule : If you point your right hand thumb in the velocity direction and your other fingers in the field direction, then your palm will point in the force direction.

For example, if the particle is moving to the right in a field directed away from you, then the force on the particle will be up ward as shown in the figure.



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Conceptual Examples

a. Is the magnetic force on a moving charge be zero in a non - zero field ?

Explanation

Yes. The force on a moving charged particle immersed in a uniform magnetic field is zero when the particle is moving parallel or anti

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parallel to the field.

18. When will the magnetic force acting on a charged particle became maximum ?

Explanation

When the particle moves perpendicular to the direction of the magnetic field.

Illustrative Examples

19. What is the magnetic force acting on a particle of change 2Nc moving at a speed of 6×10^5m/s at an angle of 30^0 to a uniform magnetic field of 100T ?

Solution:

$$F = Qv\ B \sin\theta$$

$$F = (2\times10^{-6})(6\times10^{5})\ (100)\ \text{Sin}\ (30^0),\ \text{Sin}\ 30^0 = 0.5$$

$$F = 12\times10\times0.5 = 60\text{N}$$

20. A magnetic force of 100 N is acting on a 5Nc charge moving at a speed of 8×10^6m/s at right angle to a uniform magnetic field. What is the magnetic field strength ?

Solution :

$$F = QvB \sin\theta,\ \theta = 90^0 \text{ and } \sin 90^0 = 1$$

$$F = QvB$$

$$B = \frac{F}{Qv} = \frac{100}{(5\times10^{-6})(8\times10^{6})}$$

$$= 2.5\text{T}$$

21. If a magnetic force of 80N acted on a particle of change 2Nc moving at an angle of 53^0 to a uniform field of 10T, with what speed was the particle entered the field ?

Solution:

$$F = QvB \sin\theta$$

$$V = \frac{F}{QB \sin\theta}$$

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$$V = \frac{80}{(2\times10^{-6})(10)\text{Sin}53^0} \quad \text{Sin}53^0 = 0.8$$

$$V = \frac{80}{(2\times10^{-6})(10)(0.8)}$$

$$= 5\times10^4 \text{m/s}$$

22. If a particle of charge 3Nc moving at a speed of 2×10^6m/s along the positive x axis enters a uniform field of 50 T directed in the positive y axis, what is the magnitude and direction of the force acting on the particle ?

Solution :

$$\vec{F} = Q(\vec{V}\times\vec{B})$$

$$F = QvB\sin\theta, \quad \theta = 90^0 \text{and} \sin90^0 = 1$$

$$F = (3\times10^{-6})(2\times10^6)(50)$$

$$F = 300\text{N, along the positiv z axis}$$

23. What is the charge carried by a particle that is moving at a speed of 4×10^5m/s in a uniform field of 10T at an angle of 37^0 and experienced a force of 60N

Solution:

$$F = QvB\sin\theta$$

$$Q = \frac{F}{VB\sin\theta} = \frac{60}{(4\times10^5)(10)(0.6)}$$

$$Q = 0.25\times10^{-6} = 0.25\mu c$$

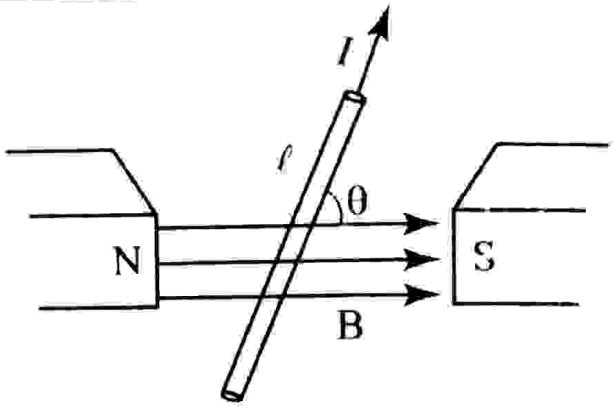
10.5 Magnetic Force on a Current Carrying Wire

A current carrying wire feels magnetic force when subjected to a magnetic field since the wire is full of many moving particles.

The magnitude of the magnetic force acting on a current wire exposed to a magnetic field is obtained as follows:

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$$F = I \ell B \sin\theta$$

Where : F = Magnetic force acting on the wire

I = Current flowing through the wire

ℓ = Length of the wire

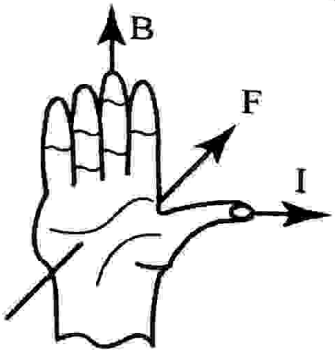
B = Magnetic field strength

θ = Angle between the length element and field

Moreover, the direction of the force acting on the wire is determined using the right hand rule

Right Hand Rule : If you point your right hand thumb in the current direction and your other fingers in the field direction,then your palm faces in the magnetic force direction .

For example, if current is moving to the right and the magnetic field is away from you, then the force on the wire will be upward - out of the page.



Illustrative Example

24. When will the magnetic force on a current wire became zero ?

Explanation

When the current flows parallel or anti - parallel to the field, the

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force on the wire is zero.

25. When will the magnetic force on a current carrying wire become maximum ?

Explanation

When the wire is perpendicular to the external magnetic field.

26. What is the magnitude of the force on a wire that carried a current of 5A and whose 25cm length element is immersed in a magnetic field of 0.8T at an angle of 37^0 ?

Solution:

$$F = I \ell B \sin\theta \quad \sin 37^0 = 0.6$$
$$F = (5)(0.25)(0.8)(0.6)$$
$$= 0.6N$$

27. What is the force per length acting on a wire that carried a current of 5A subjected to a magnetic field of 0.8T perpendicularly ?

Solution :

$$F = I \ell B \sin\theta, \quad \sin 90^0 = 1$$
$$F = I \ell B$$
$$\frac{F}{\ell} = I B = (5)(0.8)$$
$$= 4 \, N/m$$

28. What current is flowing through a wire of 50cm if it experienced a force of 20N when subjected to a magnetic field of 20T at an angle of 30^0 ?

Solution:

$$F = I \ell B \sin\theta$$
$$I = \frac{F}{\ell B \sin\theta}, \quad 30^0 = 0.5$$
$$I = \frac{20}{(0.5)(20)(0.5)}$$
$$= 4A$$

29. What is the magnitude and direction of a magnetic field in to which a

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wire of length 50cm carrying current of 2A along the positive x axis experiences a 10N magnetic force along the positive y axis ?

Solution:

$$\vec{F} = I(\vec{\ell} \times \vec{B})$$

$$F = I\,\ell\,B\sin\theta, \quad \theta = 90^0$$

$$F = I\,\ell\,B$$

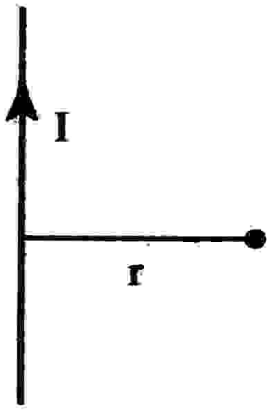
$$B = \frac{F}{I\ell} = B = \frac{10}{(2)(0.5)} = \frac{10}{1}$$

$B = 10T$ along the negative z axis

10.6 Magnetic Field of a Current Carrying Wire

A current carrying conductor produces magnetic field around it, and a magnet produces electric current.

The magnitude of the magnetic field produced due to a current carrying wire at some distance from the wire is obtained by;



$$\boxed{B = \frac{\mu o I}{2\pi r}}$$

Where : $\mu o = 4\pi \times 10^{-7} \frac{Tm}{A}$ is called *permeability of free space*

I = current flowing through the wire

r = distance from the wire

Moreover, the direction of the magnetic field of a current carrying wire is determined using the right hand rule.

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Right Hand Rule : If you grasp the current carrying wire with your right hand in such a way that your thumb points in the current direction,then your other fingers will curl in the direction of the magnetic field.

Symbols used to indicate field and current direction;

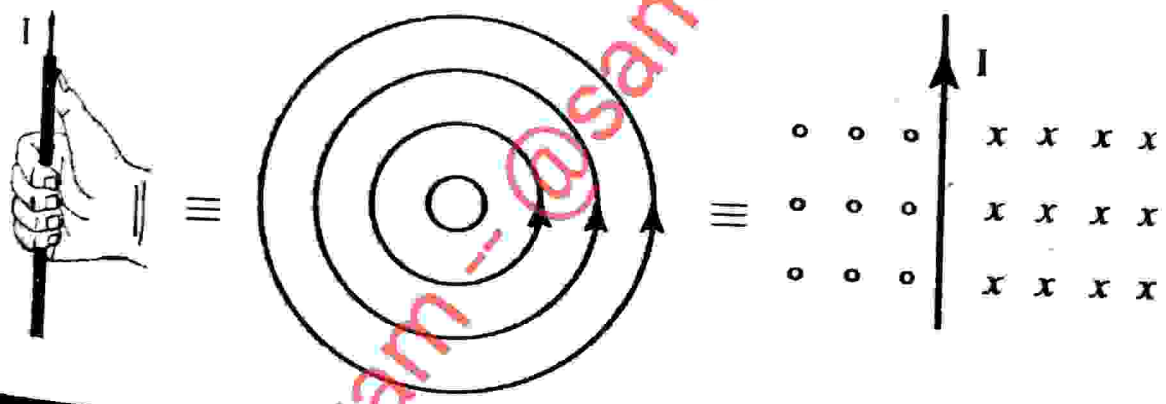
x = Magnetic field directed into the page

. = Magnetic field directed into the page

⊗ = Electric current directed into the page

θ = Electric current directed out of the page

For example, if current is flowing up ward,through a wire, then the magnetic field is counter clockwise according to the right hand rule .



Illustrative Examples

30. What is the magnitude and direction of the magnetic field at 4cm from a wire of current 5A flowing downward ?

Solution :

$$B = \frac{\mu o I}{2\pi r}$$

$$B = \frac{(4\pi \times 10^{-7})(5)}{2\pi(0.04)}$$

$$= 0.25 \times 10^{-4}\ T, \text{ Clockwise.}$$

31. At What distance from a wise of current 2A will the magnetic field strength be 0.5T

Solution:

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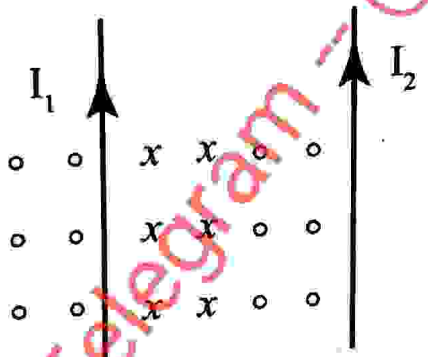
$$B = \frac{\mu o I}{2\pi r}$$

$$r = \frac{\mu o I}{2\pi B} = \frac{(4\pi \times 10^{-7})(2)}{2\pi(0.5)}$$

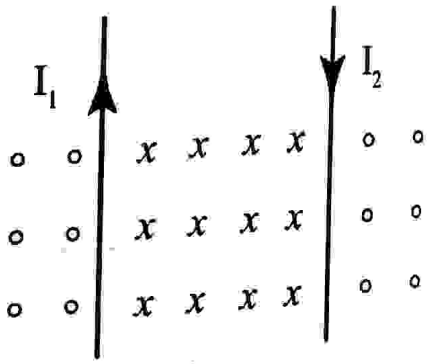
$$r = 8 \times 10^{-7} m$$

10.7 Magnetic Force between two Parallel Current Carrying wires

The nature of the magnetic force between two current carrying wires is determined using the right hand rule. As a result, two wires carrying current in the same direction attract each other, while wires carrying current in opposite directions repel each other.

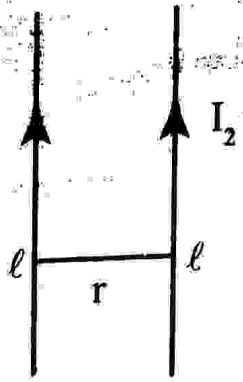


Attract each oher



Repel each oher

And, the magnitude of the magnetic force between two straight parallel wires is obtained as follows :



$$\boxed{F = \frac{\mu o I_1 I_2 \ell}{2\pi r}}$$

Where : I_1 = current flowing through wire - 1

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I_2 = current flowing through wire - 2

ℓ = length of the wires

r = separation of the wires.

Illustrative Examples

32. What is the magnetic force between two wires of each 100m long, 5cm apart and carrying currents of 15A and 30A in opposite directions ?

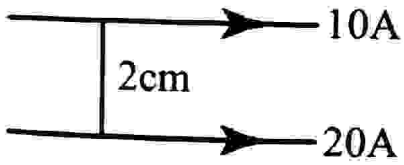
Solution :

$$F = \frac{\mu o I_1 I_2 \ell}{2\pi r}$$

$$F = \frac{(4\pi \times 10^{-7})(15)(30)(100)}{2\pi(5\times 10^{-2})}$$

$F = 0.18N$, Repulsive

33. What is the force per length of the two wires shown below ?



Solution :

$$F = \frac{\mu o I_1 I_2 \ell}{2\pi r}$$

$$\frac{F}{\ell} = \frac{\mu o I_1 I_2}{2\pi r}$$

$$\frac{F}{\ell} = \frac{(4\pi \times 10^{-7})(10)(10)}{2\pi(2\times 10^{-2})}$$

$= 10^{-3}$ N/m, Attractive

34. Two parallel wires that are 10cm apart carry same currents and interact with a force of 0.08N/m. What is the current in each wire

Solution :

$$\frac{F}{\ell} = \frac{\mu o I_1 I_2}{2\pi r} = \frac{(4\pi \times 10^{-7})(I)(I)}{2\pi(0.1)} = 2\times 10^{-6} I^2$$

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$$0.08 = 2 \times 10^{-6}\ I^2$$

$$I = 200A$$

35. Magnetic domains are :

A) The magnetic poles of a bar magnet

B) Microscopic magnets in the magnet

C) Collection of similar magnetic poles

D) Bar magnets arranged in the same direction

Hint: Magnetic domains are tiny magnets **Answer : B**

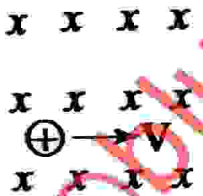
36. The Magnetic field is into the page If a proton is moving in the field at velocity v, the magnetic force on the proton is directed :

A. ←

B. →

C. ↑

D. ↓



Hint : use the right hand rule. **Answer : C**

37. The direction of the magnetic force on a current carrying wire is a shown. What is the direction of the magnetic field ?

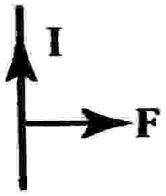
A) Into the plane of this page

B) In the direction of the force

C) In the direction of the current

D) Out of the plane of this page

Hint : use the right hand rule



Answer : D

38. A charged particle moving perpendicularly to magnetic field may experience a path with constant.

A) Velocity

B) Acceleration

C) Radius

D) Displacement

Hint: The charged particle will follow a circular path of constant radius.

Answer : B

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39. An electron moving due east in a uniform magnetic field directed into the page may experience a force towards

A) North B) South C) East D) West

Hint: Use right hand rule and then reverse direction so,in this case North to a proton, south for an electron

Answer : B

40. For what orientation between the moving charge particle and the uniform magnetic field will the force become maximum ?

A) $\theta = 0^0$ B) $\theta = 45^0$ C) $\theta = 90^0$ D) $\theta = 180^0$

Hint : $F = QvB$ in $\theta, = 90^0, \sin(90^0) = 1$

$F = QvB$, maximum

Answer : C

41. What is the magnetic field strength 10cm from a straight wire of current 3A ?

A) 3T B) 10T C) $6x10^{-6}T$ D) $6x10^{-7}T$

Solution:

$$B = \frac{\mu o I}{2\pi r} = \frac{(4\pi \times 10^{-7})(3)}{2\pi (0.1)}$$

$$B = 6\times 10^{-6}\ T$$

Answer : C

42. A current wire is put in a uniform magnetic field All the modifications will change the magnitude the magnetic force. Except

A) Doubling the current through the wire

B) Increasing the magnetic field strength

C) Rotating the wire to change its orientation

D) Reversing the direction of flow of the current

Hint : $F = ILB \sin\theta$

Answer : D

43. Two long parallel conductors that are 50cm apart carry currents $I_1 = 7A$ and $I_2 = 10A$ in opposite directions. What is the resultant magnetic field at point P ?

A) 8×10^{-6} T up

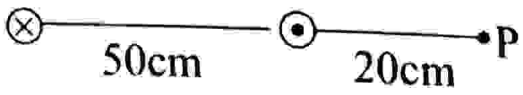
B) 8×10^{-6} T down

C) 1.2×10^{-5} T up

D) 1.2×10^{-5} T down

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Solution :



$$B_p = B_2 - B_1$$

$$B_p = \frac{\mu o I_2}{2\pi r_2} - \frac{\mu o I_1}{2\pi r_1}$$

$$B_p = \frac{4\pi \times 10^{-7}}{2\pi \times 10^{-2}}\left[\frac{10}{20} - \frac{7}{70}\right]$$

$$= 8 \times 10^{-6} \text{Tup}$$

Answer : A

44. If two long parallel wires 1m apart carry the same current and the force per unit length on each wire is 2×10^{-7}N/m, then what is the current in each wire ?

A) 2A B) 1A C) 3A D) 2×10^{-7}A

Solution :

$$F = \frac{\mu o I_1 I_2 \ell}{2\pi r}$$

$$\frac{F}{\ell} = \frac{\mu o I^2}{2\pi r}$$

$$2 \times 10^{-7} \frac{(4\pi I^2)}{2\pi(1)} 10^{-7}$$

$$2 = 2I^2 \quad I = 1A$$

Answer : B

45. A positive charge moving upward enters a magnetic field directed to the North. The force on the change will be towards ?

A) North B) South C) East D) West

Hint: using the right hand rule,

Answer : D

46. A magnetic field ;

A) Always exert a force on a changed particle

B) Never exert a force on a charged particle

C) Exert force on a charged particle moving across it

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D) Exert force on a charged particle moving along it

Hint: F Qv B sinθ

Answer : C

47. The North pole of Earth's magnet is near the geographic

A) South B) North C) East D) West

Hint: 11° tilted from the south

Answer : A

48. How much current should a wire carry to produce a magnetic field of 0.8 T at distance of 2mm ?

A) 4×10^3A B) 2×10^3A C) 8×10^3A D) 6×10^3A

Solution :

$$B = \frac{\mu o I}{2\pi r} \Rightarrow I = \frac{(B)(2\pi r)}{\mu o}$$

$$I = \frac{(0.8)(2\pi)(2\times10^{-3})}{4\pi \times 10^{-7}}$$

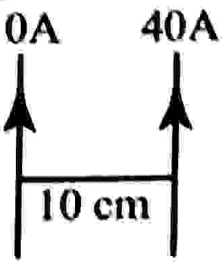
$$= 8\times10^3 A$$

Answer : C

49. What is the magnetic field strength at the mid - point of the two wires

A) 1.2×10^{-6}T into the page C) 1.2×10^{-4}T into the page

B) 1.2×10^{-6}T out of the page D) 1.2×10^{-4}T out of the page



Solution :

$$B = B_2 - B_1$$

$$B = \frac{\mu o I_2}{2\pi r_2} - \frac{\mu o I_1}{2\pi r_1}, r_1 = r_2 = 5\times10^{-2}m$$

$$B = \frac{4\pi \times 10^{-7}}{(2\pi(5\times10^{-2})}[40-10]$$

$$B = 12\times10^{-5}T = 1.2 \times10^{-4}T, \text{ out of the page}$$

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Answer : D

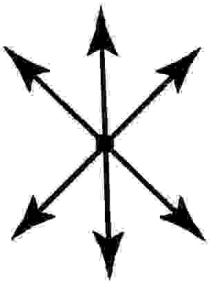
50. A permanent magnet
   A) Attract all materials
   B) Attract only magnetic materials
   C) Attract magnetic materials and repel non magnets
   D) Attract non - magnets and repel magnetic materials

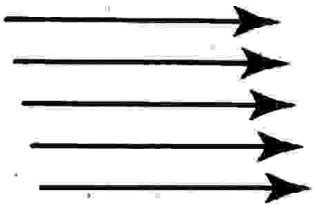
Answer : B

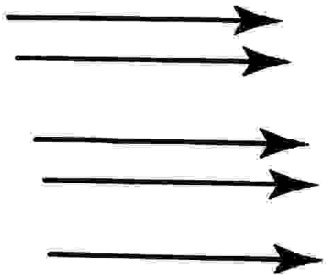
51. Out of the following which one is non - magnetic ?
   A) Zinc B) Steel C) Copper D) Cobalt

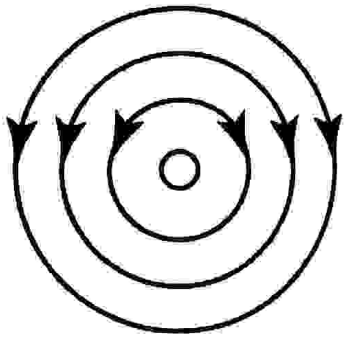
Answer : A

52. Which graph of magnetic field lines represent a uniform magnetic field ?

A. 

C. 

B. 

D. 

Answer : C

End of unit Questions and problems

I. Give short Answers to the Following Questions

1. How is a magnetic field produced ?
2. Can a magnetic monopole exist ?
3. What happens when like poles are brought closer together ?
4. Plot magnetic field lines around a bar magnet
5. How will a compass help us to plot field lines
6. How will a compass help us to know direction
7. Is there any difference between the earth's magnetic north pole and the geographic north pole ?

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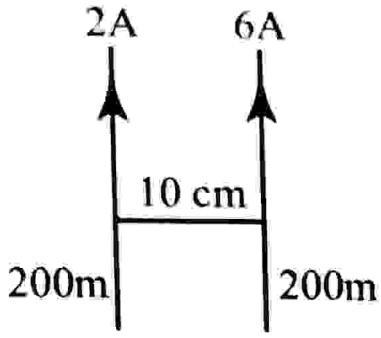
8. Can we conclude that a moving particle and a current wire always felt magnetic force when immersed in an external magnetic field ?
9. Can an electric current produce magnetic effect ?
10. What is the nature of the magnetic force between two parallel current carrying wires
11. How can we determine the direction of the magnetic field produced by a Current carrying wire ?
12. What will happen to the magnetic field strength of a straight wire if the current is doubled ?
13. What will happen to the magnetic force between two straight parallel current carrying wires torrents if ;
    a) Their separation is doubled
    b) Their separation is halved
    c) Their currents are halved

II. End of Unit Problems

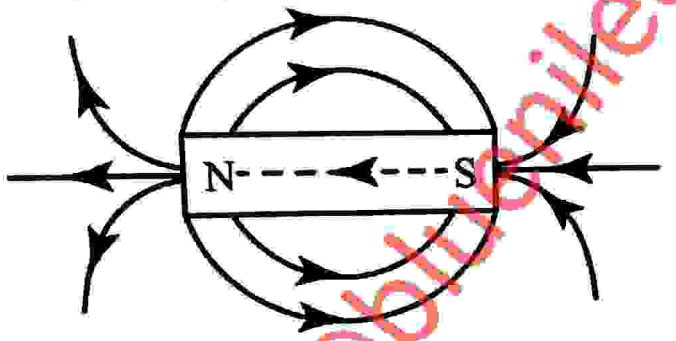
14. What amount of magnetic force will a 4Nc charge experience when exposed to a field of 0.6 T directed at 45^0 from the particle's velocity of 4×10^5 m/s ?
15. What perpendicular field to a 50cm wire carrying current of 8A exert a force of 0.4 N ?
16. What is the magnetic force on a wire of length 80cm carrying a current of 5A when immersed in a magnetic field of 0.6T perpendicularly ?
17. What is the magnetic field strength at a distance of 10cm from a wire of current 25A ?
18. Determine the magnetic force acting on an electron that is moving due east at a speed of 8.75×10^5 m/s in a magnetic field of strength 0.75T directed upward
19. For the following parallel wires, compute;
    A) The force that one wire exerts on the other
    B) The force per length

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Answers to End of unit Questions

1. It produced due to the spinning of electrons
2. No. Single pole of a magnet does not exist
3. They will repel each other
4. 

5. Because the pointer indicates the direction of the field lines
6. Because it is itself a magnet and interacts with earth's magnetic field
7. Yes. The magnetic north pole of the earth is near its geographic south pole
8. No. There orientation with the field matters
9. Yes, it can
10. They attract each other if their current directions are the same. And, repel when they carry currents in opposite directions
11. Using the right hand rule
12. $B_1 = \frac{\mu o I_1}{2\pi r}$, if $I_2 = 2I_1$

    $B_2 = \frac{\mu o I_2}{2\pi r} = 2B_1$, doubled
13. a) $F = \frac{\mu o I_1 I_2}{2\pi r}$ halved

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b.) Doubled

c) quartered

Solution to End of unit problems

14. $F = Qv\,B \sin\theta$

$F = (4 \times 10^{-6})(4 \times 10^{5})(0.6)\sin 45^{0}$

$F = 0.67\ N$

15. $F = I\ell B \Rightarrow B = \dfrac{F}{Ie} = \dfrac{0.4}{(8)(0.5)} = 0.1T$

16. $F = IB\ell = (0.8)(5)(0.6)$

$F = 2.4\,N$

17. $B = \dfrac{\mu oI}{2\pi r} = \dfrac{(4\pi\times10^{-7})}{2\pi(0.1)} = 5\times10^{-5}T$

18. $F = QvB = (1.6\times10^{-19})(8.75\times10^{5})(0.75) = 1.05\times10^{-3}N$

19. A. $F = \dfrac{\mu oI_1I_2\ell}{2\pi r}$

$F = \dfrac{(4\pi\times10^{-7})(2)(6)(200)}{2\pi(0.1)} = 4.8\times10^{-3}N$

B. $\dfrac{F}{\ell} = \dfrac{\mu oI_1I_2}{2\pi r} = \dfrac{(4\pi\times10^{-7})(2)(6)}{2\pi(0.1)}$

$\dfrac{F}{\ell} = 2.4\times10^{-6}N/m$

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UNIT 11

11. Electromagnetic Wave and Geometrical Optics

11 . 1 What is Electromagnetic Wave

11 . 2 The Electromagnetic Spectrum

11 . 3 Light as a Wave

11 . 4 Laws of Reflection and Refraction

11 . 5 Mirrors and Lenses

11 . 6 The Human Eye and Optical Instruments

11 . 7 Primary Colours of Light and Human Vision

11 . 8 Colour Addition of Light

11 . 9 Colour Subtraction of Light Using Filters

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11. Electromagnetic Wave and Geometrical Optics

Welcome to this impressive unit where you will be able to explain the nature and type of electromagnetic waves, describe light wave,state the laws of reflection and refraction of light wave, identify types of mirrors and lenses, find out the nature of images formed by mirrors and lenses, examine the human eye and decide the appropriate lens or mirror to correct eye defects, be familiar with optical devices and engaged in adding and subtraction of colours. Have a nice time.

11.1 What is Electromagnetic Wave

Wave transports energy from one place to another with out transferring matter. And, depending on their need of a medium for their production and propagation, waves can be mechanical or electromagnetic

Mechanical Wave: Is a wave that uses matter to be produced and to move through and can not travel through vacuum.

Examples of mechanical wave includes sound wave, water wave and waves on a string.

Electromagnetic Wave: Is a wave produced when an accelerating charged particle enters a region where electric and magnetic field interact and can pass through a medium and a vacuum

Examples of electromagnetic waves include, light wave, microwave, infrared, ultraviolet, x - ray ,Gamma ray

Did you Know ?

The sun emits radiant energy in the form of electromagnetic wave. And about 92% of this energy is carried by infrared which makes us feel warm. A small amount of the radiant energy is carried by ultra violet, UV light that has the power to cause sunburn, bone cancer and blindness if exposed to it for too long.

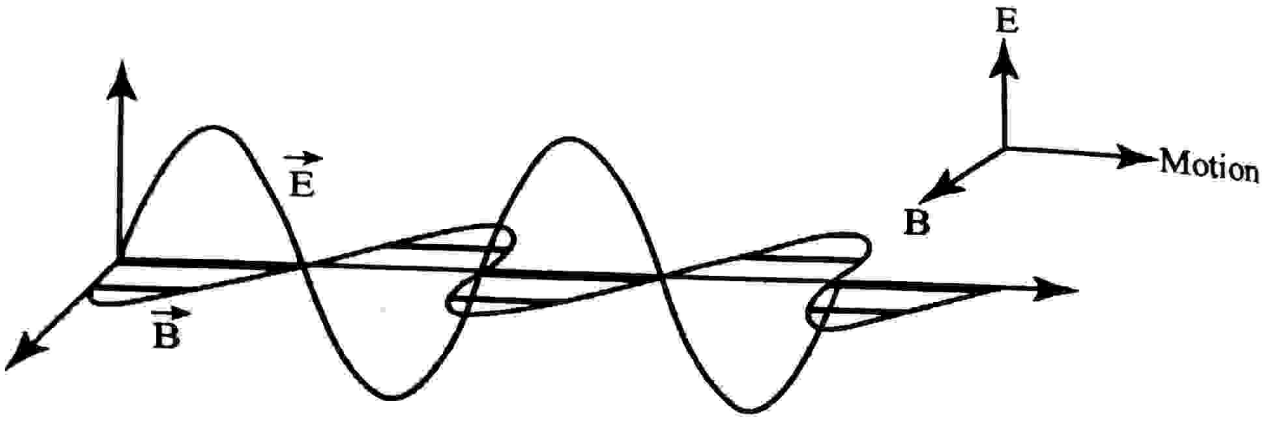
Conceptual Example

1. Describe an electromagnetic wave

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Explanation : Electromagnetic wave is a transverse wave produced by an accelerating charge in an electric and magnetic fields and can travel in vacuous at a speed of $C = 3 \times 10^8$ m/s.

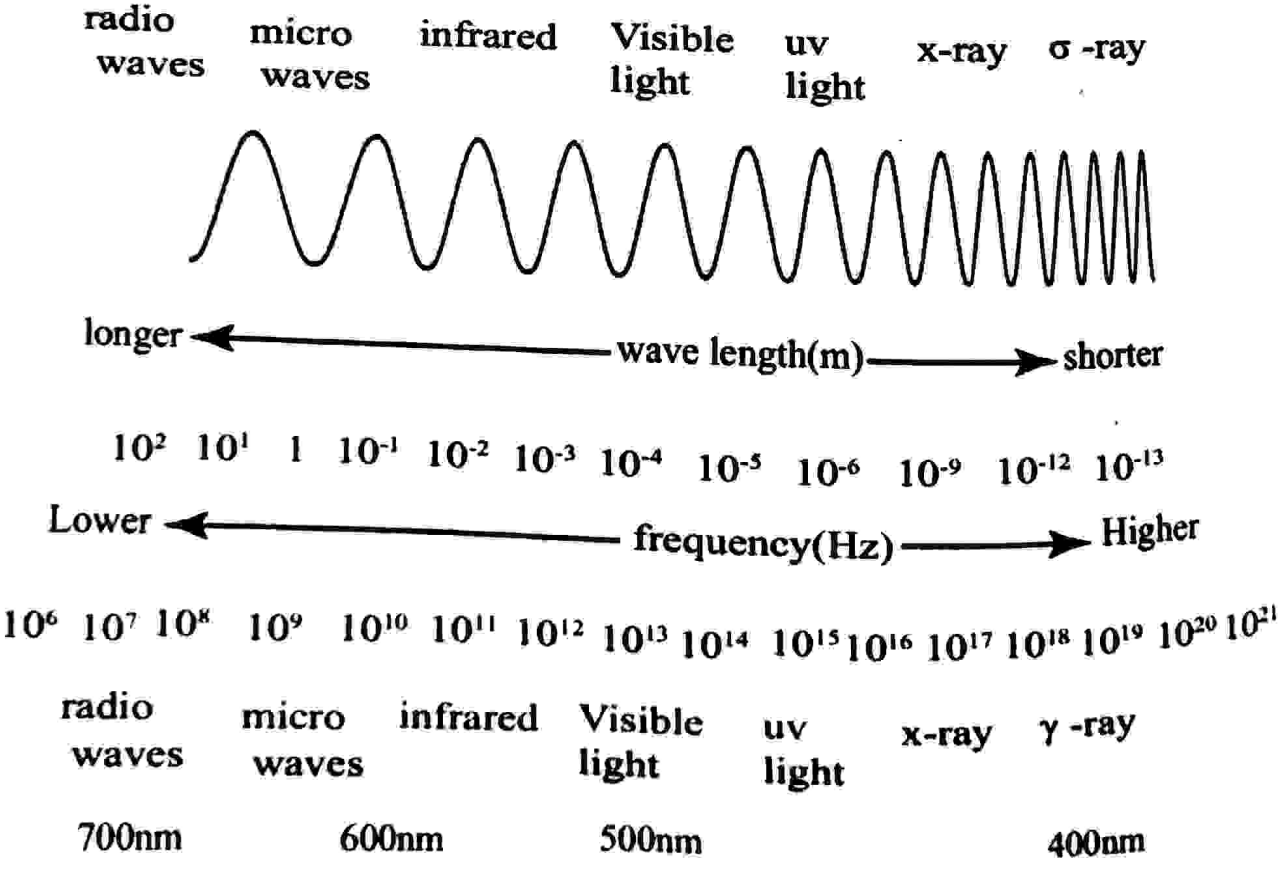
2. Plot the nature of the electric and magnetic fields that gives rise to the propagation and production of electromagnetic wave



11.2 The Electromagnetic Spectrum

The electromagnetic spectrum contains a range of frequency and wave lengths. The spectrum consists of radio wave, micro wave, infrared, visible light, ultraviolet, x-ray and gamma ray

The radiations frequency and wavelength in the spectrum has the following range



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☑ Gamma, γ ray has highest frequency, shortest wavelength and the most penetrating power. And, radio wave has the longest wavelength, shortest frequency and the least penetrating power.

Now, let us study components of the electromagnetic wave one by one.

Radio Wave

They are emitted from transmitters, stars, sparks and lightning.

They have longest wavelength ranging from 1m to thousands of meters and lowest frequency ranging from 500k Hz to 1000MHz.

They are mainly used fore communication purposes in police radios, military aircraft radios and television transmissions

☑ Over dose of radio wave cause cancer, leukaemia and other disorders

Microwave

They are emitted from electronic devices such as transmitter chip, cell phone, micro oven, antenna and stars. They are used in cell phone, micro oven for cooking, traffic speed central control, for radar air trafficking and for weather forecasting.

☑ Over does of microwave is known to cause,which is a clouding of the cornea.

Infrared Wave

They are given off by hot bodies, stars, lamps, flames and our body. They are used in remote control for TV and video records, in heat lamp to heal injuries and in cameras to see objects in the dark.

☑ Over dose of infrared causes you feel hot.

Visible Light

They are emitted by man made and natural sources of light and create a sense of vision to the eye.

They have a frequency range of 4×10^{14} Hz and 7×10^{14}Hz and a wavelength range of 400nm to 700nm

☑ Too much exposure to light ray may damage the retina in the eye.

Ultraviolet UV Light

They are emitted from lamps,the sun and other stars.

They are used to detect bank notes, to harden dental filling, to kill microbes,

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to sterilize surgical equipments, foods and to provide vitamin D.

☑ Over does of UV light may caues sunburn, skin cancer and blindness.

X- rays

They are given off by stars and nebula and they have high frequency and energy

They are used in hospitals to investigate internal part of a body, in airport security checks to examine inside a luggage and in astronomy to examine the space.

☑ Over does of x- ray can cause cell damage and cancer

Gamma - rays

They are given off by stars and radioactive elements.

They have the highest frequency, the lowest wavelength and the most penetrating power

ϑ - rays are used in radio therapy to kill cancer cells without physical sorcery, to kill microbes, sterilize food and medical equipment

☑ Over dose of ϑ - ray may cause cancer, mutation in growing tissues.

Illustrative Examples

3. What is the electromagnetic spectrum? Explain

   Explanation : Electromagnetic spectrum is a band of radiations with a range of frequencies and wavelengths.

4. List the electromagnetic spectrum in the order of

   A) Increasing wavelength

   B) Increasing frequency

Explanation

A) In the order of increasing wavelength ϑ - ray, x - ray, UV light, visible light, infrared, Micro wave, Radio wave

B) In the order of increasing frequency Radio wave, Microwave, infrared, visible light, UV light, x- ray and ϑ - ray

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Match the items in column A with those in column B

A	B
5. Radio wave	A) longest frequency
6. Micro wave	B) longest wavelength
7. Infrared	C) 700nm to 400nm
8. Visible light	D) Remote control
9. Uv light	E) Air-port security check
10. X - ray	F) source of vitamin D
11. 9 - Ray	G) Radar control
	H) Require a Medium

Answers

5. B 7. D 9. F 11. A
6. G 8. C 10. E

11.3 Light as a Wave

Light is an electromagnetic wave that can pass through vacuum as well as matter and gives the sensation of vision to the eye.

Speed of light is belived to be the fastest speed in the universe and in vacuum light travels at a speed of $3x10^8$ m/s.

And,the wavelength, the speed and frequency of light wave are related by;

$$V = \lambda f$$

Where: V = Speed of light wave

λ = Wavelength of light

f = frequency of light

Note, the eqation, $V = \lambda f$ is derived from the fact that $V = \frac{S}{t} = \frac{\lambda}{T}$

$V = \lambda\left(\frac{1}{T}\right) = \lambda f$. Ok

Illustrative Examples

12. How long will light takes to travel from the sun to the earth which is

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about 150 million kilometres ?

Solution :

$$S = Vt \Rightarrow t = \frac{s}{v} = \frac{150,000,000}{3,000,000,00}$$

$$S = 510 \text{ seconds}$$

13. An FM radio Station broad casts at a frequency of 102.1 MHz. What is the wavelength of the radio wave ?

Solution

$$V = \lambda f \Rightarrow \lambda = \frac{V}{f} = \frac{300,000,000}{102100000}$$

$$\lambda = 2.94m$$

14. What is the frequency of a gamma ray of wavelength 10^{-13} m

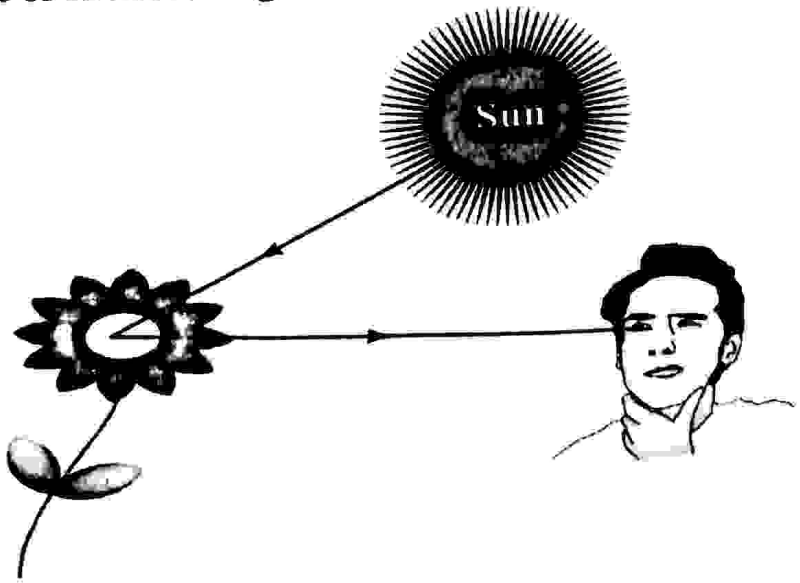
Solution

$$V = \lambda f \Rightarrow f = \frac{V}{\lambda} = \frac{300,000,000}{0.000,000,000,0001}$$

$$f = 3 \times 10^{21} \text{ Hz}$$

Did You Know ?

Light travels in a straight line and is represented by using a ray and,the group of light rays gives a light beam. What is more,we can see an object when a direct or reflected light come from the object to our eyes.



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Therefore, we see an object only when light from the object enters our eyes.

Conceptual Example

15. Give evidence to support the idea that light travels in a straight line

Explanation

The formation of a shadow is one possible evidence to support the idea that light travels in a straight line.

16. Explain how we can see objects ?

Explanation

When light came to our eyes directly or after being reflected from the object.

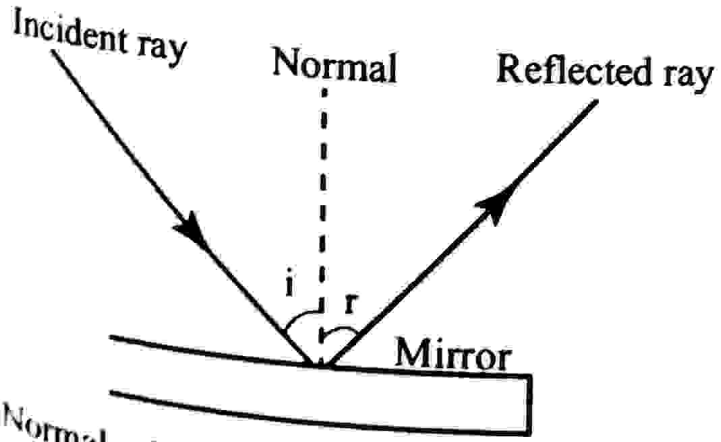
11.4 Laws of Reflection and Refraction

Like all other waves, light undergoes reflection and refraction.

Reflection of Light

It is the bouncing or turning back of light from smooth and polished surfaces such as the mirror, metal sheet or water

Using a ray diagram, we can visualize reflection of light from a mirror as follows.



Where : Normal = Is imaginary line perpendicular to the reflecting surface at the point of incidence.

i = Angle of incidence

r = Angle of reflection

Specular and Diffuse Reflection

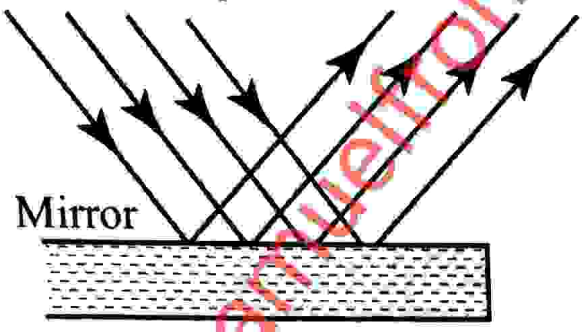
Based on the nature of the reflecting surface and the reflected rays,

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reflection can be regular or irregular.

Specular Reflection: Is the reflection of light from smooth,polished and shiny surface where the reflected rays move in the same direction parallel to each other.

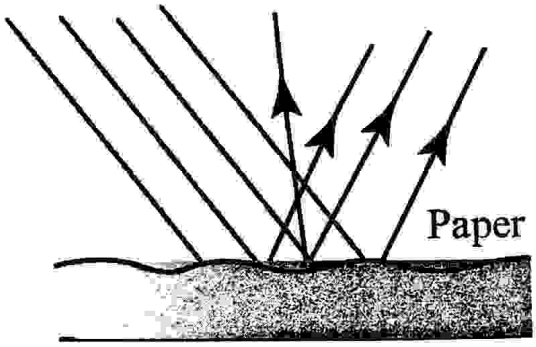
Reflection from a mirror is specular



Specular reflection is also called regular reflection

Diffuse Reflection: Is the reflection of light from rough surface such as paper where the reflected rays move randomly in all directions.

Reflection from a paper sheet is diffuse

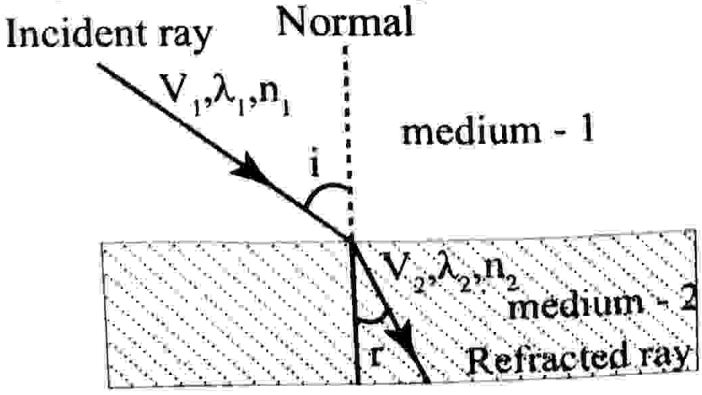


Diffuse reflection is also called irregular reflection.

Refraction of Light

It is the bending of light as it move from one medium to the other because of the change in its speed.

Using ray diagram, refraction of light can be visualized as follows:



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Where: n = refractive index

V = Speed of light

λ = Wavelength of light

The Law of Refraction

States that, the ratio of the sine of the angle of incidence to the sine of the angle of refraction is constant is summarized in Snell's law as follows

$$\frac{\text{Sin } i}{\text{Sin } r} = \frac{n_2}{n_1} = \frac{\lambda_1}{\lambda_2} = \frac{v_1}{v_2}$$

Refractive index is the measure of how difficult it is for light to get through a material and is defined as

$$\text{Refractive Index} = \frac{\text{speed of light in vacuom}}{\text{speed of light in a material}}$$

$$n = \frac{C}{V}$$

For example, the speed of light in vacuum is 3×10^8 m/s and that in ordinary glass is 2×10^8 m/s, and the refractive index of glass is then ;

$$n = \frac{c}{v} = \frac{3\times10^8}{2\times10^8}, \quad n = 1.5$$

Refractive index, n is a dimensionless quantity.

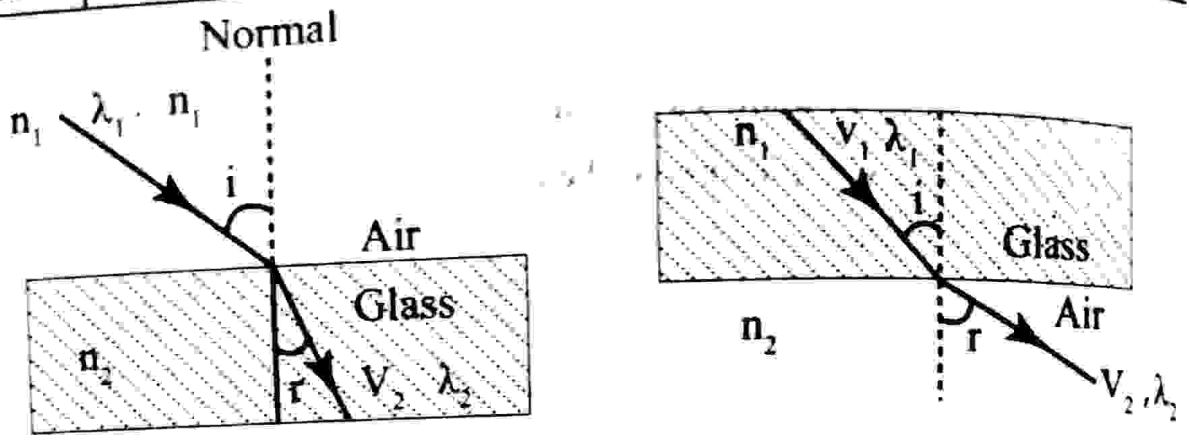
The refractive index of some materials is given below

Material	Refractive Index
Air	1.00
Water	1.33
Glass	1.50
Diamend	2.42

Moreover, when light moves from less dense medium to more dense medium, it bends toward the normal. And, if it moves from more dense medium to less dense medium it bends away from the normal.

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Illustrative Examples

17. If light of 600nm wavelength move from air into glass of refractive index 1.5 then what will be its

A) Wavelength, and B) Speed in glass

Solution

A $\frac{n_2}{n_1} \quad \frac{\lambda_1}{\lambda_2}$

$\frac{1.5}{1} = \frac{600}{\lambda_2}$

$\lambda_2 = 400\text{nm}$

B) $\frac{n_2}{n_1} = \frac{v_1}{v_2}$

$\frac{1.5}{1} = \frac{3\times10^8}{v_2}$

$v_2 = 2\times10^8\ \text{m/s}$

18. A light ray with an angle of incidence 45^0 is moved from air into water of refractive index 1.3 what will be its

A) Angle of refraction, and

B) Speed in water

Solution

A) $\frac{n_2}{n_1} = \frac{\sin i}{\sin r}$

$\frac{1.3}{1} = \frac{\sin(45^0)}{\sin r} = \frac{0.707}{\sin r^n}$

$\sin r = \frac{0.707}{1.3}$

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$$r = \sin^{-1}(0.54)$$

$$r = 34^0$$

B) $$\frac{n_2}{n_1} = \frac{v_1}{v_2}$$

$$\frac{1.3}{1} = \frac{3\times10^8}{v_2}$$

$$v_2 = 2.3 \times 10^8 \text{ m/s}$$

19. What is the difference between specular and diffuse

Explanation

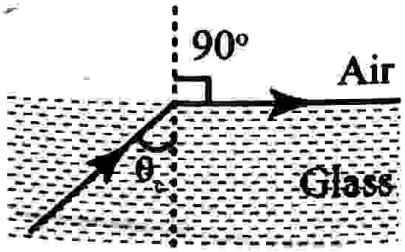
Specular reflection is also called regular reflection is the reflection of light from smooth surface and the reflected rays are parallel to each other. While, diffuse or irregular reflection task place on rough surfaces and the reflected rays move randomly and may cross - each other.

Total Internal Reflection

As its name suggests, total internal reflection involves the returning of light to its initial medium totally. The phenomena of total internal reflection to occur, two conditions must be full filled. These conditions are:

- ☑ Light must move from more dense medium to less dense medium, say from glass to air.
- ☑ The angle of incidence must be greater than the critical angle θ_c

Critical angle, θ_c : Is the angle of incidence for which the angle of refraction is 90°



Remember, when the angle of incidence is equal to the critical angle, then the refraction is equal to 90° .

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Illustrative Examples

20. What is the critical angle for total internal reflection to occur as light moves from glass to air ?

Solution:

$$\frac{n}{n} = \frac{\sin i}{\sin i}$$

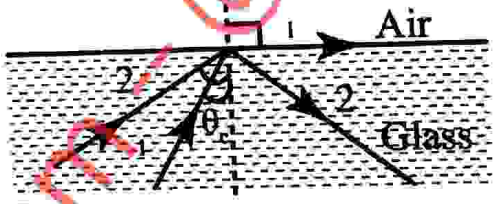
$$\frac{1}{1.5} = \frac{\sin\theta_C}{\sin 90^0}, \sin 90^0 = 1$$

$$\frac{1}{1.5} = \frac{\sin\theta_C}{1}$$

$$\sin\theta_C = \frac{1}{1.5},$$

$$\theta_C = \sin^{-1}(0.67) = 43^0$$

Now, when the angle of incidence is greater than the critical angle, total internal reflection occurs as shown below



Ray-1 represents the critical angle formation, while ray-2 indicates us the formation of total internal reflection.

Dispersion of Light

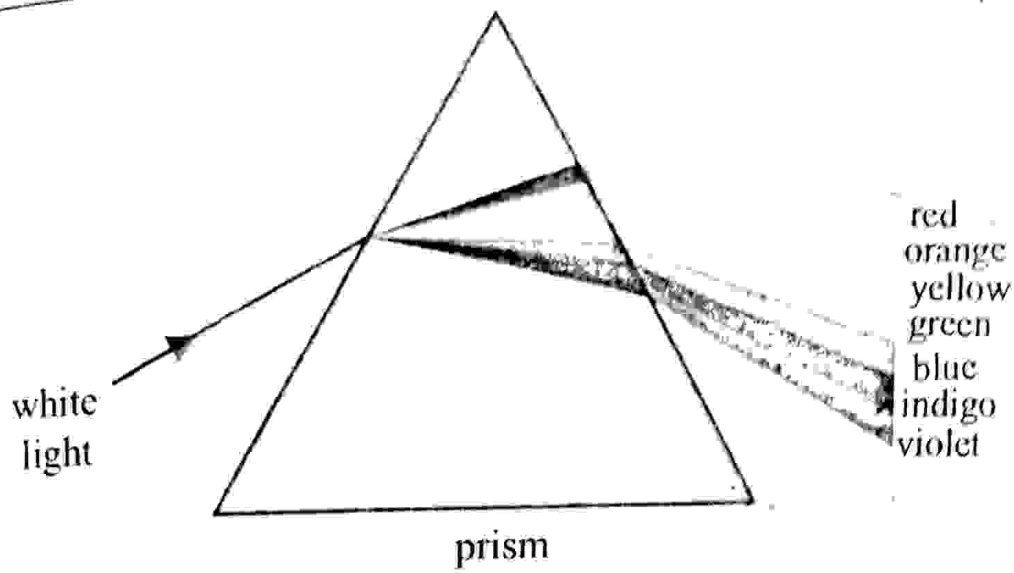
White light scatters into different colours up on passing through a prism and this scattering phenomena of light into components is called dispersion.

Dispersion : Is the splitting of white light into its component colours.

For example,when white light passes through a prism, it decomposes into seven different colours as shown below:

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The band of colours seen after dispersion is called spectrum of light and is abbreviated as VIBGYOR which stands for violet, Indigo, Blue, Green, Yellow, Orange and Red.

☑ Red light refracts the least,while violet refracts the most.

Conceptual Examples

21. How is a rainbow formed ? Explain

Explanation

It if formed due to refraction of light through droplets

22. What is refractive index ? Explain

Explanation:

It is the measure of the quality of the medium to refract light

23. A) Describe total internal reflection B) what is critical angle ?

Explanation

A) Is a condition that occurs when light moves from dense medium and incident angle the greater than the critical angle

B) It is angle incidence for which $r^n = 90^0$

24. A) What is dispersion of light ? B) Why a diamond sparkles ?

Explanation

A) Dispersion is the scattering of white light into its components as it moves through prism,

B) It sparkles since it disperses white light into different frequency

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11.5 Mirrors and Lenses

Now, let us study the types of mirrors and lenses and examine the nature of the images formed by them.

Mirrors

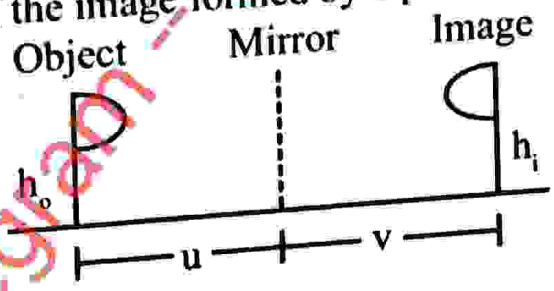
We encounter mirrors of different type in the dressing room, in supermarkets and in beauty salons and many other places.

Mirror : Is a smooth surface of glass that bounces off light at one side but silver coated at the other.

The most common types of mirrors are plane and curved.

Plane Mirror: Is a flat mirror with a reflecting surface at one side but not on the other.

The nature of the image formed by a plane mirror is the following.



The characteristics of the image formed by a flat mirror are

- ☑ Formed behind the mirror
- ☑ Virtual, can not be seen on a screen
- ☑ Same size as the object, $h_i = h_o$
- ☑ Equal distance behind the mirror as the object is in front of the mirror, $u = v$
- ☑ Erect or upright
- ☑ Laterally inverted, right side of the object became left side of the image

Number of Images formed by Two plane mirrors

The number of images formed by two plane mirrors inclined at an angle θ to each other is obtained as follows

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$$N = \left(\frac{360^0}{\theta} - 1\right), \text{ if } \frac{360^0}{\theta} \text{ is even}$$

$$N = \frac{360^0}{\theta}, \text{ if } \frac{360^0}{\theta} \text{ is odd}$$

Illustrative Examples

25. What is a mirror ?

Explanation

Mirror is a smooth glass with a reflecting surface at one side and form an image when the reflected light rays intersect

26. What is an image ?

Explanation

Image is the reproduction of an object due to reflection or refraction of light rays.

27. Describe the nature of the image formed by a plane mirror

Explanation

Virtual, erect, laterally inverted, same size as the object, same distance behind the mirror as the object is in front of the mirror.

28. How many images are formed when a body is placed between two plane mirrors that are inclined at an angle of ?

A) 60^0 B) 120^0

Solution:

A) $\theta = 60^0$

$$\frac{360^0}{\theta} = \frac{360^0}{60} = 6, \text{even}$$

use, $N = \left(\frac{360^0}{\theta} - 1\right)$

$$N = \left(\frac{360^0}{60^0} - 1\right) = (6-1) = 5 \text{ images}$$

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B) $\theta = 120^0$

$$\frac{360^0}{\theta} = \frac{360^0}{120^0} = 3, \text{ odd}$$

$$\text{use } N = \frac{360^0}{\theta} = \frac{360^0}{120^0}$$

$N = 3$ images

29. A pencil of size 10 cm is placed 20 cm in front of a thin plane mirror. What will be

A) The size of the image of the pencil

B) The distance from the pencil to its image

Solution

A) Image size = object size = 10 cm

Image of pen cile size = 10cm

B) Image is formed 20 cm behind the mirror,

Therefore, the distance from the pencie to its image is equal to

20 + 20 = 40cm

30. Which one of the following is Not True about the image formed by a plane mirror

A) diminished B) Sam size C) virtual D) same distance

Hint: The image formed by a plane mirror is not diminished or smaller Than the object. Rather, it is of the same size as the object.

Answer : A

31. Which one of following is TRUE about the image formed by plane mirror ?

A) Real C) Inverted

B) Enlarged D) Laterally Inverted

Hint : The image formed by a plane mirror is not real, diminished enlarged or inverted

Answer : D

32. The scattering of reflected light rays randomly in all directions from a rough, irregular surface is known as ;

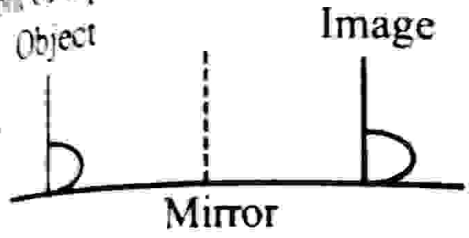
A) Regular reflection C) Diffuse reflection

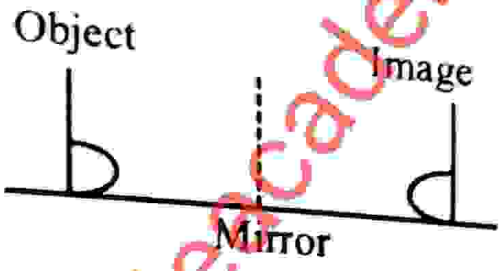
B) specular reflection D) mirror Image

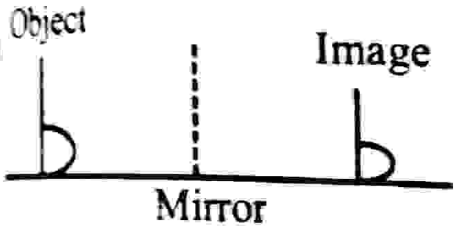
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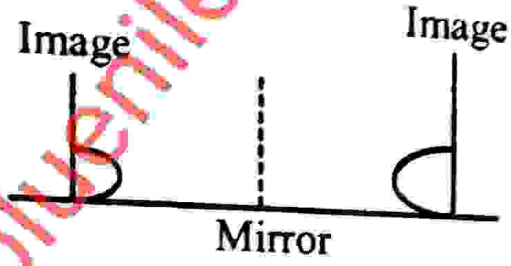
Answer : C

Which one of the following is the possible image of the body placed in front of a plane mirror

A)


B)


C)


D)


Hint: The correct nature of the image formed by a plane mirror is as in case B.

Answer : B

When letter p is placed in front of a plane mirror, its image will be

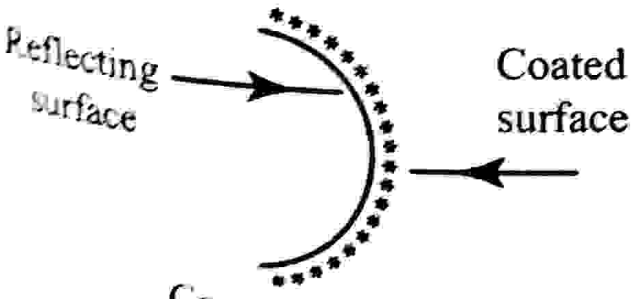
A) p B) b C) q D) d

Answer: C

Curved Mirrors

Curved mirrors are also called spherical mirrors and in this section, we are going to study two types of them concave and convex

Concave Mirror: Is formed from the inside surface of a glass sphere and is curved inward like the bowl of a spoon.

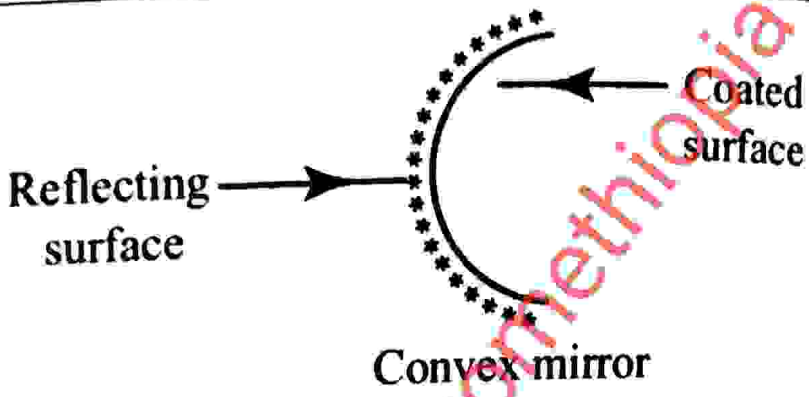


Since a concave mirror causes light rays to come together,it is also called a converging mirror

Convex Mirror : Is formed from the outer surface of a glass sphere and is curved out ward like the back of a spoon

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Convex mirror

Since a convex mirror cause light rays to spread out, it is also calle *diverging mirror*

Next, let us be familiar with the common terms that are frequently used wh studying the nature of the images formed by curved mirrors and lens

Centre of Curvature, C: Is the centre of the sphere of which the mirr is a part.

Radius of Curvature, R: Is the radius of the sphere the mirror made fro

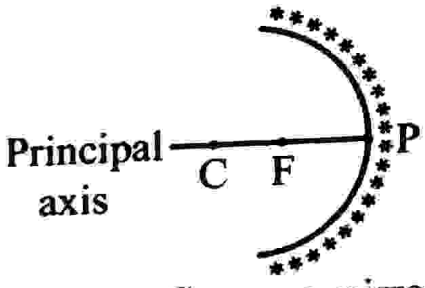
Pole, P: Is the middle point of the curved mirror

Principal Axis: Is the straight line that passes through the centre curvature and the pole of the mirror

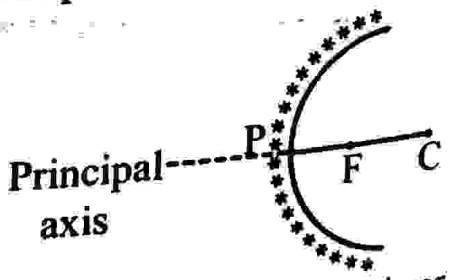
Aperture: Is the periphery or circular out line of the mirror and it is th measure of the size of the mirror.

Principal focus, F: Is the point to wards which reflected rays converge o diverge.

Focal length,F: Is the distance from the pole to the principal focus.



Concave mirror



Convex mirror

☑ The radius of conservator of a curved mirror is equal to twice the focal Length

$$R = 2f$$

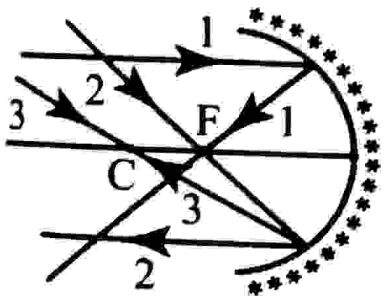
$$\boxed{f = \frac{R}{2}}$$

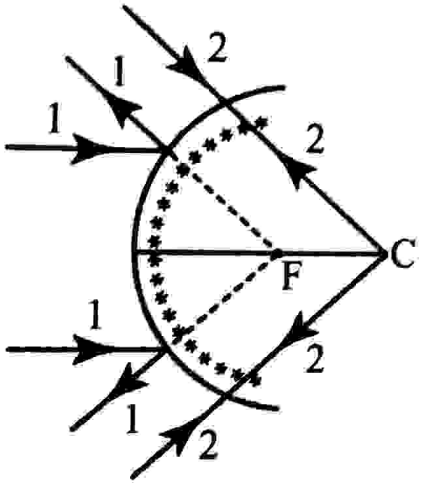
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And the three main reflected light rays that are responsible for the formation of an image are the following.

- ☑ An incident ray parallel to the principal axis passes through the principal focus after reflection in concave mirror and appears to come from the principal focus in convex mirror
- ☑ An incident ray moving through the centre of curvature come back gain through the centre of curvature after reflection .
- ☑ An incident ray through the principal focus moves parallel to the principal axis after reflection.



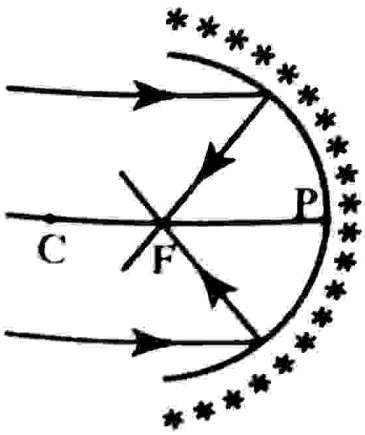


The intersection of two or more of the above reflected rays indicate the position of the image.

Now, let's examine the nature of the image formed by concave and convex mirrors.

A) Image formed by concave or converging mirror object position Ray Diagram Nature of Image

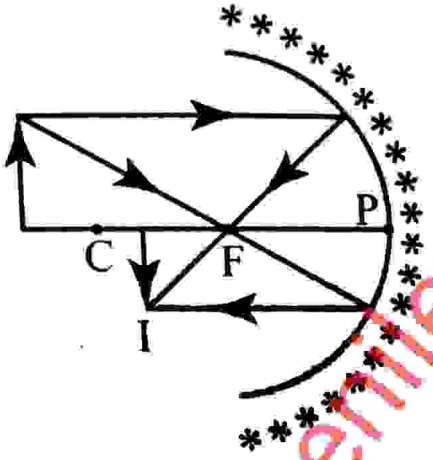
At Infinity



An F Diminished Real Inverted

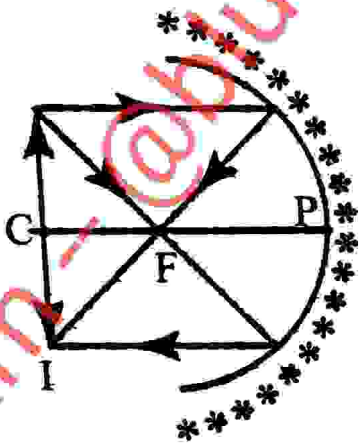
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Be Yond C



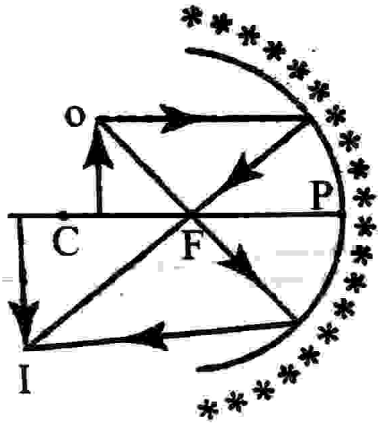
Between C and F
Diminished Real
Inverted

At C



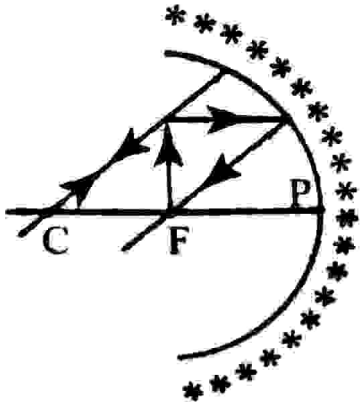
At C Same size
Real Inverted

Between C and F



Be yond C
Enlarged Real
Inverted

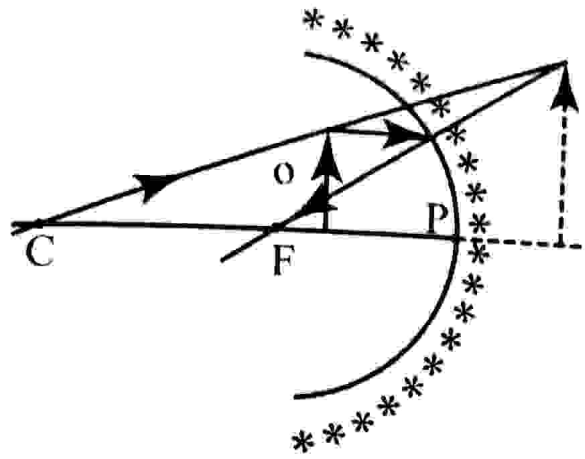
At F



At Infinity
Enlarged Real
Inverted

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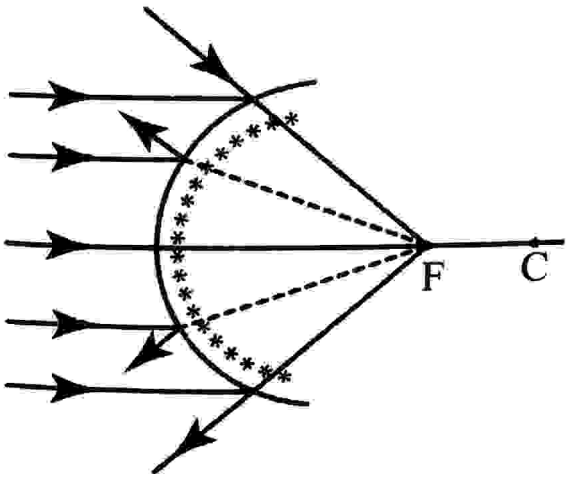
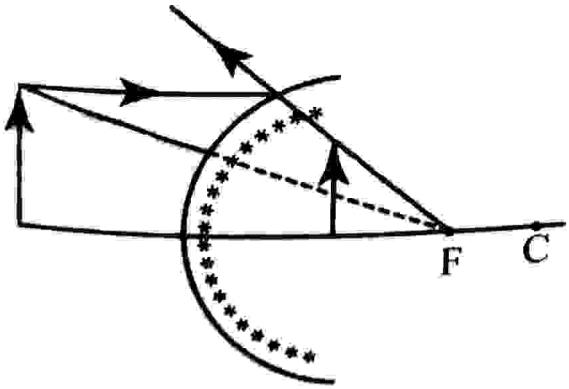


Between F and P

Behind the Mirror
Enlarged virtual
Erect

☑ Depending on the location of the object from the mirror, the image formed by concave mirror can be in front of the mirror,behind the mirror, same size, diminished, enlarged, real, virtual, inverted or erect.

B) Image formed by a convex or diverging mirror object position

Object position	Ray Diagram	Nature of Image
At Infinity		At F Behind Mirror Diminished Virtual Errect
Between Infinity and pole		Between P and F Behind Mirror Diminished Virtual Errect

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☑ For any position of the object from the mirror, the nature of the image formed by convex mirror is always diminished, virtual, erect and behind the mirror.

Illustrative Examples

35. When will the image formed by a concave mirror became, virtal ?

Explanation

When the object is placed between the focal point and the pole,the image is virtual,erect,enlarged and behind the mirror

36. Can the image formed by a concave mirror ever be

A) Same size as the object

B) Formed at infinity

Explanation

A) Yes,when the object is placed at the centre of curvature,the image size is equal to the object size

B) Yes, when the object is at the focal point

37. Can the image formed by a convex mirror ever be real, inverted and en larged ?

Explanation

Never! The image formed by a convex mirror is always arrogate, dimini shed and virtual

Mirror Equation and Magnification

The ray diagrams only give us qualitative information about the location and nature of the image formed by mirrors. Quantitatively, the object and the image distances as wel as the object and the image sizes is determined using the mirror equation

The relationship among the object distance,U,the image distance,V and the focal length,f is given by;

$$\boxed{\frac{1}{f}=\frac{1}{u}+\frac{1}{v}}$$

And,the size of the image is determined using the magnification equation

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of the mirror:

$$\text{Magnification} = \frac{\text{Image size}}{\text{object size}} = -\frac{\text{Image distance}}{\text{object distance}}$$

$$\boxed{m = \frac{hi}{ho} = \frac{-v}{u}}$$

To use the mirror equation, we need to stick to the following sign conventions

- ☑ Focal length, f is taken as positive for concave mirror and as negative for convex mirror
- ☑ Image distance V is taken as positive for real image and as negative for virtual image
- ☑ Magnification, m is taken as positive for errect image and as negative for inverted image.

Illustrative Examples

38. For which types of mirror is the magnification always become one ?

Explanation

In a flat or plane mirror, the image size is always equal to the object size and hence, the, magnification is one.

39. For which type of mirror is the magnification always less than one ?

Explanation

In a convex mirror, the image is always diminished, that is smaller than the object. hence the magnification is always less than one

40. For which type of mirror will the magnification be equal to one, less than one or greater than one ?

Explanation

In a concave mirror,the image may be same size, diminished or enlarged and hence the magnification can be one, less than one or greater than one.

Illustrative Examples

41. What is the focal length of a curved mirror of radius of curvature 20 cm ?

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Solution

$$f = \frac{R}{2} = \frac{20}{2} \qquad f = 10\,cm$$

42. A curved mirror of focal length 20 cm forms an erect image of size five times the object size. Where is the image location ?

Solution:

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$m = \frac{-v}{u} \Rightarrow -5u = v$$

$$\frac{1}{20} = \left(\frac{-v}{5}\right) + \frac{1}{v}$$

$v = -80cm$ virtual image

43. A 20cm pencil is placed 6cm in front of a concave mirror of radius of curvature 10cm . Determine

A) The image distance

B) The image size

C) Magnification of the mirror

Solution

A) $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}, f = \frac{R}{2} = \frac{10}{2} = 5cm$

$\frac{1}{5} = \frac{1}{6} + \frac{1}{v}, \frac{1}{v} = \frac{1}{5} - \frac{1}{6} = \frac{6-5}{30} = \frac{1}{30}$

$V = 30$

B) $\frac{hi}{ho} = \frac{v}{u}$

$\frac{hi}{20} = \frac{-30}{6} = hi = -100cm$, inverted image

C) $m = \frac{hi}{ho} = \frac{100}{20} = 5$

44. A pencile of 8cm tall is placed 12cm in front of a converging mirror of

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a converging mirror of magnification 4 Determine

A) The image size

B) The image distance, and

C) The focal length

Solution

A) $m = \frac{hi}{ho}$

$4 = \frac{hi}{ho} \Rightarrow hi = 4(ho) = 4\ (8)$

$h_i = 32cm$

B) $m = \frac{-v}{u}$

$4 = \frac{-v}{12}$

$v = -48cm$, virtual image

C) $m = \frac{-v}{u}$

$4 = \frac{-v}{12}$ $v = -48cm$, virtual image

$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$, $m = \frac{-v}{u}$

Subsutitute $V = -4u = -4(12)$

$v = -48cm$

45. When a body is placed in front of a concave mirror of focal length 20cm, and image is formed 30cm away. Determine

A) distance of the object B) nature of the image

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Solution

A) $\frac{1}{f}=\frac{1}{u}+\frac{1}{v}$

$\frac{1}{u}=\frac{1}{f}-\frac{1}{v}$

$\frac{1}{u}=\frac{1}{20}-\frac{1}{30}=\frac{1}{60}=60cm$

$u = 60cm$

B) Real image,because the image distance is positive.

46. When a body is placed 30cm in front of a diverging mirror, a virtual image is formed 10 cm behind the mirror. What is the focal length of the mirror?

Solution:

$\frac{1}{f}=\frac{1}{u}-\frac{1}{v}$

$\frac{1}{f}=\frac{1}{30}-\frac{1}{10}$

47. If a pen is placed 10cm infront of a convex mirror of focal length 20cm. Then,find the,

A) image distance

B) magnification of the mirror

C) nature of the image

Solution

A) $\frac{1}{f}=\frac{1}{u}-\frac{1}{v}$

$\frac{-1}{20}=\frac{1}{10}-\frac{1}{v}$

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$$\frac{1}{v} = -\frac{-1}{20} - \frac{1}{10} = \frac{-3}{20}$$

v = -6.7cm, virtual image

B) $m = \frac{hi}{ho} = \frac{-v}{u} = -\left(\frac{-6.7}{10}\right) = 0.67$

C) The image is virtual and diminished. since the magnification is less than one and the image distance is negative.

48. What is the focal length and the type of the mirror if an image is formed 10cm from the mirror when an object is placed 30cm in front of it?

49. In a plane mirror, an object is 0.5m in front of the mirror. What will be the distance between the object and image?

A) 0.5m B) 0.25m C) 1m D) 0.75m

Hint: S = u + v = 0.5 + 0.5

S = 1m

Answer: C

50. A person standing in front of a mirror finds his image larger than himself. The mirror must be?

A) Plane
B) Convex
C) Concave
D) There is no such mirror

Hint: concave mirror forms a varions images depending on the object distance.

Answer: C

51. Convex mirror always forms an image that is

A) Real and inverted
B) Virtual and inverted
C) Virtual and erect
D) Real and enlarged

Hint: Image of convex mirror is always virtual, erect and diminished.

Answer: C

52. To get three images of an object, one should have plane mirrors at an angle of

A) 30^0 B) 60^0 C) 90^0 D) 120^0

Hint: $n = \frac{360^0}{\theta} - 1 \Rightarrow \theta = \frac{360^0}{n+1} = \frac{360^0}{3+1} = 90^0$

Answer: C

53. A object of size 4cm is kept infront of convex mirror at the focal point.

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The image size is?

A) 2cm B) 8cm C) 6cm D) 4cm

Hint: $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$, $u = f$

$$-\frac{1}{f} = \frac{1}{f} = +\frac{1}{v} \Rightarrow v = \frac{f}{2}$$

$$-m = \frac{hi}{ho} = \frac{-v}{u} \Rightarrow hi = \frac{ho}{2} = \frac{4}{2}$$

$h_i = 2cm$

Answer: A

54. A mirror of focal length 18cm forms a virtual image of magnification $\frac{1}{3}$ What is the object distance?

A) 18cm B) 48cm C) 36cm D) Infinity

Hint: $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$, $m = \frac{v}{u} = \frac{1}{3} \Rightarrow v = \frac{u}{3}$

$$\frac{-1}{18} = \frac{1}{u} - \frac{3}{u} \Rightarrow u = 36cm$$

Answer: B

55. A con cave mirror of focal length f forms an image of the same site as the object. The object distance is?

A) f B) 2f C) f/2 D) 4f

Hint: $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$, $m = \frac{v}{u} = 1 \Rightarrow v = u$

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{u} = \frac{2}{u} \Rightarrow u = 2f$$

Answer: B

56. A concave mirror has a radius of curvature of 1m. Light from a distance star is incident on the mirror. The image of the star is at

A) 0.5m B) 1m C) 2m D) 0.25m

Hint: $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$, $u = \infty$

$$\frac{1}{f} = \frac{1}{\infty} + \frac{1}{v}, \; f = \frac{R}{2} = \frac{1}{2} = 0.5m$$

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$$\frac{1}{0.5} = 0 + \frac{1}{v} \Rightarrow v = 0.5m$$

Answer: A

57. A convex mirror forms an image one - forth the size of the object. If the object is 0.5m from the mirror, the focal length of the mirror is
A) 0.17m B) 0.4m C) 1.5m D) 0.5m

Hint: $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$, $u = 0.5m$

$$m = \frac{-v}{u} = \frac{hi}{ho} = \frac{1}{4} \Rightarrow v = \frac{-u}{4}$$

$$\frac{1}{f} = \frac{1}{u} - \frac{4}{u} = \frac{3}{u}$$

$$f = \frac{u}{3} = \frac{0.5}{3} = 0.17m$$

Answer: A

58. The refractive index of diamond is 2. What is the speed of light through it?
A) 3×10^8 m/s C) 2×10^8 m/s
B) 1.5×10^8 m/s D) 1×10^8 m/s

Hint: $\frac{n_2}{n_1} = \frac{v_1}{v_2} \Rightarrow v_2 = \left(\frac{n_1}{n_2}\right) v_1$

$$v_2 = \left(\frac{1}{2}\right)(3 \times 10^8) = 1.5 \times 10^8 \, m/s$$

Answer: B

59. The critical angle for medium is 45^0. What is its refractive index?
A) $\frac{\sqrt{3}}{2}$ B) $\frac{2}{\sqrt{3}}$ C) $\sqrt{2}$ D) $\frac{1}{\sqrt{2}}$

Hint: $\frac{\sin i}{\sin r} = \frac{n_2}{n_1} \Rightarrow \frac{\sin 45^0}{\sin 90^0} = \frac{1}{n_1}$

$$\frac{\frac{1}{\sqrt{2}}}{1} = \frac{1}{n_1}$$

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$n_1 = \sqrt{2}$

Answer: C

60. It is possible to observe total internal reflection when a ray travels from
A) Air in to water
B) Air in to glass
C) Water in to glass
D) Glass in to water

Hint: Total internal reflection occurs when light approaches a lighter medium

Answer: D

61. As an object gets closer to the focal point of a convex lens, its image
A) Be come smaller
B) Remains constant
C) Gets farther from the lens
D) Gets closer to the lens

Hint: When the object is at the focal point, the image is at infinity

Answer: C

62. The image formed by a lens is always virtual and diminished. The lens is
A) Convex
B) Concave
C) Plane
D) No such lens

Hint: Concave lens forms virtual, erect and diminished image

Answer: B

63. At what distance from a concave lens of 20cm focal length and half the object site is formed?
A) 20cm B) 10cm C) 30cm D) 60cm

Hint: $\frac{hi}{ho} = \frac{v}{u} \Rightarrow \frac{1}{2} = \frac{v}{u} \Rightarrow v = \frac{u}{2}$

$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} \Rightarrow \frac{1}{20} = \frac{1}{u} - \frac{2}{u}$

$u = 20cm$

Answer: C

64. Refraction of light wave results from a difference in
A) Frequency
B) Amplitude
C) Speed
D) Unknown variable

Hint: difference in refractive index causes refraction.

Answer: C

65. The colour of an object is the same as the light that is
A) Reflected
B) Refracted

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C) Diffracted D) Interfered

Hint: We see objects when light is reflected from them.

Answer: A

67. All of the waves listed are electromagnetic except
A) Sound wave C) X - ray
B) Radio wave D) Light wave

Hint: Sound wave is mechanical wave

Answer: A

68. Light has a wavelength of 500nm in air and 400nm in a material. What is the index of refraction of the material?
A) 0.65 B) 0.85 C) 1.5 D) 1.25

Hint: $\frac{n_2}{n_1} = \frac{\lambda_1}{\lambda_2}$

$n_2 = \frac{500}{400} = 1.25$

Answer: D

69. Light refracts when traveling from air into glass because light
A) Travels at the same speed in air and in glass
B) Has greater frequency in air than in glass
C) Has greater frequency in glass that in air
D) Travels slower in glass than in air

Hint: $\frac{n_2}{n_1} = \frac{v_1}{v_2} = \frac{\lambda_1}{\lambda_2}$

Answer: D

70. An object 15cm high is placed 10cm from a converging lens and its image is formed 25cm from the lens. What is the size of the image?
A) 25cm B) 15cm C) 30.5 cm D) 37.5cm

Hint: $m = \frac{hi}{ho} = \frac{-v}{u}$

$\Rightarrow hi = ho\left(\frac{v}{u}\right) = 15\left(\frac{25}{10}\right)$

$hi = 37.5cm$

Answer: D

71. A convex lens of focal length $\frac{1}{3}$ m forms an image twice in site the

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object. What is the object distance?

A) 0.5m B) 0.3m C) 0.2m D) 0.8m

Hint: $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

$\frac{hi}{ho} = \frac{v}{u} \Rightarrow v = 2u$

$3 = \frac{1}{u} + \frac{1}{2u} = \frac{3}{2u}$

$u = 0.5m$

Answer: A

Solution:

$\frac{1}{f} = \frac{1}{u} - \frac{1}{v}$

$\frac{1}{f} = \frac{1}{30} - \frac{1}{10}$

$\frac{1}{f} = -\frac{1}{15}$

f = -15cm, convex mirror

Lenses

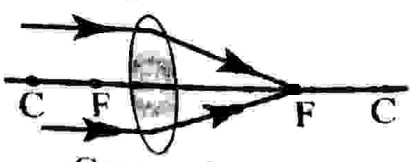
A curved transparent material that is smooth and regularly shaped so that when light strikes it the light refracts through it is called *lens*

Lens: Is a transparent material bound by two surfaces of which one or both of them are spherical.

The most common types of lenses are two. convex and concave.

Convex Lenc: Is a lens bulging outward, and is thicker at the middle than at the edges.

This lens causes light rays to come together and for this reason, convex lens is also called converging lens

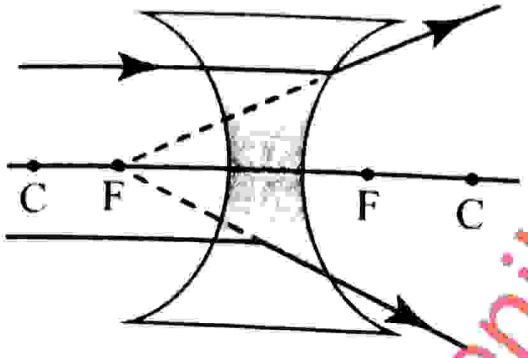


Convex lens

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Concave Lens: Is a lens curved inwards, and is thicker at the edges than at the middle.

This lens causes refracted light rays to spread out, and for this reason, concave lens is also called *diverging lens*.

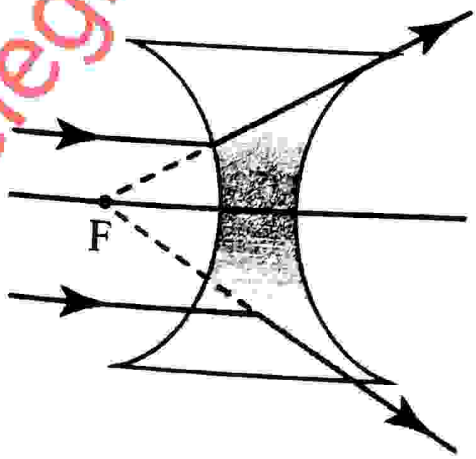
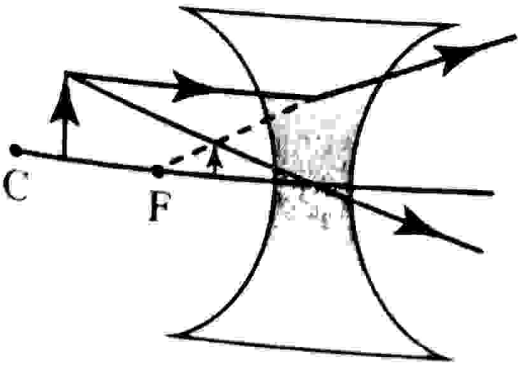


Concave Lens

☑ Unlike mirrors,lenses have two focal points and form images by refraction of light.

Now, let us examine the nature of the images formed by concave and convex lenses.

A) Image formed by concave lens

Object position	Ray Diagram	Nature of Image
At Infinity		At F diminished Virtual Errect
Between Infinity and pole		Between P × F Diminished virtual Errect

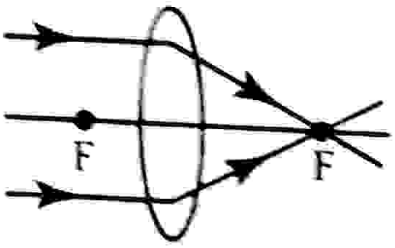
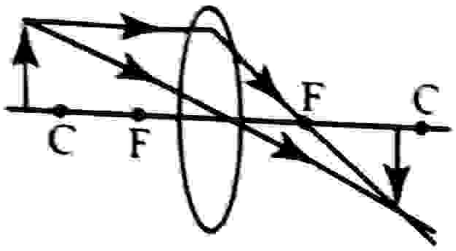
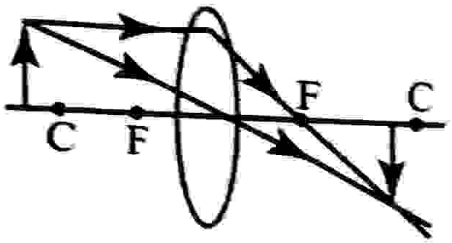
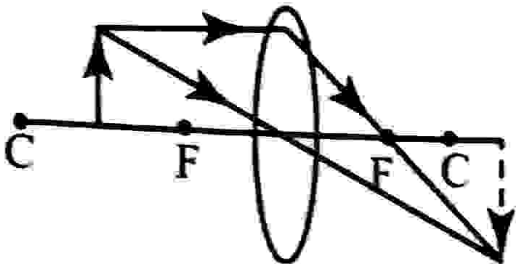
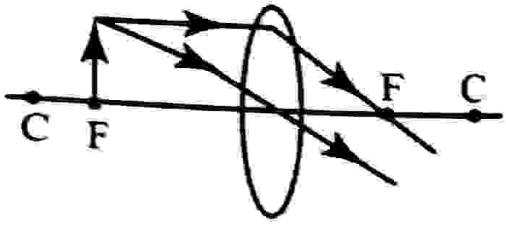
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B) Image formed by a convex Lens

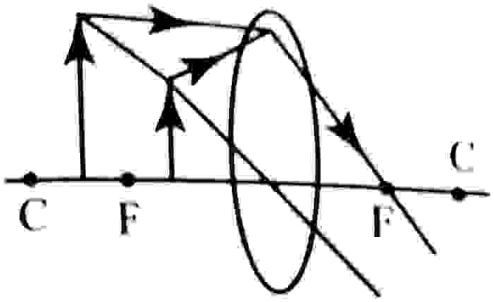
Unlike the concave lens that always forms a virtual, erect, and diminished image regardless of the object position,

convex lens formed real, virtual, erect, inverted, diminished the object from the mirror

Object position	Ray Diagram	Nature of Image
At Infinity		At F Diminished Real Inverted
Be Yond C		Between C and F Diminished Real Inverted
At C		At C same site Real Inverted
Between C and F		Be Yond C Enlarged Real Inverted
At F		At Infinity, Enlarged

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Between F and P



Enlarged, virtual

The ray diagrams give us only the qualitative aspect of the image natures formed by the corved mirrors.

Quantitatively however, we can describe the size and distance of the image formed using the lens equation.

The Lens Equation and Magnification

The relationship among the image distance (V) object distance (u) and the focal length (f) is given by the lens equation as follows

$$\boxed{\frac{1}{f} = \frac{1}{u} + \frac{1}{v}}$$

And,magnification of the lens is obtained by :

$$\text{Magnification} = \frac{\text{Image size}}{\text{object size}} = \frac{\text{Image distance}}{\text{object distance}}$$

$$m = \frac{h_i}{h_o} = \frac{-v}{u}$$

To use the lens and the magnification equations, appropriately we need to stick to the following sign conventions.

Sign conventions

- ☑ Focal length, f is taken as positive for convex lens and as negative for concave lens
- ☑ Image distance, v is taken as positive for real image and as negative for virtual image
- ☑ Magnification, m is taken as positive for erect image and as negative for inverted image

Illustrative Examples

71. If an object is pleased 24 cm in front of a concave lens of focal length

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12cm, where will the image be formed ?

Solution:

$$\frac{1}{f}=\frac{1}{u}+\frac{1}{v}$$

$$\frac{1}{v}=\frac{1}{f}-\frac{1}{u}=-\frac{1}{12}-\frac{1}{24}=-\frac{1}{8}$$

$v = -8cm$, virtual image

72. If a body is placed 40 cm in front of a converging lens of focal length 30cm, how far from the lens will the image formed ?

Solution

$$\frac{1}{f}=\frac{1}{u}+\frac{1}{v}$$

$$\frac{1}{v}=\frac{1}{f}-\frac{1}{u}=\frac{1}{30}-\frac{1}{40}=120$$

$v = 120$, real image

73. A 5cm nail is placed 10cm from a convex lens of focal length 15 cm. Determine the;

A) Image distance

B) Image size

C) Nature of the image

D) Magnification of the lens

Solution

A) $$\frac{1}{f}=\frac{1}{u}+\frac{1}{v}$$

$$\frac{1}{v}=\frac{1}{f}-\frac{1}{u}=\frac{1}{15}-\frac{1}{10}=-\frac{1}{30}$$

$v = 30cm$ virtual image

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B) $\frac{hi}{ho} = \frac{-v}{u}$

$\frac{hi}{5} = \frac{-30}{10}$

hi = -15 cm, inverted image

C) The image is virtual,inverted and enlarged

D) $m = \frac{hi}{ho} = \frac{15}{5}$ $\quad m = 3$

74. When a 10 cm tall pencil is placed 20 cm in front of a convex lens, an image is formed 20 cm from the Determine

A) Focal length of the lens

B) Size of the image

Solution

A) $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

$\frac{1}{v} = \frac{1}{20} + \frac{1}{30} = \frac{2}{10} = \frac{1}{10}$

$f = 10cm$

B) $\frac{hi}{ho} = \frac{-v}{u}$

$\frac{hi}{10} = \frac{-20}{20}$ $\quad hi = -10cm$, inverted image

75. A body placed in front of a concave lens of magnification 0.2 forms an image 5cm from the lens. Determine the

A) Object distance

B) Focal length of the lens

Solution

A) $m = \frac{v}{u} \Rightarrow 0.2 = \frac{5}{u}$, $u = 25\,cm$

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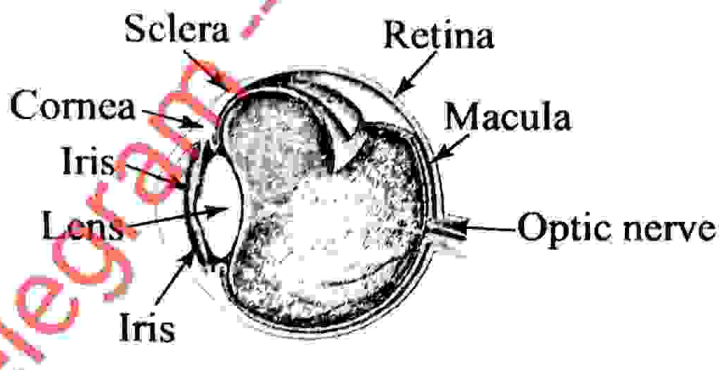
B) $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{25} - \frac{1}{5} \Rightarrow f = 6.25\text{cm}$$

11.6 The Human Eye and Optical Instruments

The Human Eye

The eye may a converging lens and it is like a camera light enters the eye through a thin membrane called the cornea where most refraction occurs and reaches the lens. The lens forwards the rays to the retina where an inverted image is formed.



Behind the cornea,there is the Iris, which is a dark muscular diaphragm that controls the size of the pupil. The pupil is a structure that regulates and controls the amount of light entering the eye.

The retina is a delicate membrane with many light sensitive calls. The light sensitive calls get activated when illuminated and generate electrical signals which are then sent to the brain via the optic nerves and get interpreted.

☑ The eye consists of a convex lens made of living tissues that focuses images on the retina where the optic nerve transfers the message to the brain where the message get interpretation .

Power of Accommodation

The eye lens is made of fibrous, jelly like living tissues and its curvature is adjusted by the cilia muscles. When these muscles are relaxed, the lens become thin and its focal length increases and enables us to see distant objects clearly. When the mussel contract,the lens become thick and its

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focal length decreases. and enables us to see near by objects clearly.

To see an object comfortably and distinctly with out strain and be ling blurred it must be about 25 cm from the eye.

☑ For the normal human eye, the near point is about 25 cm and the far point is infinity.

The minimum distance at which objects can be seen most distinctly without strain is called the least distance of distinct vision or the near point of the eye and is equal to 25 cm. And, the farthest point up to which the eye can see objects clearly is called the far point of the eye and is eqal to infinity.

For many reasons, the eye may be defected and malfunction .

Defects of vision and Their corrections

The eye many lose its power of accommodation and the vision becomes blurred due to the refractive defects of the eye.

The following are the main defects of the eye and their corrections.

Myopia - Near sightedness

A short - sighted person can see nearby objects clearly but can not see distant objects distinctly.

In a myopic eye, the image of distant object is formed in front of the retina, Not on the retina. The defect may arise due to excessive curvature of the eye lens or elongation of the eyeball.

☑ This defect can be corrected by using spectacles with a concave lens.

Hypermedia - far Sightedness

A long -sighted person can see distant objects clearly but cannot see nearby objects distinctly.

In a hypermic eye, the image of nearby object is formed behind the retina, Not on the retina. The defect may arise due to either too long focal length of the eye lens or too small eyeball size.

☑ This defect can be corrected by using spectacles with convex lens.

Presbyopia - Old Sight

The eye loses its power of accommodation as we get aged. At old ages, the near point recedes away.

This defect arises due to the weak ending of the biliary muscles and lack of

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flexibility of the eye lens as we get aged.

☑ This defect can be corrected using spectacles with bi-focal lenses. A concave lens, to facilitate near vision.

Astigmatism

This defect is caused when the eye has two different focal length,

☑ It is corrected with a pair of spectacles fitted with cylindrical lenses.

Illustrative Examples

Match the items in column A with those in column B

A	B
76. Myopia	A) long sighted
77. Hypermedia	B) short sighted
78. presbyopia	C) Age factor
79. Astigmatism	D) same facal length
	E) Different focal length

Answers

53. B 55. C 54. A 56. E

OPtical Instruments

So many optical devices are made using the reflective and refractive natures of mirrors, lenses and prisms.

Microscope, telescope, binocular, camera, eye-glasses are some examples of optical instruments

Microscope

It is used to see very small objects; such as bacteria, virus --- etc.

Microscope has an objective lens of convex lens and an eyepiece lens of convex lens both are designed to form an enlarged image.

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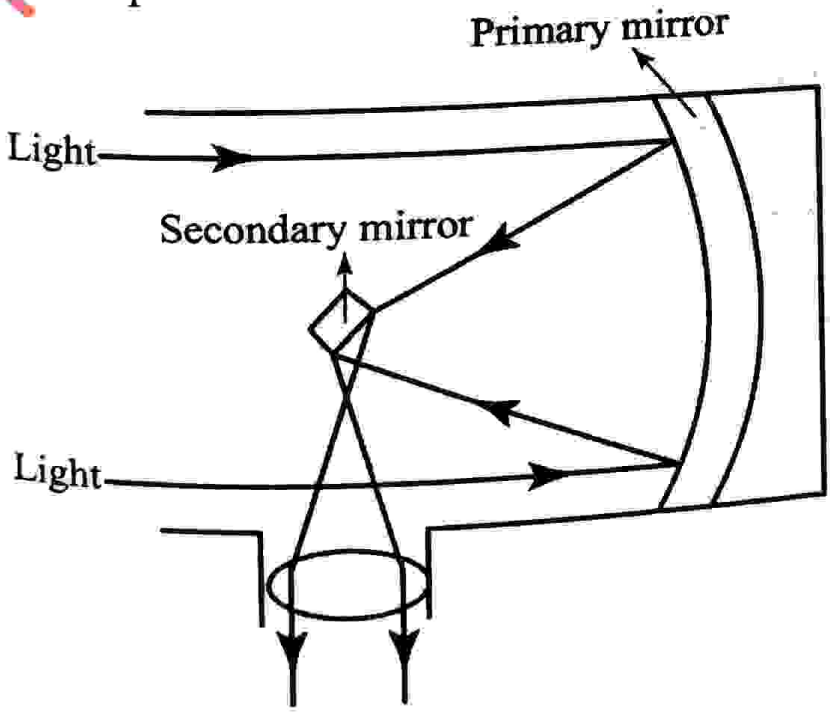
elescope

is used to see objects that are very far away such as the stars, planets and ie space

he refracting telescope uses two convex lenses to form an image of a istant object. It has an objective and eye piece lenses.



The reflecting telescope has a concave mirror and has much larger size than the refracting telescope



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Illustrative Examples

80. What is the power of accommodation ?

Explanation

It is the ability of the eye lens to adjust its focal length

81. What is the corrective lens for a my Myopia ?

Explanation

A concave lens to diverge the light rays

82. What is the near point and the far point of the human eye for normal vision ?

Explanation

The near point is about 25cm and the far point is infinity.

83. A student has difficulty reading the blackboard when sitting at the last seat. What eye defect he has and how can it be corrected ?

Explanation

Myopia, can be corrected by a concave lens.

84. List some optical instruments

Explanation

Camera, magnifying glass,microscope,telescope

85. What is the difference between simple microscope and compound microscope ?

Explanation

Simple microscope has only one lens and a compound microscope has at least two lenses.

86. Distinguish between the refractive and reflective telescope

Explanation

Refracting telescope has lenses to form image while reflecting telescope uses mirrors to form images.

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87. What type of telescope is there in Entoto observatory centre ?

Explanation

It is a reflecting telescope

11.7 Primary Colors of light and Human Vision

The retina in the human eye where image is formed is full of millions of light receptive cells called cones which are sensitive to colors and rods which are sensitive to light intensity.

When the cones and rods detect light, they send signal to the brain for interpretation. The brain, name the type of object and its colour. so you see, we are able to see an object when light from the object enters the eye and strikes these photoreceptor cells.

The cones are of three sets. one set sensitive to red light,another set sensitive to green light and a third set sensitive to blue light. These colours, red, green and blue are fundamental to the human vision are called **primary colurs.**

☑ Red, green and blue are primary colours. All other colours of the visible light spectrum can be produced by adding these three colours. And, if equal amount of the primary colours are added, white light is produced.

Illustrative Examples

88. What are the cones and rods cells found one the retina sensitive

Explanation

They are light colour and light intensity sensitive cells usually called the photoreceptor.

89. Why do we call the red,green and blue colours primary

Explanation

Because,their combination gives all the other colours of the spectrum.

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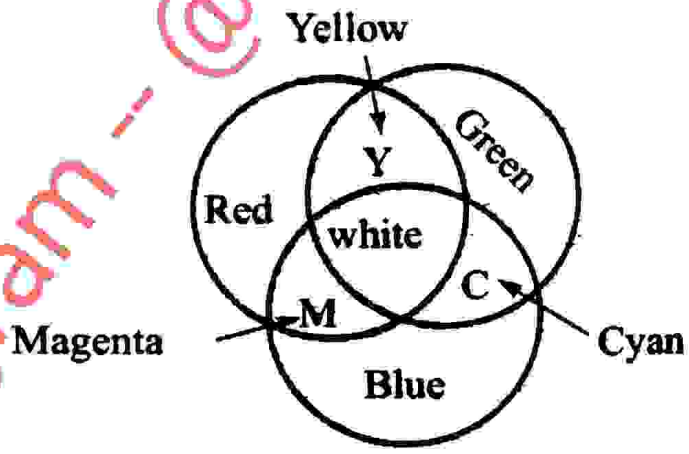
11.8 Colour Addition of Light

The red, green and blue colours are often described as the primary additive colours since their addition gives all the other colours. Appropriate combination of the primary colours, red, green and blue gives the secondary colours, cyan, magenta and yellow.

Red + Green = Yellow
Red + Blue = Magenta
Blue + Green = Cyan

☑ Yellow, Magenta and Cyan are called secondary colours of light

Remember: when added in equal proportions, red, green and blue gave white light, and the absence of all three colours results in Black !



Illustrative Examples

90. What colors will you get if you add proportionally;
A) Red and blue B) Red and green C) Green and blue

Explanation

A) Red + Blue = Magenta
B) Red + Gree = Yellow
C) Green + Blue = Cyan

91. What applications did color addition has ?

Explanation

The principles of color addition have important applications in color television,color computer monitors and on stage light systems, digital projector --- etc.

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Match the items in column A with those in column B

A	B
92. Yellow	A) Red + Blue
93. Magenta	B) Green + Blue
94. Cyan	C) Red + Green
	D) Yellow + Green

Answers

92. C 93. B 94. A

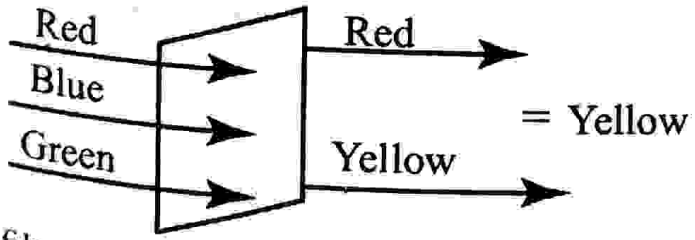
11.9 Colour Subtraction of Light Using Filters

The colours that are absorbed are subtracted from the reflected light. Colour subtraction is achieved using pigments or color filters.

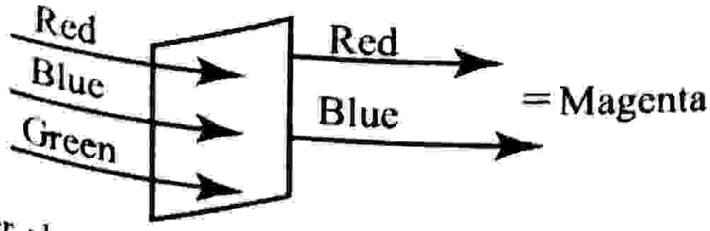
☑ Pigments are substances which give an object its color by absorbing certain frequencies of light and reflecting other frequencies. For example, a red pigment absorbed all colours of light except red which it reflects. color to pass through by absorbing all the rest. for example, when white light passes through a red filter, only red light passes and all the rest are absorbed.

The Following shows color subtraction using filters or pigments.

i. Yellow filter or pigment absorbs Blue and transmits Red and Green and gives yellow Light



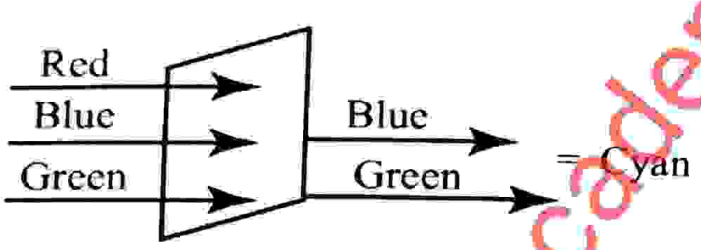
ii. Magenta filter absorbs Green and transmits Red and Blue and gives magenta light



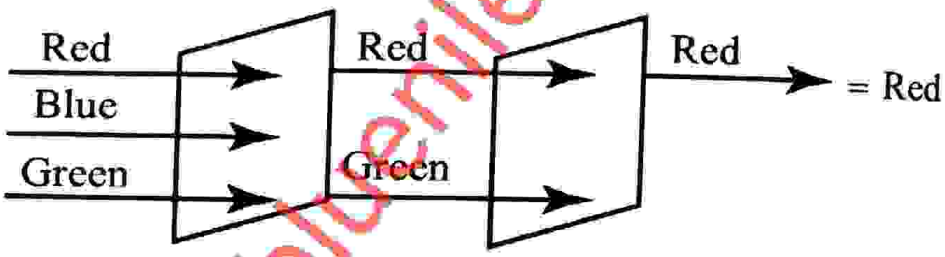
iii. Cyan filter absorbs Red and transmits Blue and Green and gives cyan

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light

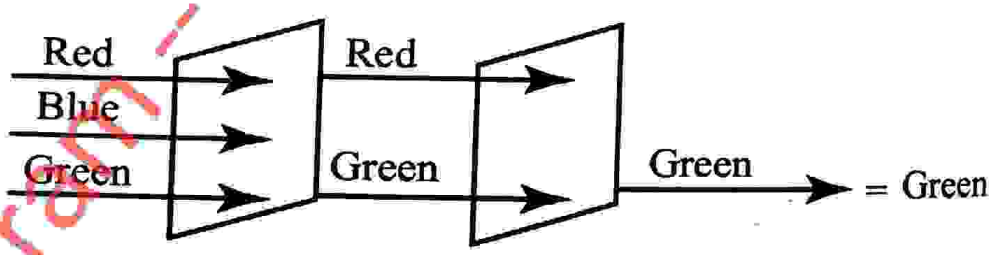


iv. Yellow filter absorbs Blue and magenta filter absorbs Green and gives Red light



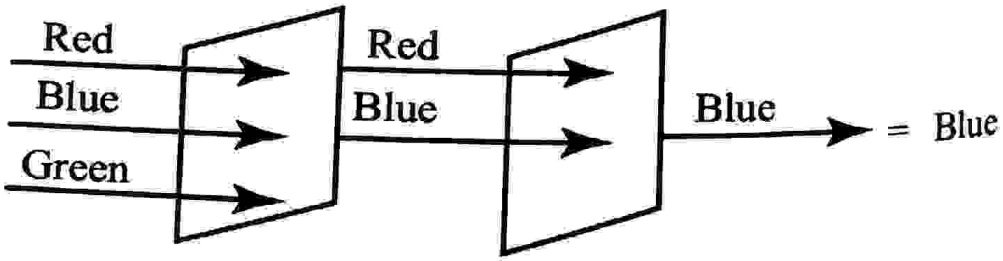
Yellow + Magenta Filters

v. Yellow filter absorbs Blue and cyan filter absorbs Red and gives Green light



Yellow + Cyan Filters

vi. Magenta filter absorbs Green and cyan filter absorbs Red and gives Blue light



Magenta + Cyan Filters

95. What are the primary additive colors ?

Explanation

Red,Blue and Green

What are the primary subtractive colors ?

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Explanation

Yellow, Magenta and Cyan

96. What are the complimentary colors ?

Explanation

Complimentary colors are the colors that are absorbed by the subtractive primaries. There Fore, cyan's compliment is Red, magenta's compliment is Green and Yellow's compliment is Blue.

Match the items in column A with those in column B

A	B
97. Green	A) Cyan - Blue
98. Red	B) Magenta - Red
99. Blue	C) Yellow - Green
	D) Yellow + Blue

Answers

97. A 98. B 99. C

End of unit Questions and problems

I. Give short answers to the following Questions

1. What is the speed of light in vacuum ?
2. Why do we protect ourselves from UV light exposure ?
3. Give advantage and disadvantage of x - rays
4. List the electromagnetic spectrum in order of increasing wavelength ?
5. Mirrors form image from reflection of light and lenses form image via refraction of light. True or False ?
6. Name the type of mirror used in the following situation
   A) Headlights of a car
   B) Side or rear - view mirror of a vehicle
   C) Solar furnace
7. The magnification of a plane mirror is 1. What does this mean ?
8. Compare and contrast primary light colours and primary pigment colour.
9. Determine the colours that are reflected from an object that appears black.

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10. What colour do the following Shirts appear to the human eye when the lights in a room are turned off.
    A) Red
    B) Blue
11. If the cover of a book appears to have a magenta colour, then what colours of light does it reflect and what colours does it absorb?

II. End Of unit Problems

12. A concave lens of focal length 15 cm forms an image 10cm from the lens. How far is the object placed from the lens?
13. An object is placed at a distance of 10 cm from a convex mirror of focal length 15cm. Find the position and nature of the image.
14. An object 5cm in length is placed at a distance of 20cm in front of a convex mirror of curvature 30cm. Find the position of the image, its nature and size.
15. An object of size 7cm is placed at 27cm in front of a concave mirror of focal length 18cm. Find the size, position and nature of the image.
16. An object 5cm in length is held 25cm away from a converging lens of focal length 10cm. Find the position, size and nature of the image.
17. A convex lens produces a virtual image which is four times larger than the object. The image is 15cm from the lens. What is the focal length of the lens ?

Answers to End of unit Questions

1. $C = 3\times10^8$ m/s
2. Because, it cause sun burn and cancel
3. It helps to diagnose to moues with out surgery but over dose of it causes cancer.
4. γ- ray, x- ray, UV light, visible light, infrared, microwave and radio wave
5. Yes It is True statement
6. a) concave mirror b) convex mirror c) concave mirror
7. It means,the image size is equal to the object size.
8. The primary colours of light are red, blue and green, and the primary colours of pigment are red, blue and yellow
9. An object that appears black absorbs all colours and no colour is

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reflected from it

10. No colour can be seen for both shirts

11. Magenta is a combination of red and blue. Hence, the cover of the book must be reflecting blue and red lights and absorbing green.

Solution to the End of unit problems

12. $\frac{1}{f}=\frac{1}{u}+\frac{1}{v}$

$\frac{1}{v}=\frac{1}{f}-\frac{1}{u}=\frac{-1}{15}+\frac{1}{10}=\frac{1}{30}$

$u = 30cm$, one option.

Object should be placed between infinity and lens centre

13. $\frac{1}{f}=\frac{1}{u}+\frac{1}{v}$

$\frac{1}{v}=\frac{1}{f}-\frac{1}{u}=\frac{-1}{15}-\frac{1}{10}=-6$

$v = -6cm$, virtual, erect,diminished

14. $\frac{1}{f}=\frac{1}{u}+\frac{1}{v}$, $f=\frac{R}{2}=\frac{30}{2}=15cm$

$\frac{1}{v}=\frac{1}{f}-\frac{1}{u}=\frac{-1}{15}-\frac{1}{20}=8.6$

$v = 8.6cm$, virtual, erect,diminished

$m=\frac{-v}{u}=\frac{hi}{ho}$

$hi=\frac{vho}{u}=\frac{(8.6)(5)}{20}=2.14cm$

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15. $\frac{1}{f}=\frac{1}{u}+\frac{1}{v}$

$\frac{1}{v}=\frac{1}{f}-\frac{1}{u}=\frac{1}{18}-\frac{1}{27}=-54$

$v=-54cm$, virtual, errect, enlarged

$m=\frac{-v}{u}=\frac{54}{27}=2$

16. $\frac{1}{f}=\frac{1}{u}+\frac{1}{v}$

$\frac{1}{v}=\frac{1}{f}-\frac{1}{u}=\frac{1}{10}-\frac{1}{25}=\frac{50}{3}$

$v=\frac{50}{3}cm$, real, inverted, and diminished

$m=\frac{-v}{u}=\frac{50/3}{25}=0.67$

17. $\frac{1}{f}=\frac{1}{u}+\frac{1}{v}$

$m=\frac{hi}{ho}=\frac{-v}{u}=4$

$u=\frac{15}{4}cm$

$\frac{1}{f}=\frac{1}{u}+\frac{1}{v}=\frac{1}{15/4}-\frac{1}{15}=\frac{3}{15}$

$f=5cm$,

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